

THE TANK GUIDE

A product of FishKeepingLifeCo | Research reference series

Feeding and Diet for Freshwater Aquarium Fish: Amount, Frequency, Variety and Waste

How much, how often, what kind — and how to feed well without turning food into a water-quality problem

Working topic: *"How Much Should I Feed My Fish? Understanding Amount, Frequency, Variety & Diet"*

Document type: Research compilation and data dump (not an episode script, outline, or title)

Intended use: Source material for NotebookLM retrieval during Tank Guide Life episode development

Scope: Freshwater aquarium fish. Ponds, marine systems, and commercial food-fish production appear only as comparison material.

What this document does and does not do

This is an evidence audit, not a rulebook. Every widely repeated aquarium feeding rule — the two-minute rule, the pinch, the weekly fasting day, the pea cure, the bloodworm warning — is traced to its origin where possible and graded for evidentiary support. Where a popular recommendation cannot be traced to reliable evidence, the document says so explicitly rather than repeating it.

It deliberately does **not** replace “feed what they eat in two minutes” with a new universal rule. The central finding of the research is that no single quantity can be correct across species, body sizes, life stages, temperatures, food types, and social structures. What can be specified is a **decision process**, and Part C sets one out.

Executive Research Summary

Twenty findings that should shape the eventual episode. Each is expanded, sourced, and caveated later in the document.

#	Finding
1	There is no defensible universal feeding quantity. Published maintenance energy requirements for ornamental species differ by roughly an order of magnitude across common aquarium fish — a neon tetra measured at about 3.8 mg of feed per day for maintenance versus about 28.5 mg for a moonlight gourami at 26 °C, and roughly 11–26 mg for goldfish of 3.6–11.7 g at 20 °C (Pannevis and Earle 1994b, reported in Sales and Janssens 2003). Any rule expressed as “a pinch per tank” ignores this range entirely.
2	Feeding rate scales with biomass and body size, not fish count. Across aquaculture species, optimal daily ration expressed as a percentage of body weight falls as fish get larger — a standard feeding chart shows about 3.6 % of body weight daily for 0.9 g fish versus about 0.7 % for 450 g fish at the same temperature (World Aquaculture 2009). Ten small tetras and one large cichlid are not comparable feeding problems.
3	Aquatic-veterinary guidance sits near 3 % of body weight daily for adult maintenance. The Merck Veterinary Manual gives roughly 3 % of body weight per day for adult maintenance and up to 5 % for fry and fingerlings, while noting that some large-meal carnivores may be fed only once or twice weekly. This is the most directly transferable published figure, and it is still a starting estimate rather than a prescription.
4	Optimal feeding frequency is set by gut clearance, not by convenience. Appetite returns as the stomach empties. In Nile tilapia at 28 °C, appetite returned roughly four hours after a satiation meal, and feeding at 2–3 hour intervals produced gastric overload (Riche et al. 2004). A meta-analysis across species found the frequency–growth relationship is quadratic, with benefit levelling off and then reversing.
5	For many adult aquarium fish, once or twice daily is defensible; more is usually not. In <i>Betta splendens</i> , growth and fecundity rose with feeding frequency up to two meals per day and then <i>declined</i> at three (James and Sampath 2004). Feed consumed rose while flesh gained fell — the extra food became waste, not fish.
6	Roughly a quarter of teleost fishes have no stomach at all. Agastric species — including the cyprinids that give us goldfish, danios, rasboras, and barbs — have no acid digestion phase and no gastric storage organ. Advice built on the image of a fish “stomach the size of its eye” does not even apply anatomically to a large share of the hobby’s fish.
7	The herbivore-long-gut rule is only partly true. Reported relative gut lengths span 0.2–2.5 times body length in carnivores, 0.6–8.0 in omnivores, and 0.8–15.0 in herbivores — overlapping ranges, not clean categories (FAO). Some herbivores have shorter guts than some carnivores.
8	Food is the dominant nutrient input to a closed aquarium. Nutrient-balance work in aquaculture consistently finds that only about 25–40 % of feed nitrogen and a similar or smaller share of feed phosphorus is retained in fish tissue; the remainder leaves the animal as dissolved ammonia and particulate faeces. Most nitrogen waste is dissolved (60–90 %), while a larger share of phosphorus is particulate.
9	Uneaten food and digested food both load the system, but differently. Uneaten food adds its full nutrient content plus a decomposing organic particle. Digested food adds ammonia promptly and faecal solids that decay more slowly. Overfeeding is therefore not only about visible leftovers.
10	“Overfeeding causes algae” is too simple, but the nutrient pathway is real. Feeding raises nitrogen, phosphorus, and dissolved organic carbon availability. Whether that produces visible algae depends on light, CO ₂ , plant biomass, stability, and existing nutrient ratios. Food is one input among several, not a single cause.
11	Snail population booms track available food. Bladder, ramshorn, and Malaysian trumpet snail populations are food-limited in most aquaria. A population surge is a reasonable prompt to audit feeding, but it is evidence of surplus food, not proof of fish overfeeding specifically.
12	Fish can become obese, and it is a liver problem before it is a cosmetic one. High-fat and high-energy feeding produces visceral fat deposition and hepatic steatosis across many finfish species, with downstream effects on immune function and lipid metabolism. Low-activity, readily fed species in home aquaria are plausible candidates.

#	Finding
13	Fasting has good evidence as a feed-management tool and thin evidence as a health requirement. Two fasting days per week in salmon produced full compensatory growth, lower feed consumption, improved conversion, and reduced nitrogen discharge with no welfare deficit. That is a strong case for fasting as a waste-reduction and portion-discipline strategy. It is not evidence that fish require a fast for digestive health, and there is no aquarium study demonstrating that a weekly fast prevents constipation.
14	Fasting is not equally safe across life stages. Early-stage feed restriction in some species produced long-term growth inhibition rather than full catch-up. Fry, small-bodied species with minimal reserves, gravid livebearers, and sick or recovering fish are the groups where routine fasting is least justified.
15	Variety is nutritional insurance and enrichment more than nutritional necessity. Modern complete diets are formulated against known requirements, and ornamental-species requirement data are thin enough that a single formulation is an educated approximation. Variety hedges that uncertainty and provides behavioural benefit. It also creates the most common accidental overfeeding pathway in the hobby, because treats are usually added to the normal ration rather than substituted for it.
16	Water-soluble vitamins leach fast. A large proportion of vitamin C, vitamin B ₁₂ , choline, and pantothenic acid was lost from some commercial flake diets within 30 seconds of contact with water (Pannevis and Earle 1994a). This is a strong argument for food that is eaten quickly — and against food that sits.
17	Storage is an underrated determinant of diet quality. Vitamin C is heat-, oxygen-, and light-labile; stability ranks freezer > refrigerator > opaque container at room temperature > clear container at room temperature (Soliman et al. 1987). Merck advises discarding a room-temperature container two months after opening. Oversized containers are a quiet nutritional problem.
18	Some very common hobby claims do not survive scrutiny. No fish species has been shown to digest cellulose efficiently, which undercuts wood-and-fibre reasoning applied loosely across plecos. Green peas contain roughly the same crude fibre as many prepared fish foods, which undercuts the pea-as-laxative mechanism. “Bottom feeders eat the leftovers” is a husbandry failure restated as a plan.
19	Begging is a conditioned response, not a hunger readout. Fish entrain strongly to feeding schedules; food-anticipatory activity is a documented circadian phenomenon that persists even when food is withheld. A tank full of fish rushing the glass is evidence of learning, not of need.
20	The practical answer is a calibration loop, not a number. Start below what looks right, observe consumption and body condition, watch nitrate trend and detritus accumulation, and adjust. This is well supported as a method even though the specific starting quantity cannot be specified in advance.

How to Read This Document

This document is written for machine retrieval first and human reading second. Headings are explicit, subheadings are often phrased as questions, key concepts are intentionally restated in more than one section, and evidence strength is labelled inline.

Evidence classification system

Substantive claims carry a bracketed label. The labels are ordered here from strongest to weakest and should be read as a statement about *how much confidence the underlying evidence supports*, not about how often the claim is repeated in the hobby.

Label	What it means	Example
[ESTABLISHED SCIENTIFIC PRINCIPLE]	Basic physiology, chemistry, or biology with no meaningful scientific dispute.	Ammonia is the primary nitrogenous excretory product of most teleost fish.
[STRONG EVIDENCE]	Multiple independent studies, or a well-replicated finding, pointing the same way.	Feeding rate as a percentage of body weight declines as fish grow.
[REASONABLE INFERENCE]	A conclusion that follows logically from established principles but has not been tested directly in the aquarium context.	Reducing feed input reduces nitrogen input to a closed aquarium.
[AQUACULTURE PRACTICE]	Standard practice in commercial production. Often well quantified, but optimised for growth and profit rather than pet longevity.	Feeding to a fixed percentage of biomass adjusted every two weeks.
[AQUATIC-VETERINARY GUIDANCE]	Recommendation from clinical aquatic veterinary literature or reference works.	Approximately 3 % of body weight daily for adult maintenance.
[COMMON AQUARIUM PRACTICE]	Widely done in the hobby with broadly acceptable outcomes; not formally validated.	Feeding once or twice daily at a consistent time.
[USEFUL RULE OF THUMB]	Imprecise but directionally helpful; works because it errs in a safe direction.	“If food is still sitting on the substrate, you fed too much.”
[MANUFACTURER RECOMMENDATION]	From product literature. Useful for composition and particle size; not independent evidence, and commercially interested.	Feeding directions printed on a flake canister.
[ANECDOTAL OBSERVATION]	Reported experience from keepers, breeders, or retailers without systematic measurement.	“My angelfish colour up after frozen food.”
[CONTESTED CLAIM]	Genuine disagreement among informed sources, or evidence pointing both ways.	Whether freeze-dried food must be soaked before feeding.
[UNSUPPORTED AQUARIUM FOLKLORE]	Repeated confidently, traceable to no reliable source, and sometimes contradicted by available data.	“Plecos eat fish waste.”
[INSUFFICIENT EVIDENCE]	A reasonable question that the literature simply has not answered for aquarium species.	Optimal feeding frequency for adult <i>Corydoras</i> .

Structural conventions used throughout

- **Question-formatted subheadings.** Many subsections are titled with the question a hobbyist would actually type or ask, so that retrieval matches natural phrasing.
- **Intentional repetition.** Core concepts — biomass over fish count, food as nutrient input, calibration over prescription — are restated in several sections. This is deliberate for retrieval quality, not an editing oversight.

- **Separation of survival from husbandry.** Where a fish *can* tolerate something is treated as a distinct question from whether it *should* be asked to.
- **Explicit uncertainty.** Where the ornamental-fish literature is absent and the claim rests on extrapolation from food-fish aquaculture, that extrapolation is named.
- **Cross-references.** Related sections are named in text so a retrieval system can follow the thread.

The aquaculture translation problem

Most quantitative fish-nutrition data comes from commercial aquaculture. That literature is enormously useful and is cited heavily here, but it was generated under a different objective function. Commercial trials optimise growth rate, feed conversion ratio, and cost per kilogram of harvested fish over production cycles that end at slaughter. Home aquarium keeping optimises something quite different: long life, sustained condition, colour, behaviour, and stable water quality in a closed box, usually with fish that are already adult and often with several species sharing one meal.

Dimension	Commercial aquaculture	Home aquarium	Consequence for translation
Goal	Maximum growth at minimum feed cost	Long-term health and stable water	Growth-optimal rations are usually above aquarium needs
Life stage	Predominantly juvenile / growing	Predominantly adult / maintenance	Published % body weight figures skew high
Species mix	Monoculture	Mixed community, several trophic types	One ration cannot be optimal for all tank residents
Water exchange	Flow-through, ponds, or heavy RAS turnover	Closed system with periodic partial changes	Waste tolerance is far lower in the aquarium
Measurement	Fish are weighed regularly	Fish are almost never weighed	% body weight is scientifically sound but practically unusable
Time horizon	Months to one production cycle	Years	Chronic effects (obesity, liver) matter more

The single translation rule that matters most

When an aquaculture number is quoted in this document, read it as *an upper bound generated under growth-maximising conditions in growing fish*, not as a target for an adult pet fish in a closed glass box. The direction of the error, if the number is applied literally, is almost always toward overfeeding.

A note on manufacturer information

Feed labels and technical sheets are cited in this document for composition, guaranteed analysis, particle size, physical behaviour in water, and storage direction. They are not treated as independent evidence for health claims. A feeding direction printed on a container is a **[MANUFACTURER RECOMMENDATION]** and is discussed as such in Part LXXIII.

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Feeding Terminology Used in This Document

Definitions are given here once and used consistently. Where hobby usage and scientific usage differ, both are noted.

Term	Definition as used here
Ration	The quantity of food offered over a defined period, usually per day. Distinct from intake, which is what the fish actually eats.
Feeding rate	Ration expressed as a percentage of fish body weight per day (% BW/day). The standard scientific unit.
Satiation / apparent satiation	Feeding until fish visibly stop actively taking food. Standard in feeding trials; the observable endpoint, not a physiological one.
Maintenance ration	The intake at which body weight is held constant — no growth, no loss. Almost always far below satiation.
FCR (feed conversion ratio)	Feed offered divided by weight gained. Lower is more efficient. Useful as a waste proxy: a high FCR means more of the food became waste rather than fish.
Gastric evacuation	The emptying of the stomach after a meal. In stomached species, its rate sets the return of appetite and therefore the sensible interval between meals.
Agastric / stomachless	Having no true stomach and no acid digestion phase. Roughly a quarter of teleost fishes, including cyprinids.
Relative gut length (RGL)	Gut length divided by body length. Broadly higher in herbivores and detritivores, but with wide overlap between dietary categories.
Compensatory growth	Accelerated growth after a period of feed restriction, often with hyperphagia on refeeding.
Aufwuchs	The mixed community of algae, biofilm, microfauna, and trapped detritus growing on submerged surfaces. A real dietary category, not a synonym for algae.
Detritus	Particulate organic matter in decomposition, together with its associated bacteria, fungi, and microfauna. The microbial load, not the dead plant matter, is usually the nutritional value.
Biofilm	The bacterial and microalgal surface layer that forms on any wetted surface in an established aquarium. A genuine food source for grazers and for fry.
Nutrient leaching	Loss of water-soluble nutrients from food into the water before ingestion. Fastest for water-soluble vitamins and for finely divided or long-immersed food.
Bioload	Loose hobby term for the waste-generating burden placed on a system. In this document it is used only where the underlying quantity — nitrogen input, oxygen demand, organic loading — is specified.
Target feeding	Deliberate delivery of food to a specific animal or location, usually with a pipette, tube, tongs, or feeding station.
Feeding zone	The vertical stratum where a species preferentially takes food: surface, midwater, or benthic (bottom).
Body condition	Visual or measured assessment of an animal's energy reserves and muscle mass relative to its skeletal frame.
Food-anticipatory activity (FAA)	Increased locomotor activity before an expected feeding time. A documented circadian phenomenon in fish. Commonly misread by keepers as hunger.

Part I — What Do Freshwater Aquarium Fish Actually Eat?

The hobby compresses an enormous range of feeding ecology into three words. Understanding why those three words fail is the foundation for everything that follows about quantity and frequency.

Why “herbivore, carnivore, omnivore” is not enough

[ESTABLISHED SCIENTIFIC PRINCIPLE] Trophic categories in fishes are descriptive conveniences applied to a continuum. They compress at least four independent variables: what the animal eats, where in the water column it eats, when it eats, and what its gut can actually process. Two fish labelled “omnivore” may share almost nothing operationally.

Consider two fish sold in the same store on the same day. A hillstream loach and a common platy are both reasonably called omnivores. One is a benthic grazer adapted to scraping aufwuchs from current-swept rock, with a mouth positioned ventrally and a feeding rhythm tied to surface films. The other is a midwater-to-surface opportunist that takes particulate food from the water column and grazes soft plant matter. A single floating flake serves one of them adequately and the other barely at all. The shared label predicted nothing useful.

Dietary strategies found in freshwater aquarium fish

The categories below overlap, and most species occupy more than one. They are listed separately because each implies a different practical feeding problem.

Strategy	Typical natural foods	Typical adaptations	Captive feeding consequence	Aquarium examples
Carnivore	Animal tissue — fish, crustaceans, insects, worms	Typically shorter gut, true stomach, acid phase, high protease activity	Larger discrete meals; longer intervals tolerated; high protein requirement	Oscar, pike cichlid, many predatory catfish
Insectivore	Aquatic and terrestrial insects and their larvae	Surface- or midwater-oriented; often upturned mouth	Small, frequent, particulate meals; floating and slow-sinking food suits them	Betta, many killifish, hatchetfish, some rasboras
Piscivore	Whole fish	Large gape, distensible stomach, slow evacuation	Infrequent large meals; the group where multi-day intervals are genuinely normal	Arowana, large pike cichlids, snakeheads
Planktivore	Suspended zooplankton and phytoplankton	Fine gill rakers; continuous small-particle intake	Frequent tiny particles; a single large pellet may be unusable	Many small tetras and rasboras in the wild
Herbivore	Macrophytes, filamentous algae, plant tissue	Often long gut, sometimes agastric; hindgut microbial fermentation in some taxa	Near-continuous intake; benefits from grazeable material, not just meals	Silver dollar, some mbuna, tinfoil barb
Algivore	Algae scraped from surfaces	Ventral mouth, scraping dentition, long gut	Needs supplemental food when tank algae is limited	Otocinclus, Siamese algae eater, many <i>Hypostomus</i>
Aufwuchs grazer	The whole surface community: algae, biofilm, microcrustaceans, detritus	Grazing dentition; near-continuous feeding rhythm	Is <i>not</i> a strict herbivore — the animal fraction matters	Mbuna, many rift lake cichlids
Detritivore	Decomposing organic matter and its microbial load	Long gut, high throughput, low assimilation per unit intake	High-volume, low-value intake; supplemental feeding still required in captivity	Many <i>Panaque/Panaqolus</i> , some loaches
Benthic / substrate feeder	Invertebrates and organic particles in or on the substrate	Barbels, taste receptors, sifting behaviour	Requires food that reaches the bottom intact and can be found by smell and touch	<i>Corydoras</i> , kuhli loach, many catfish
Surface feeder	Items at or near the air-water interface	Superior (upturned) mouth	Sinking food is functionally invisible to them	Hatchetfish, many anabantoids, guppies

Strategy	Typical natural foods	Typical adaptations	Captive feeding consequence	Aquarium examples
Midwater feeder	Items in the water column	Terminal mouth	Slow-sinking food maximises access	Most tetras, danios, barbs
Nocturnal feeder	Whatever is available after dark	Reduced eye reliance; chemoreception and lateral line	Daytime feeding may never reach them at all	Kuhli loach, many <i>Corydoras</i> to a degree, most <i>Ancistrus</i> , <i>synodontis</i>

Table I-1. Dietary strategies and their practical consequences. Note that the fourth column — the captive consequence — is what actually changes a keeper's behaviour, and that it varies even within a single label.

How do wild diets differ from captive diets?

[STRONG EVIDENCE] Wild diets are seasonal, variable, spatially patchy, and typically far more dilute in energy than prepared aquarium food. Captive diets are consistent, concentrated, predictable, and delivered without foraging cost.

Four differences matter for feeding decisions:

- **Energy density.** Aufwuchs, detritus, and most wild plant matter are bulky and low in digestible energy. A grazer that spends much of its active day feeding in the wild is not consuming the caloric equivalent of a day's worth of extruded pellets. Reproducing wild *time spent feeding* with modern prepared food would massively overfeed the animal.
- **Foraging cost.** Wild fish expend energy searching, chasing, scraping, and evading. Aquarium fish expend almost none. The same intake therefore produces more surplus in captivity.
- **Seasonality and scarcity.** Many aquarium species originate from systems with pronounced wet and dry seasons, flood pulses, and months of relative scarcity. Periods without abundant food are a normal part of their evolutionary history — a point that matters in Part X on fasting.
- **Nutrient profile.** Wild invertebrate prey differs in fatty acid and micronutrient profile from cultured prey. Cultured *Artemia*, for example, varies in n-3 HUFA content by strain and batch to the point that enrichment is a standard commercial step, not an optional refinement.

The wild-diet fallacy in aquarium feeding

It is tempting to reason: “this fish grazes all day in nature, therefore it should have food available all day.” The inference is unsound, because it holds feeding *duration* constant while quietly multiplying feeding *value* by a factor of many. The defensible version is: “this fish is adapted to frequent small intake, therefore its daily ration is better divided into several small portions, or supplied partly as low-value grazeable material such as an established biofilm or a vegetable.”

This distinction recurs throughout the document. **[REASONABLE INFERENCE]**

What does a species label actually tell a keeper?

A useful diagnostic sequence for any new fish, in order of how much it changes feeding behaviour:

1. **Where does it feed?** Surface, midwater, or bottom. This determines food *form* and is the single most common cause of an animal going underfed in a well-fed tank.
2. **When does it feed?** Diurnal, crepuscular, or nocturnal. This determines feeding *time*.
3. **What particle size can it handle?** Gape and mouth morphology. This determines food *size*.
4. **What is its dietary strategy?** Protein-dominated, plant-dominated, or mixed. This determines food *type*.
5. **How fast and how competitively does it feed?** This determines feeding *method* — broadcast, multiple stations, or target feeding.

Note that the trophic label — the thing on the store sign — is fourth on this list, not first. Cross-reference Part LV on community tank feeding and Part LIII on bottom dwellers.

Part II — Digestive Physiology in Plain Terms

What a fish can do with food constrains how much and how often it should be offered. This section covers the anatomy and biochemistry in beginner-accessible terms and then tests two common hobby generalisations against it.

Do all fish have a stomach?

[ESTABLISHED SCIENTIFIC PRINCIPLE] No. Roughly a quarter of fish species lack a true stomach. These agastric fishes have no acid phase of digestion at all, even where the anterior midgut forms a stomach-like bulb. The generalisation holds without known exception: no stomach means no gastric acid phase (FAO, *Digestion in Teleost Fishes*; FISHBIO).

This is not an obscure technicality for aquarists. The Cyprinidae — goldfish, koi, danios, rasboras, barbs, White Cloud Mountain minnows — are agastric, possessing an intestinal bulb rather than a stomach. Between the cyprinids and the loricariids, a large fraction of a typical freshwater aquarium may have no stomach.

Group	Stomach?	Practical implication
Cyprinids (goldfish, koi, danios, rasboras, barbs)	No (agastric)	No gastric storage or acid phase. Continuous or frequent small intake suits them better than one large meal. “Stomach size” arguments do not apply.
Characins (tetras, pencilfish, silver dollars)	Yes, usually small	Small meals, moderate frequency. Rapid transit.
Cichlids	Yes	True gastric digestion; large-meal capacity varies enormously by species from aufwuchs grazers to piscivores.
Anabantoids (bettas, gouramis)	Yes, small	Discrete small meals; capacity genuinely limited by body size.
Loricariids (plecos, otocinclus)	Yes, modified in some genera	Very long gut in detritivores; high throughput, low assimilation efficiency.
Callichthyids (<i>Corydoras</i>)	Yes	Also capable of aerial respiration; benthic foraging with high search time.
Cobitids and related loaches	Yes	Benthic, often nocturnal or crepuscular; food must reach the substrate and persist.
Poeciliids (guppies, mollies, platies, swordtails)	Yes, small	Near-continuous grazers with small capacity; frequent small portions suit them.

Table II-1. Stomach presence across common aquarium families. Assignments are family-level generalisations; species-level exceptions exist.

How does gut length relate to diet?

[STRONG EVIDENCE] Gut length correlates with diet in a broad statistical sense but does not sort species cleanly. Reported relative gut lengths span roughly 0.2–2.5 times body length in carnivores, 0.6–8.0 in omnivores, and 0.8–15.0 in herbivores. The longest guts are in herbivores, but some herbivores have shorter guts than some carnivores (FAO).

Published relative gut length (RGL) values illustrate the range: common carp feeding on algae and detritus around 15.5, a *Barbus* plant-feeder around 2.8–3.1, a zooplanktivore around 0.7, and carnivorous rainbow trout around 0.9 (De Silva and Anderson 1995, as reproduced in teaching material). Part of the scatter is explained by ingested indigestible material: a detritivore swallowing substrate and mud needs gut volume for reasons unrelated to the chemistry of what it is digesting.

What do digestive enzymes tell us?

[STRONG EVIDENCE] Amylase activity — the starch-digesting enzyme — separates dietary groups more reliably than protease activity does. Herbivorous and omnivorous cyprinids show α -amylase activity greater than, and protease activity similar to or greater than, that of unrelated carnivores. Hidalgo et al. (1999) concluded that protease activity gives little insight into dietary habit, while amylase activity separates carnivores with stomachs from omnivores without them.

Work on stomachless beloniform fishes reinforced how flexible the simple gut is: herbivorous halfbeaks showed significantly greater α -amylase activity than carnivorous needlefish, but trypsin, lipase, aminopeptidase, and maltase activity were not substantially different between the two families (Day et al. 2011). In other words, the protein- and fat-handling machinery is broadly shared; the carbohydrate machinery is what specialises.

Nutrient class	Where digested	Species variation	Aquarium relevance
Protein	Stomach (acid + pepsin) where present, then intestine (trypsin, chymotrypsin, peptidases)	Protease activity is broadly similar across dietary types	Nearly all aquarium fish digest protein well. Excess protein is not wasted biologically so much as converted to ammonia.
Carbohydrate	Intestine (α -amylase, maltase); hindgut microbial fermentation in some herbivores	Large. Warmwater omnivores and herbivores handle starch far better than coldwater and carnivorous species, which use it poorly or not at all	The main axis on which a “cichlid food” can be wrong for a given cichlid.
Lipid	Intestine, bile-assisted; lipase activity broadly conserved	Moderate; requirement for specific fatty acids varies	Digestibility is high, which is exactly why high-fat feeding becomes an obesity problem rather than passing through.
Cellulose	Not efficiently digested by any fish species studied	Effectively none — this is a general finding	Undercuts fibre-based reasoning applied to wood-eating and plant-eating fish alike. See Part XLII.
Chitin	Partially digestible via chitinase in some species	Substantial; digestibility falls at higher dietary inclusion	Relevant to insect-based foods and to crustacean prey. See Part LXXVI.

Is it true that herbivores need to eat constantly?

[REASONABLE INFERENCE] The claim is directionally sound for wild herbivores eating bulky, low-energy plant material, and it is the reason many herbivores have long guts and high throughput. It does not follow that an aquarium herbivore fed concentrated prepared food needs constant access to that food.

The mechanism behind the claim is real: low energy density plus poor cellulose digestion means a wild herbivore must process large volumes to extract adequate energy. But the aquarium keeper has substituted a food whose energy density may be five or ten times higher. Constant access to *that* is overfeeding, not natural feeding.

The husbandry translation that survives scrutiny: herbivores and grazers benefit from having *something to graze* — established biofilm, algae, a vegetable clip, soft plants — alongside a modest prepared ration, rather than from more prepared food more often. This distinguishes behavioural need from caloric need. Cross-reference Part XXVIII on fibre, Part XXXIX on vegetables, and Part XCVII on natural foods.

Is it true that carnivores should eat less often?

[REASONABLE INFERENCE] Partly true, and the reason is mechanical rather than categorical. Large-meal predators have distensible stomachs and slow gastric evacuation, so their appetite genuinely returns slowly. The Merck Veterinary Manual reflects this in noting that carnivorous fish taking large meals may be fed once or twice weekly. But a betta is a carnivore too, and it is an insectivore taking many tiny prey items — a completely different feeding rhythm.

The generalisation fails because “carnivore” conflates prey size with prey type. The operative variable is **meal size relative to gut capacity**, not whether the meal was animal or plant. Restated in a defensible form: *fish that naturally take large infrequent prey tolerate and often prefer larger, less frequent meals; fish that naturally take small frequent prey do better on small frequent meals, regardless of whether that prey is animal or vegetable.*

Two hobby generalisations, restated

Instead of: “Herbivores need to eat constantly.”

Use: “Grazers benefit from continuously available low-value grazing material plus a modest measured ration, not from constant access to concentrated food.”

Instead of: “Carnivores should eat less often.”

Use: “Large-prey predators tolerate infrequent large meals. Small-prey insectivores — also carnivores — do not.”

How fast does digestion actually happen?

[STRONG EVIDENCE] Gastric evacuation is temperature-dependent, meal-size dependent, and species-specific, and it governs the return of appetite. In Nile tilapia at 28 °C the return of appetite after a satiation meal was approximately four hours; fish fed at 4–5 hour intervals consumed roughly what they had evacuated, while fish fed at 2–3 hour intervals showed gastric overload (Riche et al. 2004).

Two consequences follow directly. First, meaningful feeding intervals for a warmwater aquarium fish are measured in hours, not minutes — which is why several feedings crammed into an evening are worse than the same total split across morning and night. Second, because evacuation is temperature-driven, the same fish at 22 °C and 28 °C is not the same feeding problem. Cross-reference Part LXXXV.

[STRONG EVIDENCE] Wood-eating loricariids pass wood through the gut in under four hours and cannot assimilate significant amounts of cellulose from it (German and Bittong 2009). No fish species has been shown to efficiently digest cellulose. This is the single most useful physiological fact for correcting pleco folklore, and it is expanded in Part XLII.

Part III — How Much Should You Feed?

This is the question the eventual episode exists to answer, and it is the question with the weakest evidentiary foundation in the hobby. This section traces each common rule to its origin, identifies what it is actually measuring, and specifies the conditions under which it breaks.

Where do aquarium feeding rules come from?

None of the common hobby quantity rules traces to a published feeding trial in ornamental fish. They appear to have arisen as practical retail and hobby heuristics, propagated through aquarium literature and later through forums and video, and to have drifted in specifics over time. The clearest evidence of drift is that the same rule circulates simultaneously as 30 seconds, one minute, two minutes, three minutes, and five minutes, with each version stated confidently.

[UNSUPPORTED AQUARIUM FOLKLORE] No primary source has been located for the origin of the “two-minute rule” or of “a pinch.” Contemporary retail and hobby sources state the interval variously as 2–3 minutes, 4–5 minutes, and 5 minutes, within the same segment of the industry. That inconsistency is itself the strongest evidence that the specific number is not derived from measurement.

Rule-by-rule audit of common quantity advice

Feeding rule	What people are actually using it for	Where it breaks down	Verdict
“Feed what they eat in 2 minutes”	A visible stopping signal that prevents open-ended feeding and keeps food from sitting	Fast surface feeders can consume a week’s worth of energy in two minutes. Slow, timid, benthic, and nocturnal fish cannot eat anything meaningful in two minutes. The rule measures competitive feeding speed, not nutritional need.	[USEFUL RULE OF THUMB] for a single-species tank of moderate feeders; [CONTESTED CLAIM] for mixed communities
“Feed a pinch”	A portion small enough to be hard to get catastrophically wrong	A pinch is not a unit. It varies by hand size, by food form (flake versus pellet mass differs several-fold at equal volume), and by tank population. A pinch for a 10-gallon shrimp-and-tetra tank and a pinch for a 75-gallon cichlid tank are the same gesture and wildly different rations.	[USEFUL RULE OF THUMB] only as a lower bound
“One eye-sized amount” / “stomach the size of its eye”	Constraining portion size for small fish, especially bettas	The anatomical claim is not accurate as stated, and it does not apply at all to the many agastric species. As a portion heuristic for a small insectivore it is conservative and therefore relatively safe.	[UNSUPPORTED AQUARIUM FOLKLORE] as anatomy; [USEFUL RULE OF THUMB] as portion discipline
“What they eat in 30 seconds”	A stricter version for small tanks and bettas	Same failure mode as the 2-minute rule, applied harder. Very likely to underfeed slow feeders in a community.	[USEFUL RULE OF THUMB] for single small fish
“What they eat in 3–5 minutes”	Giving slower feeders a chance	Over five minutes, water-soluble vitamins have largely leached and fast feeders have long since finished. Extending the window usually increases total food rather than improving distribution.	[CONTESTED CLAIM]
“Feed until the belly rounds slightly”	A body-based rather than clock-based endpoint	Belly profile is confounded by gravidity in livebearers, by body shape in fancy varieties, and by gas. It is also a single-meal signal being used to judge a chronic variable.	[USEFUL RULE OF THUMB] for fry and for target-fed individuals; unreliable in groups
“Feed a percentage of body weight”	The scientifically correct formulation	Requires knowing fish weight. Almost no hobbyist weighs fish, and weighing them requires handling that is itself stressful.	[STRONG EVIDENCE] as science; impractical as day-to-day method

Feeding rule	What people are actually using it for	Where it breaks down	Verdict
“Feed until they stop being interested”	Using the animal's own signal	Many aquarium fish are opportunistic feeders whose feeding reflex triggers on food presence rather than on satiety. Interest can outlast need.	[CONTESTED CLAIM]

Table III-1. Audit of common quantity rules. Note the pattern: the rules that survive do so because they fail safe, not because they are accurate.

Why does the two-minute rule persist if it is unreliable?

Because it usually errs in the right direction and because it solves a real behavioural problem. The rule's genuine function is to stop a keeper from continuing to add food while fish continue to appear enthusiastic — which, given food-anticipatory behaviour and opportunistic feeding reflexes (Part XXII), is otherwise an open loop. Any stopping rule beats no stopping rule.

Its failure is not that it is too strict but that it is the *wrong kind of measurement*. It measures the aggregate consumption speed of whichever fish are fastest, and then infers from that a quantity intended to feed every fish in the tank. In a single-species tank of similar-sized moderate feeders, that inference is roughly acceptable. In a community tank with hatchetfish at the surface, tetras in midwater, and *Corydoras* and kuhli loaches on the bottom, the fastest feeders set a timer that the slowest feeders can never meet.

What should replace it — and what should not

The research does not support replacing the two-minute rule with a different single number. Every candidate replacement fails for the same structural reason: quantity depends on species, biomass, temperature, food energy density, competition, and life stage simultaneously.

What the evidence *does* support is replacing a single measurement with a **multi-signal calibration loop**: a conservative starting portion, direct observation of consumption and distribution, periodic body-condition assessment, and monitoring of nitrate trend and detritus accumulation. This is developed fully in Parts C, CI, and CII.

What is actually known about small-fish maintenance quantity?

[STRONG EVIDENCE] Maintenance feeding requirements have been directly measured for several popular ornamental species. Pannevis and Earle (1994b), as reported by Sales and Janssens (2003), give: neon tetra (0.18 g) about 3.8 mg feed/day (68 J digestible energy); leopard danio (0.30 g) under 7.2 mg/day; kribensis (1.02 g) under 10.2 mg/day; moonlight gourami (1.87 g) under 28.5 mg/day — all at 26 °C. Goldfish of 3.59–11.66 g at 20 °C required approximately 11.5–25.8 mg feed/day.

Species	Body weight	Maintenance feed	Maintenance energy	Water temp
Neon tetra (<i>Paracheirodon innesi</i>)	0.18 g	3.8 mg/day	68 J DE/day	26 °C
Leopard danio (<i>Brachydanio rerio</i>)	0.30 g	<7.2 mg/day	<128 J DE/day	26 °C
Kribensis (<i>Pelvicachromis pulcher</i>)	1.02 g	<10.2 mg/day	<182 J DE/day	26 °C
Moonlight gourami (<i>Trichogaster microlepis</i>)	1.87 g	<28.5 mg/day	<508 J DE/day	26 °C
Goldfish (<i>Carassius auratus</i>)	3.59 g	14.36 mg/day	239 J DE/day	20 °C
Goldfish	4.78 g	11.47 mg/day	191 J DE/day	20 °C
Goldfish	8.06 g	25.79 mg/day	429 J DE/day	20 °C
Goldfish	11.66 g	18.33 mg/day	306 J DE/day	20 °C

Table III-2. Measured maintenance feeding requirements for five popular ornamental species (Pannevis and Earle 1994b, as tabulated in Sales and Janssens 2003). Goldfish diet 19.38 kJ/g gross energy, 16.65 kJ/g digestible energy; other species 19.66 kJ/g GE, 17.83 kJ/g DE. Note that the goldfish values are not monotonic with body weight — individual variation is real and substantial.

What these numbers mean for a real aquarium

Maintenance is the intake at which weight is held constant. It is a *floor*, not a target: fish that are still growing, breeding, recovering, or simply active need more. But the magnitudes are sobering. A neon tetra's measured maintenance ration is under four milligrams of dry feed per day. A school of twelve neon tetras therefore sits near 45 mg per day for maintenance — well under a tenth of a gram.

For scale, a single standard flake is on the order of a few milligrams, and a level small pellet scoop can easily exceed half a gram. It is entirely possible to deliver ten times a small school's maintenance requirement in one casual pinch, and to do it twice a day.

This is the strongest available quantitative answer to why overfeeding is so easy and so common. [\[REASONABLE INFERENCE\]](#)

What variables make any quantity rule break down?

Variable	Direction of effect	Magnitude	Notes
Body size	Smaller fish need more per unit weight	Large — roughly 5× across the size range in feeding charts	The dominant driver of % BW figures
Life stage	Fry > juvenile > adult > senior	Large	Merck: up to 5% BW for fry vs ~3% adult maintenance
Temperature	Higher temp raises metabolism and intake	Large within a species' range	Also raises oxygen demand; see Part LXXXIV
Food energy density	Denser food means smaller volume needed	Large — flake and pellet differ severalfold by volume	The reason volume-based rules are unreliable
Activity level	Active swimmers need more	Moderate	A danio and a bristlenose of equal mass are not equal
Reproductive status	Gravid and conditioning fish need more	Moderate	Livebearers especially
Competition	Changes distribution, not total requirement	Large in practice	The single most common cause of simultaneous over- and underfeeding in one tank
Available natural food	Established tanks supply some intake	Small to moderate; species-dependent	Real but routinely overestimated; see Part XCVII
Health status	Illness usually reduces intake	Variable	Appetite loss is diagnostic; see Part LXXXI

Part IV — Feeding by Body Weight

Percentage of body weight per day is how fish nutrition science actually expresses ration. This section reports the numbers, explains what drives them, and then explains carefully why the method is scientifically sound and practically unusable in a home aquarium.

What feeding rates does the literature report?

[STRONG EVIDENCE] Optimal feeding rates in controlled trials typically fall between roughly 1 % and 5 % of body weight per day for growing fish, with the optimum falling as fish grow and rising with temperature within the species' tolerance range.

Species / context	Reported optimal or used rate	Conditions	Source context
Starry flounder juveniles	1.6–2.4% BW/day optimised growth, nutrient deposition and immune function; 2.4% for growth, 1.7% for innate immunity	183 g initial weight, 10-week trial, fed twice daily	Peer-reviewed feeding trial
Mandarin fish juveniles	Growth increased up to 2.5% BW/day, no further gain above; feed efficiency and protein retention <i>fell</i> at 3.0%	18.4 g initial, RAS, 10 weeks, 55% protein diet	Peer-reviewed feeding trial
Juvenile meagre	Optimum thermal growth coefficient at 2.2% and 1.73% BW/day for 53 g and 188 g fish respectively	Two size classes	Peer-reviewed; shows size effect within one species
Largemouth bass fingerlings	3% treatment gave lowest FCR (0.83) with no difference in final biomass versus 5% and 7%	30 °C, 28 days	Higher rates added waste, not fish
Generic warmwater culture chart	3.6% BW/day for 0.9 g fish; 0.7% BW/day for 450 g fish	Same temperature band (4.4–5.6 °C in the cited chart)	Illustrates the size effect starkly
Tilapia pond culture	Commonly ~2–5% BW/day	Varies with size, temperature, oxygen, feed quality	Industry practice
Adult ornamental maintenance	~3% BW/day; fry and fingerlings up to 5% BW/day	Aquarium context	Merck Veterinary Manual

Table IV-1. Reported feeding rates. Note the recurring pattern in the trial data: growth plateaus while feed intake keeps rising, so the rate above the plateau converts almost entirely into waste.

Why does feeding percentage fall as fish grow?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Metabolic rate scales with body mass at an exponent below one. A fish twice as heavy does not need twice the maintenance energy. Simultaneously, the proportion of intake devoted to growth falls sharply as an animal approaches adult size. The combined effect is that ration as a percentage of body weight declines steeply with size.

For the aquarium keeper the practical version is blunt: **a mature fish needs proportionally far less food than the same fish did as a juvenile**, and a tank that was fed correctly when its occupants were young is overfed once they are grown. This is a slow, invisible drift, and it is a plausible contributor to the common pattern of a tank that “suddenly” develops nitrate and algae problems after a year or two with no change in routine.

How do the other variables shift the percentage?

Factor	Effect on % BW/day	Mechanism
Decreasing body size	Increases, substantially	Allometric scaling of metabolic rate plus higher growth fraction

Factor	Effect on % BW/day	Mechanism
Rising temperature (within tolerance)	Increases	Ectotherm metabolic rate rises with temperature; digestion and gastric evacuation accelerate
Falling temperature	Decreases, sometimes to zero	Digestion slows or effectively stops at low temperatures — the basis for cold-weather pond feeding protocols
Higher feed energy density	Decreases	Requirement is for energy and nutrients, not mass
Growth or reproduction	Increases	Tissue and gamete synthesis has a real energy cost
High activity	Increases	Locomotor energy expenditure
Approaching satiation	Diminishing returns then reversal	Feed efficiency and protein retention fall above the growth plateau

Why is percentage-of-body-weight feeding impractical at home?

The method requires an estimate of total live biomass, which requires either weighing the fish or estimating their weight. Neither is realistic for most keepers:

- **Weighing requires handling.** Netting, transferring, and weighing fish is stressful and carries injury and infection risk. Commercial operations accept that cost because the economics demand it; a pet keeper has no equivalent justification.
- **Estimation error compounds.** Length-to-weight relationships in fish are approximately cubic, so a 20 % error in estimated length becomes roughly a 70 % error in estimated weight.
- **Food mass is not measured either.** Even a keeper with an accurate biomass figure would need to weigh food to a fraction of a gram. A 0.01 g resolution scale is required for a small tank, and flake food at that scale is difficult to handle.
- **Mixed communities have no single correct rate.** A tank containing juveniles and adults of different species has several different correct percentages simultaneously.

How to use body-weight thinking without weighing anything

The value of % BW/day for a hobbyist is **conceptual, not arithmetic**. Three usable takeaways:

1. **Small fish are not small eaters proportionally.** Fry and tiny species need a much higher share of their body weight daily than adults do — which is why they need more frequent, not just larger, feedings.
2. **Adults need remarkably little.** Roughly 3 % of body weight per day for maintenance means an adult fish weighing 10 g needs on the order of 0.3 g of food daily — and most aquarium fish weigh far less than 10 g.
3. **The total is set by mass in the tank, not by the number of animals.** This is developed in Part V.

What about feed conversion ratio as a waste proxy?

[REASONABLE INFERENCE] FCR is the cleanest single expression of the relationship between feeding and waste. Commercial salmonid systems achieve FCR near 1.0; a trial FCR of 0.83 was reported in largemouth bass at the lowest feeding rate tested, rising as rate rose. Every unit by which FCR exceeds the achievable minimum represents feed that became waste rather than fish.

An aquarium's effective FCR is, for adult fish at maintenance, essentially infinite — the fish are not gaining weight, so *all* of the nutrient content of the food eventually leaves the animal as waste. This is an underappreciated point. In a tank of grown fish, feeding is not building anything; it is sustaining metabolism and delivering nitrogen and phosphorus to the water column. Cross-reference Parts XIV, LXVII, and LXVIII.

Part V — Fish Count Versus Biomass

“One pinch per tank” is the most common quantity rule in circulation and the easiest to falsify. Feeding scales with the total mass of living fish, not with how many animals there are.

Why is fish count the wrong unit?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Nutritional requirement is a function of metabolic mass. Two tanks with identical fish counts can differ in total biomass by more than an order of magnitude, and therefore in required ration by a similar factor.

The comparison the hobby needs is simple arithmetic. Using approximate adult weights for common aquarium species:

Stocking scenario	Approx. individual mass	Approx. total biomass	Relative daily ration	Practical feeding form
10 small tetras (neon / ember size)	~0.15–0.3 g	~1.5–3 g	Baseline (1×)	Very small quantity of fine particulate food; midwater delivery
10 adult livebearers (platy / molly size)	~2–5 g	~20–50 g	Roughly 10–20× baseline	Larger volume; surface and midwater; plant matter component
10 adult rasboras or danios	~0.5–1.5 g	~5–15 g	Roughly 3–6× baseline	Small particulate, fast-consumed
1 adult mid-size cichlid (e.g. firemouth)	~40–70 g	~40–70 g	Roughly 15–30× baseline	Discrete pellets; can be delivered as countable units
1 adult oscar	~350–1200 g	~350–1200 g	Very large multiple of baseline	Large pellets; meal frequency may drop to alternate days
6 adult <i>Corydoras</i>	~3–8 g	~18–48 g	Roughly 8–18× baseline	Sinking food delivered where they forage — not leftovers

Table V-1. Fish count versus biomass. Individual masses are order-of-magnitude estimates for illustration, not measured values; they vary with strain, age, and condition. The point of the table is the ratio between rows, not the precision of any single figure.

Is biomass a complete answer either?

[REASONABLE INFERENCE] No. Biomass is a better first approximation than count, but it is not sufficient on its own, because equal biomass can have very different requirements.

Three reasons biomass alone still misleads:

- **Many small fish need more than one large fish of equal total mass.** Because per-unit-mass metabolic rate is higher in small animals, twenty 1-gram fish require more total food than one 20-gram fish. The size effect visible in feeding charts — roughly 3.6 % versus 0.7 % of body weight across the size range — is the same phenomenon.
- **Trophic type changes the food, not just the amount.** Thirty grams of mbuna and thirty grams of oscar want different protein levels, different plant fractions, and different feeding frequencies.
- **Access is not distribution.** Total requirement is a tank-level number, but delivery is an individual-level problem. Cross-reference Part XX and Part LV.

The replacement heuristic worth teaching

Instead of: “How many fish do I have?”

Ask: “How much fish do I have, what sizes are they, and which of them can actually reach the food I am about to put in?”

That question cannot be answered with a single pinch, which is precisely the point. [\[REASONABLE INFERENCE\]](#)

Part VI — How Often Should Fish Be Fed?

Frequency has better evidence behind it than quantity does, because it has a measurable physiological driver: the rate at which the gut clears and appetite returns.

What determines the right feeding interval?

[STRONG EVIDENCE] Appetite recovery is closely associated with gastric evacuation rate. The optimal feeding interval aligns with the time required for substantial gastric emptying; a conventional target in the literature is roughly 80 % gastric evacuation. Feeding faster than evacuation produces gastric overload, reduced digestion and absorption efficiency, poorer feed utilisation, and increased body fat.

The tilapia data give a concrete anchor. At 28 °C, appetite returned approximately four hours after a satiation meal. Fish fed at 4–5 hour intervals consumed about what they had evacuated; fish fed at 2–3 hour intervals showed gastric overload (Riche et al. 2004). Because evacuation is strongly temperature-dependent, the equivalent interval is longer at lower temperatures and shorter at higher ones.

Note also that gastric evacuation rate appears to be independent of the feeding frequency the fish is accustomed to — instantaneous evacuation rates in the tilapia work did not differ significantly between three-meal and five-meal regimes. The gut does not speed up to accommodate a more generous schedule.

What does the comparative evidence say about frequency?

Regime	What the evidence shows	Best suited to	Main risk
Once per day	Adequate for many adult fish. In one snapper trial, no significant difference in weight gain between one and two meals daily, and the once-daily group had higher nitrogen retention efficiency, lower FCR, and lower total nitrogen waste output than either twice-daily or alternate-day groups	Adult community fish; keepers prioritising water quality	Slow, timid, or nocturnal fish may get nothing
Twice per day	Optimal in several species. Two meals daily maximised growth and fecundity in <i>Betta splendens</i> , with a decline at three meals. Twice daily to apparent satiation gave optimum-to-maximum weight gain and FCR in juvenile snapper	Growing fish; species with small gut capacity; mixed communities where splitting the ration improves distribution	Easy to double the ration rather than split it
Three to four small meals	A meta-analysis across species found the frequency–average daily gain relationship is quadratic, with many species peaking around four feedings per day. But benefits reversed beyond the optimum in several individual studies	Fry, juveniles, very small species, hospital or conditioning situations	Very difficult to reduce portion size proportionally; the usual outcome is 3–4× the intended ration
Every other day	In a snapper trial, feed intake and weight gain fell relative to daily feeding, and nitrogen waste and FCR were worse than once-daily	Large-prey carnivores; short-term portion recalibration	Not a general improvement; can encourage larger, messier meals
Continuous grazing	Appropriate where the food source is genuinely low-value (established biofilm, algae, soft plants). Not appropriate with concentrated prepared food	Aufwuchs grazers, algivores, herbivores — as a supplement	Automatic-feeder or 'always available' interpretations produce severe overfeeding
Once or twice weekly	Aquatic-veterinary guidance notes that carnivorous fish which ingest large meals may be fed once or twice weekly	Large piscivores and large-meal predators only	Applied to non-predatory species this is simple underfeeding

Table VI-1. Feeding frequency regimes and the evidence for each. The nitrogen-waste findings are particularly relevant to closed aquarium systems.

Is more frequent feeding better?

[STRONG EVIDENCE] No, not beyond a species- and stage-specific optimum. The relationship between feeding frequency and growth is quadratic: growth improves with frequency up to a point and then plateaus or declines. Increasing frequency beyond that point does not improve growth or feed utilisation and can increase body fat.

The betta data illustrate the reversal unusually clearly, because the study reported both feed consumed and flesh gained (James and Sampath 2004, 77-day trial on 30-day-old juveniles):

Feeding regime	Feed consumed (dry wt)	Flesh gained (dry wt)	Feed per unit gain	Reading
1 meal per 3 days	266 mg	61 mg	~4.4	Underfed
1 meal per 2 days	384 mg	116 mg	~3.3	Underfed
1 meal per day	543 mg	175 mg	~3.1	Adequate
2 meals per day	1000 mg	279 mg	~3.6	Best growth
3 meals per day	886 mg	242 mg	~3.7	Worse than 2 meals on both intake and gain

Table VI-2. Feeding frequency in juvenile *Betta splendens*. The three-meal group both ate less and gained less than the two-meal group — the classic signature of exceeding the optimum. Growth and fecundity rates both rose to two meals per day and then declined.

How does frequency interact with total quantity?

[REASONABLE INFERENCE] Frequency and quantity are independent decisions that hobbyists routinely confuse. Splitting a fixed daily ration into more meals is a distribution improvement. Adding meals without reducing portion size is a ration increase disguised as a technique.

The FAO aquaculture manual makes the relationship explicit: for optimum growth and feed conversion, each individual feed should ideally be about 1 % of body weight, so a 5 % daily ration implies five feedings. The number of feedings is derived from the ration, not chosen independently of it.

The frequency decision, stated cleanly

Step 1. Decide the total daily ration (Parts III–V and Part CII).

Step 2. Decide how many meals to divide it into, based on species gut capacity, life stage, and — importantly — how well food reaches every fish in one sitting.

Step 3. Divide. Do not add.

For most adult community aquariums, one or two feedings per day is defensible and well supported. Two feedings usually improves distribution in a mixed community; one feeding usually produces less nitrogen waste. Both are reasonable.

[COMMON AQUARIUM PRACTICE]

Does feeding frequency affect water quality directly?

[STRONG EVIDENCE] Yes, and not only through total quantity. In a snapper trial, the once-daily group had relatively high nitrogen retention efficiency, low FCR, and low total nitrogen waste output compared with twice-daily or alternate-day feeding. Feeding frequency is therefore a water-quality variable in its own right, independent of ration.

The mechanism is efficiency: a gut that is neither overloaded nor empty digests and retains more of what passes through it, so less nitrogen exits as ammonia. Cross-reference Part LXVII.

Part VII — Fry and Juveniles

This section covers why young fish need different feeding, not how to breed fish. Spawning triggers, conditioning, and raising specific species are out of scope.

Why do fry need more frequent feeding?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Three factors compound. First, per-unit-mass metabolic rate is far higher in small animals. Second, growth consumes a much larger share of intake in early life than at any later stage. Third, absolute gut capacity is minimal, so a fry physically cannot hold a day's worth of energy in one meal.

The requirement gap is large and measurable in both directions. On the quantity axis, aquatic-veterinary guidance places fry and fingerlings at up to 5 % of body weight daily against roughly 3 % for adult maintenance. On the nutrient-quality axis, the gap is starker still: goldfish larvae showed a protein requirement of about 53 % of the diet, compared with about 29 % for juveniles of the same species (Fiogbé and Kestemont 1995; Lochmann and Phillips 1994). Essential amino acid requirements for goldfish larvae were reported at roughly two to three times the values published for juvenile tilapia.

Requirement	Goldfish larvae	Juvenile goldfish	Interpretation
Dietary crude protein	~53%	~29%	Larval diets are not simply smaller adult diets
Arginine (g/100 g protein)	7.8	— (tilapia: 2.8)	Amino acid density, not just protein mass, differs
Lysine (g/100 g protein)	11.8	— (tilapia: 3.8)	Same pattern
Histidine (g/100 g protein)	4.1	— (tilapia: 1.1)	Same pattern
Leucine (g/100 g protein)	9.1	— (tilapia: 3.4)	Same pattern

Table VII-1. Larval versus juvenile nutrient requirements in goldfish, with juvenile tilapia values as the available comparison for amino acids (Fiogbé and Kestemont 1995; Jauncey 1983, as tabulated in Sales and Janssens 2003). The authors attribute the elevated larval values to the intense growth phase compressed into about 20 days, and possibly to differences in protein absorption mechanisms in larvae.

What is the water-quality trade-off?

[REASONABLE INFERENCE] Fry feeding is the one situation in normal aquarium keeping where the nutritional requirement genuinely conflicts with water-quality management, and where the resolution is more water changes rather than less food.

Fry need four to six small meals a day of highly nutritious food in a small volume of water, usually without a mature filter running at full capacity because fine food and small fish constrain flow. Every element of that description increases nitrogen loading. The standard resolution in both hobby practice and hatchery practice is aggressive water exchange, not ration restriction — because restricting fry ration has consequences that may not be recoverable.

[STRONG EVIDENCE] Early-stage feed restriction does not reliably produce full compensatory growth. In turbot, early-stage fasting led to long-term growth inhibition and body composition changes. The literature is mixed — compensatory effects have been demonstrated in yellowfin seabream and Sobaity bream after juvenile-stage restriction — but reviews consistently caution that restriction applied at juvenile stages must be evaluated carefully rather than assumed benign.

What food form do fry require?

Particle size is the binding constraint, and it is a hard one. A larva whose mouth gape is too small for the offered food will starve in a tank full of food. Work on angelfish illustrates how fine the margin is: albino angelfish larvae, with a smaller mouth gape than common angelfish, survived at about 59 % when fed rotifers or a microdiet in the first three days post-hatch versus about 2 % when fed solely *Artemia*.

- **Live first foods** — infusoria, rotifers, *Paramecium*, and newly hatched *Artemia* — are standard where gape allows, and have the practical advantage of remaining alive and available in the water column rather than decaying.
- **Decapsulated *Artemia* cysts** were found to be a more hygienic, off-the-shelf alternative to hatched nauplii for growth of adult guppies and fry of guppies, platies, swordtails, mollies, and black neon tetras (Lim et al. 2002a).
- **Microdiets and powdered foods** work but leach rapidly and foul quickly; frequent small deliveries are essential.
- **Established green water and biofilm** supply continuous low-level microfauna. This is genuinely useful for fry in a way that it is not for adults. Cross-reference Part XCVII.

The fry principle worth stating plainly

For fry, the correct answer to “am I feeding too much?” is usually “probably, and you should increase water changes rather than reduce food.” This is the one life stage where the ordinary aquarium advice to err on the side of less is wrong.

[REASONABLE INFERENCE]

Part VIII — Adult Fish

Most fish in most home aquariums are adults at maintenance. This is the population for which the hobby's feeding advice is least well matched to the underlying science, because the science was mostly done on growing fish.

How many meals does a healthy adult fish need?

[CONTESTED CLAIM] There is no universal answer, and the honest position is that one and two meals per day are both defensible for most adult community fish, with the choice depending on species, community composition, and the keeper's priorities.

Adult feeding pattern	Argument for	Argument against	Where it fits best
One meal daily	Lower nitrogen waste output and better nitrogen retention efficiency in trial data; simpler to keep portion-disciplined; adequate growth in species tested	Single delivery window disadvantages slow, shy, benthic, and nocturnal fish	Single-species tanks; tanks with a strong water-quality priority; species with adequate gut capacity
Two meals daily	Improved distribution across a mixed community; better matched to small-capacity species; optimal in several published trials	Requires halving portions, which keepers often fail to do; slightly higher waste output at equal total ration	Mixed communities; small-bodied species; tanks with both surface and bottom feeders
Three or more small meals	Matches natural rhythm for continuous grazers and micro-prey feeders	Growth benefit reverses beyond the optimum in several species; portion control becomes very difficult; waste rises	Rarely justified for adults outside conditioning or hospital contexts
Alternate-day feeding	Reduces total input; can be a deliberate recalibration tool	Worse FCR and higher nitrogen waste per unit gain in trial data; risks larger, messier meals	Large-meal predators; temporary correction of an overfed tank
Once or twice weekly	Appropriate for large-prey piscivores per veterinary guidance	Inappropriate for essentially every small community fish	Large predatory species only

Why are adults so easy to overfeed?

Four compounding reasons, none of which is the keeper's fault:

1. **Requirement fell and routine did not.** A tank stocked with juveniles and fed accordingly is overfed once those fish mature, because % BW/day falls steeply with size. Nothing visibly changes on the day the requirement drops.
2. **Maintenance is far below satiation.** An adult fish at maintenance will still eat enthusiastically well past its requirement; appetite is not calibrated to need in an opportunistic feeder.
3. **Feeding is the keeper's main interaction.** The act of feeding is the relationship, and reducing it feels like neglect.
4. **Conditioned begging reads as hunger.** Food-anticipatory activity is a learned circadian response, and it is extremely convincing. Cross-reference Part XXII.

Do adults need less food than the label suggests?

[REASONABLE INFERENCE] Almost certainly, in many cases. Commercial feeding directions are typically framed around growth-appropriate rations, and manufacturer recommendations are a commercially interested source. They remain useful for particle size and composition and should be treated as an upper bound rather than a target for adult

maintenance feeding. See Part LXXIII.

Part IX — Senior and Older Fish

The thinnest evidence base in this document. Almost no published work addresses geriatric nutrition in ornamental fish, because commercial aquaculture has no geriatric population.

What is actually known about feeding aging fish?

[INSUFFICIENT EVIDENCE] There is no substantial body of research on nutritional requirements of senescent ornamental fish. Recommendations in circulation derive from general animal-nutrition reasoning, clinical observation by aquatic veterinarians, and keeper experience rather than from controlled study in aging aquarium fish.

Sales and Janssens (2003) identified the underlying gap explicitly: research on nutrition of ornamental fish is hampered by a lack of suitable measurement other than growth rate. A senior fish is not growing, so the standard experimental endpoint does not apply. The authors argued that measurement of nutrient requirements needs to be correlated with animal welfare — a field they described as still unexploited in aquatic species. Two decades later, that is still broadly true.

What changes are plausible, and on what basis?

Dimension	Plausible change with age	Basis	Confidence
Portion size	Reduced, in line with reduced growth and activity	Growth fraction of intake reaches zero; activity often declines	Reasonable inference
Feeding frequency	Possibly more, smaller meals if intake capacity or feeding competence declines	Clinical reasoning; parallels other species	Low — untested
Food texture and size	Softer, smaller particles where dentition, jaw function, or capture ability have declined	Direct observation of feeding difficulty	Reasonable inference; observable case by case
Energy density	Possibly lower, to avoid fat deposition in a less active animal	High-fat diets produce hepatic steatosis across many finfish	Reasonable inference
Protein	No clear basis for either raising or lowering	Requirement data are for growth, which no longer applies	Insufficient evidence
Digestibility	Prefer highly digestible ingredients if intake falls	General nutrition principle	Reasonable inference

What is the practical position for an older fish?

The defensible approach is observational rather than prescriptive. Watch for the functional signs — food missed on the strike, food taken and spat, slower approach to the feeding station, declining body condition despite apparently normal feeding — and adjust particle size, softness, delivery location, and portion in response to what is actually observed. Do not preemptively restrict an old fish's ration on the theory that it needs less; confirm from body condition first. Cross-reference Part XXI.

An honest statement for the episode

“We don't really know how an old fish's nutritional needs change, because nobody studies old fish — commercial aquaculture harvests them long before they get old. What we can do is watch the individual and respond to what we actually see.”

This kind of explicit uncertainty is more useful to the audience than confident invention, and it is consistent with the evidence. [INSUFFICIENT EVIDENCE]

Part X — Fasting

The weekly fasting day is one of the most confidently repeated recommendations in freshwater fishkeeping. The evidence supports it, but for a substantially different reason than the one usually given.

What does the fasting literature actually show?

[STRONG EVIDENCE] Controlled fasting cycles are well studied in aquaculture and are generally well tolerated by healthy adult fish. In rainbow trout, fish subjected to fasting cycles of two days per week showed no differences from continuously fed fish in growth parameters or whole-body composition — a full compensatory response. The fasted group also showed lower weekly feed consumption (74.8 versus 87.9 g per fish), improved feed conversion ratio (1.00 versus 1.08), higher protein retention efficiency (30.5 % versus 27.6 %), and reduced nitrogen discharge to the environment (45.5 versus 51.1 kg per tonne of fish produced).

That single result is the strongest quantitative case for fasting available, and it is worth reading closely. The fasted fish were not healthier by any measured physiological index. They grew the same and were composed the same. What changed was **efficiency and waste**: less feed used, better converted, and roughly 11 % less nitrogen discharged per unit of fish produced.

[STRONG EVIDENCE] A separate study of an eight-week fasting period in Atlantic salmon reported that the fast did not reduce welfare status in the short or long term, assessed by external morphology scoring, bone health, and social behaviour monitoring. The authors went so far as to suggest that formulating welfare guidelines for allowable fasting periods may be redundant, because the time required to initiate severe starvation is much longer than any realistically encountered fasting period in that production system. Catch-up growth nonetheless took several months.

Does fasting improve fish health?

[CONTESTED CLAIM] The evidence does not show a direct health benefit from fasting in adequately fed fish. It shows tolerance, efficiency gains, and reduced waste. Claims that fasting improves health generally conflate three different things: fasting being harmless, fasting reducing overfeeding, and fasting being actively beneficial.

Common fasting claim	What the evidence supports	What it does not support	Verdict
“Fish need a fasting day for digestive rest”	Healthy adult fish tolerate 1–2 fasting days per week without measurable growth or composition cost	No study demonstrates a physiological requirement for periodic digestive rest in fish. The gut does not appear to need scheduled downtime the way the phrase implies	[UNSUPPORTED AQUARIUM FOLKLORE] as stated
“Fasting prevents constipation”	Nothing directly. No aquarium study has tested this	No published evidence links a weekly fasting day to reduced incidence of digestive impaction in aquarium fish	[UNSUPPORTED AQUARIUM FOLKLORE]
“Fasting prevents obesity”	Indirectly and substantially — a fasting day removes roughly 14% of weekly intake, which is a meaningful reduction in energy delivery	That the benefit comes from the fast specifically rather than from the reduced total ration	[REASONABLE INFERENCE]
“Fasting reduces waste and improves water quality”	Directly. Reduced feed input, improved FCR, reduced nitrogen discharge in trial data	That the water-quality benefit requires a full fasting day rather than a proportional daily reduction	[STRONG EVIDENCE]
“Fasting is a natural condition for fish”	Wild fish in seasonal systems do experience periods of scarcity, and fish are physiologically equipped to mobilise glycogen and lipid reserves	That naturalness alone justifies the practice, or that it applies equally to all species and life stages	[REASONABLE INFERENCE] with large caveats

Is fasting really a portion-control strategy in disguise?

[REASONABLE INFERENCE] For most home aquariums, yes. A weekly fasting day removes about 14 % of weekly food input without requiring the keeper to measure anything, resist begging behaviour on six other days, or make any judgment about portion size. It is an administratively simple way to reduce a ration that is probably too large.

This reframing is more useful than the digestive-rest story and it is also more honest. It also implies something the standard advice does not: **a keeper who is genuinely feeding appropriately gains very little from a fasting day, and a keeper who is substantially overfeeding gains a great deal.** The practice is a correction, not a requirement — which explains why experienced keepers disagree about it so consistently. They are describing different starting conditions.

Is fasting species-dependent?

[STRONG EVIDENCE] Yes. Compensatory growth responses depend strongly on species, life stage, environmental conditions, and feeding management during recovery. Outcomes range from full compensation through partial compensation to long-term growth inhibition.

Group	Routine weekly fasting?	Reasoning
Healthy adult community fish (tetras, rasboras, barbs, danios, adult livebearers)	Reasonable; low risk	Adequate reserves; tolerance well supported; the ration is often excessive to begin with
Large-prey carnivores and piscivores	Already the normal pattern	Veterinary guidance places some large-meal carnivores at one or two feedings weekly regardless
Adult goldfish and koi	Reasonable	Prone to overfeeding and to fat deposition; frequently overfed in practice
Bettas and small anabantoids	Commonly practised; low risk	Small capacity, frequently overfed; but the fish is small enough that reserves are limited — one day, not several
Fry and small juveniles	No	Early-stage feed restriction has produced long-term growth inhibition in some species; compensatory response is not reliable at this stage
Very small-bodied adult species (chili rasbora, ember tetra, least killifish)	Use caution; short duration only	Minimal absolute energy reserves. Tolerance data come from far larger fish
Gravid livebearers and conditioning breeders	Not advisable	Active gamete and embryo production is an additional energy demand
Sick, recovering, or newly imported fish	No	Already energy-depleted and often already off feed; see Parts LXXXI–LXXXIII
High-metabolism species at high temperature	Use caution	Faster energy turnover; shorter tolerance window
Herbivorous grazers	Partial fasting only	In a planted or biofilm-rich tank these fish continue grazing regardless; a 'fast' from prepared food is not a true fast

Table X-1. Fasting suitability by group. Note that in several rows the answer is not about tolerance but about whether there is anything to gain.

What are the risks of fasting done badly?

- **Compensatory hyperphagia on refeeding.** Fish that have been fasted eat more aggressively afterward. A keeper who fasts and then feeds a larger meal to “make up for it” has increased peak loading without reducing weekly total — the worst of both outcomes.
- **Increased competition.** Hungrier fish compete harder. In tanks with an existing distribution problem, fasting can widen the gap between dominant and subordinate feeders. Cross-reference Part XCIV.

- **Masking an underlying problem.** A fish that is thin for reasons unrelated to ration — parasites, competition, dentition, illness — is made worse by a scheduled fast, and the fast makes the real cause harder to see.
- **Escalation.** Weekly fasting sometimes drifts into multiple fasting days as a water-quality intervention. At that point the keeper is managing water chemistry by restricting an animal's food, which inverts the priority order.

The fasting position this research supports

A weekly fasting day for healthy adult fish is **safe, well-tolerated, and a genuinely effective way to reduce food input and waste output**. It is not a biological requirement, it has no demonstrated effect on constipation in aquarium fish, and it is not appropriate for fry, gravid, sick, or very small-bodied fish.

The most defensible framing for an audience: “A fasting day won't hurt healthy adult fish, and for most tanks it quietly fixes a portion problem. But if you need a fasting day to keep your nitrate under control, the fasting day is treating a symptom.” [\[REASONABLE INFERENCE\]](#)

Part XI — How Long Can Fish Safely Go Without Food?

This section deliberately refuses to produce a universal number, and explains why the number circulating in the hobby is the wrong kind of answer to the question being asked.

Why is “fish can go a week without food” the wrong answer?

[CONTESTED CLAIM] The claim is broadly true as a *survival* statement for healthy adult fish of moderate size, and it is a poor *husbandry* statement because survival is not the standard being asked about. The person asking is usually planning a trip, not conducting a tolerance study.

The distinction matters enough to be stated formally:

Question being asked	What it is really about	Answer type
“Will my fish die if I don't feed them for a week?”	Survival threshold	Usually no, for healthy adults — but this is a low bar
“Is it fine to not feed my fish for a week?”	Husbandry acceptability	Depends entirely on species, size, life stage, and condition
“How long before there are consequences I can't see?”	Sublethal cost	Poorly characterised in aquarium species; reserves are mobilised, condition declines, and recovery is not always complete

What happens physiologically during food deprivation?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Fish respond to food deprivation by mobilising stored energy — glycogen first, then lipid, then protein — while down-regulating metabolic rate to conserve reserves. Metabolic homeostasis is initially maintained through glycogen and lipid mobilisation. As deprivation extends, structural impairments can occur, and severe nutrient deficiency can cause irreversible muscular damage.

Temperature modulates the whole process: low temperatures slow gastrointestinal evacuation and reduce metabolic activity and nutrient turnover, conserving reserves. A fish at 22 °C endures a period without food more comfortably than the same fish at 28 °C.

How long, by group?

[REASONABLE INFERENCE] The ranges below are inferences from physiological principles, aquaculture tolerance data, and clinical aquatic-veterinary practice. They are **not** measured tolerance limits for aquarium species, and they should not be presented as such.

Group	Typically tolerated without harm	Where the limit comes from	Notes
Healthy adult mid-size community fish	Commonly cited as up to ~1 week; longer periods are survivable but increasingly costly	Reserve mobilisation capacity; broad aquaculture tolerance data	Condition declines progressively; there is no bright line
Large-bodied adult fish (large cichlids, adult goldfish and koi)	Longer than small fish — weeks in some documented aquaculture cases	Greater absolute reserves relative to metabolic rate	The eight-week salmon fast is the outer documented case, in a production context
Small-bodied adult fish (nano species)	Substantially shorter than a week is prudent	Minimal absolute reserves; high mass-specific metabolic rate	Rarely studied directly; the safe assumption is low tolerance
Juveniles	Short; days at most	Growth demand plus limited reserves	Risk is stunting, not death

Group	Typically tolerated without harm	Where the limit comes from	Notes
Fry	Very short — hours to a day or two depending on species and yolk absorption	No meaningful reserves after yolk absorption	Fry mortality from missed feedings is fast and often total
Gravid / pregnant livebearers	Shorter than for non-gravid adults	Embryo provisioning is an additional continuous demand	Also at elevated risk of being outcompeted; see Part XX
Sick fish	Assume no reserve margin	Already energy-depleted; often already anorexic	Fasting a sick fish is rarely the intervention it appears to be
Newly imported / recently shipped fish	Short	Shipping depletes reserves and imposes stress	See Part LXXXIII

Survival is not the standard

A fish can survive conditions that shorten its life, reduce its condition, and compromise its immune function. The hobby's reliance on “they'll be fine for a week” answers a question about mortality with a recommendation about husbandry, and the two are not the same. The correct framing for an episode is: “*Most healthy adult fish will survive a week unfed. That does not make a week unfed a plan.*” [\[REASONABLE INFERENCE\]](#)

Part XII — Vacation Feeding

The most common real-world application of the previous section. The evidence base here is largely practice-based rather than experimental, and it is labelled accordingly.

What are the actual options?

Option	How it works	Strengths	Failure modes	Best fit
Leave healthy adults unfed	No intervention	Zero risk of overfeeding; zero equipment failure risk; no water-quality degradation during absence	Not suitable for fry, juveniles, small-bodied species, gravid or sick fish; condition declines over longer absences	Absences of a few days to about a week, established tank, healthy adults
Automatic feeder	Timed mechanical dispenser	Maintains routine; adjustable frequency; good for longer absences	Dispensing too much; humidity causing food to clump and either jam or dump; battery failure; food type incompatibility (flake especially)	Absences beyond a week where a sitter is unavailable
Vacation block / slow-release feeder	Bound food matrix that dissolves gradually	No power or mechanism to fail; simple	Adds binder and often calcium-bearing material to the water; dissolution rate is not controllable; fish may ignore it; if ignored it becomes a large decaying organic mass	Short absences where an auto-feeder is not available; least preferred option in most discussions
Fish sitter	A person feeds on a schedule	Responsive to problems; can spot equipment failure and mortality	Overfeeding by the sitter is the classic failure. Untrained helpers interpret begging as hunger and feed generously	Longer absences, valuable livestock, complex systems
Pre-portioned meals	Sitter is given individual sealed portions	Eliminates sitter overfeeding, the single biggest risk	Requires preparation; sitter may still add extra	Any sitter arrangement — this is the highest-value modification
Reduced feeding before departure	Lower ration in the days before leaving	Reduces waste accumulation while unattended; fish enter the period with a clean system	Not a substitute for any of the above	Preparation step for any strategy

What is the single biggest vacation risk?

[COMMON AQUARIUM PRACTICE] Overfeeding by a well-intentioned helper is the most frequently reported vacation problem in hobby discussions — more commonly cited than starvation. The mechanism is straightforward: the sitter sees the same conditioned begging behaviour the owner sees, has no calibration reference, and errs generously because underfeeding feels cruel.

Pre-portioning into sealed containers, one per feeding, removes the judgment call entirely and is the highest-leverage single change available to a departing keeper. A written instruction to feed less often than normal is also useful; a written instruction that says “do not add more even if they look hungry” is more useful still.

Automatic feeder practicalities

- **Calibrate before departure, not on the day.** Run the feeder for several days while still at home and observe actual dispensed quantity and fish response.
- **Pellets generally dispense more reliably than flake.** Flake bridges, clumps, and dispenses inconsistently in many mechanisms.
- **Humidity is the main enemy.** Mount the feeder away from the water surface where possible, and consider a desiccant if the design allows. Damp food in a hopper can either jam the mechanism or release as a single mass.

- **Set it to underfeed.** A week of slight underfeeding in healthy adults is recoverable. A single hopper dump into an unattended aquarium is not.
- **Check power and battery state.** Battery-powered units are preferable to mains-powered where outages are plausible, since a feeder that stops is far safer than a filter that stops.

Do vacation blocks affect water quality?

[CONTESTED CLAIM] Vacation blocks are widely reported in hobby discussion to affect water parameters, most often through the binder and any calcium-bearing carrier material. Formulations vary substantially between products, and independent testing is not available for most of them. What can be stated confidently is that a block adds a slowly dissolving organic and mineral mass to a closed system with no mechanism to stop if the fish are not eating it.

Vacation feeding hierarchy, most to least defensible

1. Sitter with pre-portioned sealed meals and written instructions.
2. Calibrated automatic feeder, tested before departure, set to underfeed.
3. No feeding at all, for healthy adults, over a short absence.
4. Untrained sitter with a container of food and general instructions.
5. Vacation block.

Note that option 3 outranks option 4. An unfed healthy adult fish for five days is a lesser problem than an aquarium that received five days of enthusiastic feeding from someone who wanted to help. **[COMMON AQUARIUM PRACTICE]**

Preparation checklist before an absence

1. Perform a water change and substrate clean shortly before leaving, so the system starts with minimal accumulated organics.
2. Clean or check mechanical filtration, since an unattended clogged filter compounds every other problem.
3. Reduce feeding modestly for two or three days beforehand rather than feeding heavily to “top up” the fish — which does not work and adds waste.
4. Remove or address anything at risk of dying unattended: an ailing fish, an unstable plant mass, a large uneaten vegetable.
5. Test and calibrate any feeding device across several days of normal presence.
6. Leave written instructions that specify quantity in absolute terms and explicitly instruct against adding extra.

Part XIII — Overfeeding: What It Actually Means

“Overfeeding” is used in the hobby to describe at least three different failures with different mechanisms and different fixes. Separating them is the most useful single contribution this section can make.

What are the three distinct kinds of overfeeding?

Type	Definition	Primary consequence	How you detect it	How you fix it
Type 1: Uneaten food	More food enters the tank than the fish consume	Direct organic loading; decomposition; oxygen demand; visible detritus	Food visible on substrate after feeding; detritus accumulation; filter clogging	Reduce portion; match food form to feeding zone; improve distribution
Type 2: Excess consumed food	Fish eat more than they metabolically require	Higher ammonia and faecal output per fish; fat deposition; possible hepatic effects	Rounded body profile over time; increased faeces; rising nitrate with no visible leftovers	Reduce portion; reduce frequency; reduce energy density
Type 3: Misdirected food	The right total quantity, distributed wrongly	Simultaneous overfeeding and underfeeding within one tank	Some fish rounded, others thin; food missing before it reaches the bottom	Multiple feeding zones; target feeding; food-form change; timing change

Table XIII-1. Three types of overfeeding. Type 2 is the one most often missed, because a tank with no visible leftovers looks like a well-fed tank.

The critical insight for the episode

A tank with no uneaten food can still be badly overfed. If the fish eat everything but eat more than they need, all of that surplus nitrogen and phosphorus still enters the water — just via the fish rather than via decomposition. The classic advice “feed only what they can eat” addresses Type 1 completely and Type 2 not at all.

This is why nitrate can climb steadily in a tank where the keeper has never seen a single leftover flake. [\[REASONABLE INFERENCE\]](#)

What is the full consequence chain?

Each link below is separately supported; the chain as a whole is a reasonable synthesis rather than a single studied pathway.

Consequence	Mechanism	Evidence strength	Caveat
Uneaten food accumulation	Food not consumed settles and decomposes	[ESTABLISHED SCIENTIFIC PRINCIPLE]	Direct observation; no ambiguity
Increased faecal solids	Higher intake produces proportionally higher undigested output	[ESTABLISHED SCIENTIFIC PRINCIPLE]	Amount depends on diet digestibility
Ammonia production	Protein catabolism produces ammonia as the primary nitrogenous waste in teleosts; decomposition of uneaten food adds more	[ESTABLISHED SCIENTIFIC PRINCIPLE]	A cycled filter converts it; ammonia itself may never be measurable
Nitrate accumulation	Nitrification converts ammonia to nitrite to nitrate, which accumulates without export	[ESTABLISHED SCIENTIFIC PRINCIPLE]	Source water and substrate can contribute; not all nitrate is from food
Phosphate accumulation	Feed phosphorus is poorly retained; most exits the animal or the uneaten food into the system	[STRONG EVIDENCE]	Reported dietary P excretion of roughly 69–86% in aquaculture contexts

Consequence	Mechanism	Evidence strength	Caveat
Dissolved organic compounds	Leached nutrients, mucus, and partial decomposition products enter solution	[REASONABLE INFERENCE]	Rarely measured by hobbyists; associated with yellowing and surface film
Heterotrophic bacterial growth	Dissolved and particulate organics are a bacterial substrate	[ESTABLISHED SCIENTIFIC PRINCIPLE]	Usually invisible unless it blooms
Increased biological oxygen demand	Bacterial respiration on organic load consumes dissolved oxygen	[ESTABLISHED SCIENTIFIC PRINCIPLE]	Most consequential at high temperature and in low-circulation tanks
Algae pressure	Elevated N and P availability alongside light and CO ₂	[CONTESTED CLAIM]	One input among several; see Part XV
Cloudy water	Heterotrophic bacterial bloom on excess organics; suspended fine particulate	[REASONABLE INFERENCE]	Several unrelated causes exist; see Part XVI
Pest snail population growth	Snail populations are food-limited; more food supports more snails	[REASONABLE INFERENCE]	Reflects surplus food generally, not fish overfeeding specifically; see Part XVII
Substrate detritus	Settled uneaten food and faeces accumulate in low-flow zones	[ESTABLISHED SCIENTIFIC PRINCIPLE]	Some detritus is normal and biologically useful
Mechanical filter clogging	Increased particulate load saturates mechanical media faster	[REASONABLE INFERENCE]	A useful and often-overlooked diagnostic; see Part LXVI

Separating direct evidence from hobby assumption

It is worth being precise about which parts of the overfeeding story are solidly established and which are inference:

- **Well established:** food is the primary nutrient input to a closed aquarium; nitrogen and phosphorus retention in fish tissue is partial; the remainder enters the water; decomposition of organics consumes oxygen and supports bacterial growth.
- **Reasonable but not directly demonstrated in aquaria:** the quantitative relationship between a given degree of overfeeding and a given rate of nitrate rise; the contribution of feeding to any particular algae outbreak; the threshold at which organic loading produces visible cloudiness.
- **Assumption rather than evidence:** that overfeeding is *the* number one cause of aquarium problems; that any given tank problem can be attributed to feeding without investigating stocking, filtration, maintenance interval, and source water.

Part XIV — Overfeeding and Water Quality

The nutrient accounting. This is the section that makes the “feeding is a water-quality decision” argument quantitatively rather than rhetorically.

Why is food the primary nutrient input to a closed aquarium?

[ESTABLISHED SCIENTIFIC PRINCIPLE] A sealed aquarium is a closed system for nutrients except for three routes in — food, source water, and any added fertiliser or supplement — and three routes out: water changes, harvested biomass (plant trimmings, removed snails, exported fish), and gaseous loss. For most freshwater aquariums without heavy fertiliser dosing, food is the dominant nitrogen and phosphorus input.

Food is one of the primary nutrient inputs into a closed aquarium

Every gram of food added to an aquarium carries nitrogen and phosphorus into a box from which those elements do not leave on their own. Fish retain some of it as tissue. Plants take up some of it and hold it until they are pruned and removed. Everything else accumulates until it is exported by a water change.

This single sentence reframes the feeding question. It is not “how much food do my fish want?” It is “how much nitrogen and phosphorus am I willing to import per week, and how much am I prepared to export?” **[ESTABLISHED SCIENTIFIC PRINCIPLE]**

How much of the food's nitrogen and phosphorus actually stays in the fish?

[STRONG EVIDENCE] Not much. Nutrient-balance studies across aquaculture species report average nitrogen retention in fish biomass of roughly 25–30 %, with a reported range of about 10–49 %, and phosphorus retention of roughly 17–40 % (Piedrahita 2003, as summarised by the Global Seafood Alliance). Commercially fed Nile tilapia has been estimated to retain about 35 % of feed nitrogen and 28 % of feed phosphorus; the remainder becomes waste.

Nutrient	Retained in fish	Excreted / egested	Dominant form of waste	Aquarium implication
Nitrogen	~25–40% typical; range ~10–49%	The remainder	Mostly dissolved — 60–90% as ammonia	Enters the water column promptly; converted by the filter to nitrate, which accumulates
Phosphorus	~17–40% typical	The remainder; dietary P excretion often cited at 69–86%	A larger share is particulate, in faeces (reported 25–85%)	Accumulates in substrate and detritus as well as in solution; removed by substrate cleaning as well as by water change

Table XIV-1. Nutrient fate in fed fish. These figures come from growing production fish; adult aquarium fish at maintenance retain essentially nothing, so the excreted fraction approaches 100 %.

That last point deserves emphasis because it is the single biggest divergence between the aquaculture data and the aquarium situation. A production fish is growing, so a meaningful share of feed nitrogen is locked into new tissue and leaves the system at harvest. An adult ornamental fish at maintenance is not building tissue. Over a year, essentially all of the nitrogen and phosphorus in everything it eats ends up in the water or in the substrate.

How do uneaten food and digested food differ in effect?

Dimension	Uneaten food	Digested food (faeces + excretion)
Nutrient content released	Full nutrient content of the food, including the portion the fish would have retained	The fraction not retained — typically 60–75% of N and 60–83% of P in growing fish; approaching all of it in adults

Dimension	Uneaten food	Digested food (faeces + excretion)
Timing of release	Immediate leaching of solubles; slower release as the particle decomposes	Ammonia released rapidly and continuously via gills; faecal solids release more slowly
Oxygen demand	High — a concentrated organic particle undergoing microbial decomposition	Moderate — faeces are already partially degraded
Physical behaviour	Settles in low-flow zones, lodges in substrate and decor, or is trapped by mechanical filtration	Similar settling behaviour but generally finer and more dispersed
Visibility to the keeper	High — this is what people look for	Low — the reason Type 2 overfeeding goes undetected
Ease of removal	Removable by siphon or net if caught promptly	Ammonia cannot be removed except by dilution or biofiltration; solids are siphonable

Does this mean the goal is a sterile aquarium?

No, and the distinction is important for the brand's framing. Detritus and biofilm are functional parts of an established aquarium: they support microfauna, feed grazers, and host bacterial communities. The objective is not to eliminate organic matter but to **avoid excessive and accelerating accumulation** — a trajectory rather than a state. Cross-reference Parts LXIV and XCVII.

Part XV — Overfeeding and Algae

The most frequently asserted and least rigorously supported link in aquarium feeding advice.

Does overfeeding cause algae?

[CONTESTED CLAIM] Overfeeding increases nutrient availability, and nutrient availability is one determinant of algal growth. It does not follow that overfeeding causes any given algae outbreak, and reducing feeding alone frequently fails to resolve one.

Algae growth in a planted or unplanted aquarium is governed by the interaction of several variables. Food influences one of them:

Variable	Role in algae growth	Relationship to feeding
Light intensity and photoperiod	Primary energy driver; often the binding constraint	None
Nitrogen availability	Required macronutrient	Directly increased by feeding
Phosphorus availability	Required macronutrient; often limiting in unfertilised tanks	Directly increased by feeding
CO ₂ availability and stability	Determines whether higher plants can outcompete algae in a planted system	None directly
Competing plant biomass	Healthy growing plants consume the same nutrients	Indirect — the same nutrients are contested
Dissolved organic carbon	Supports some algal and bacterial growth; associated with biofilm-type problems	Increased by feeding, especially by uneaten and leaching food
Stability of the system	Fluctuation favours opportunistic algae	Erratic feeding contributes to nutrient fluctuation
Surface area and existing algal inoculum	Determines how fast a bloom can express	None

How to state the algae relationship honestly

Do not say: “Overfeeding causes algae.”

Do say: “Overfeeding raises the nutrient and organic load in the tank, which removes one of the brakes on algae. Whether algae actually takes off depends on light, CO₂, plant growth, and stability. Cutting food is worth doing on its own merits, but if you cut food and the algae stays, the problem was never primarily food.”

This framing is consistent with system-level thinking and avoids the single-variable intervention that usually disappoints the viewer. **[REASONABLE INFERENCE]**

Part XVI — Overfeeding and Cloudy Water

A narrower and better-mechanised link than the algae one. This section connects to the existing cloudy-water content cluster without duplicating it.

Can overfeeding cause a bacterial bloom?

[REASONABLE INFERENCE] Yes, plausibly and commonly. Heterotrophic bacteria multiply in response to available dissolved and particulate organic carbon. A sudden increase in organic input — a large feeding, a dead fish, a disturbed substrate, a dumped auto-feeder hopper — provides exactly that substrate. Heterotrophic bacteria can reproduce on the order of hours, which matches the observed speed of aquarium cloudiness.

Cloudy water appearance	Likely mechanism	Feeding involvement	Distinguishing observation
Milky / white haze appearing within hours to a day	Heterotrophic bacterial bloom on dissolved organics	Often — a recent large feeding, decaying food, or dosed organic	Appears suddenly; not green; often follows a specific event
Green water	Free-floating phytoplankton bloom	Contributory via nutrients, but light is usually the driver	Distinctly green; usually associated with strong light
Fine suspended particulate	Disturbed substrate, fine detritus, or inadequate mechanical filtration	Contributory — more food means more fine detritus	Settles when flow stops; clears with mechanical filtration
Yellow or tea-coloured tint	Dissolved organic compounds, tannins	Partly — DOC accumulates with organic loading	Water is clear but coloured; not hazy

The practical diagnostic value is high: a tank that goes cloudy within a day of a feeding change, a new fish food, or a period of unattended automatic feeding is giving a strong signal about organic input. A tank that has been persistently cloudy for weeks with stable feeding is more likely telling a story about filtration, maintenance, or stocking. For the full differential, see the existing cloudy-water article cluster; this document covers only the feeding pathway.

Part XVII — Overfeeding and Pest Snails

A useful diagnostic signal that is routinely over-interpreted in both directions.

Do snail populations track available food?

[REASONABLE INFERENCE] Yes. Bladder snails (*Physella*), ramshorns (*Planorbella*), and Malaysian trumpet snails (*Melanoides*) reproduce in proportion to available resources in a closed system. Population size in an aquarium is primarily food-limited rather than space-limited, so a sustained increase in available organic matter supports a sustained increase in snail numbers.

The population responds with a lag — reproduction, egg development, and growth to visible size take time — which is why a snail boom often appears weeks after the feeding change that enabled it, and why the connection is easy to miss.

Does a snail boom prove overfeeding?

[CONTESTED CLAIM] No. A population increase indicates increased available food, which has several possible sources. Attributing every snail boom to fish overfeeding is a common error.

Possible driver of a snail population increase	Relationship to fish feeding
Uneaten fish food accumulating	Direct
Increased fish faeces and detritus from higher intake	Direct
Decaying plant matter after a trim, a melt, or a plant die-off	None
A dead fish or invertebrate decomposing unnoticed	None
Algae growth providing grazing surface	Indirect at most
Newly introduced snails arriving on plants or decor	None — this is a founder effect, not a growth response
Reduced maintenance allowing detritus to build up	None
Blanched vegetables left in the tank for extended periods	Direct — and frequently overlooked
A recently established tank reaching biofilm maturity	None

Using snails as an instrument rather than a verdict

Snails are a reasonable **prompt to audit**, not a proof of guilt. The productive response to a population boom is to look at what changed: feeding quantity, feeding form, maintenance interval, plant health, vegetable feeding, or the presence of something dead. Shrimp and snails are livestock in their own right in this brand's framing, and a population response is information about the system, not a pest event to be suppressed. **[REASONABLE INFERENCE]**

Part XVIII — Fish Obesity

Under-discussed in the hobby relative to how well documented the underlying physiology is in fish generally.

Can aquarium fish become obese?

[STRONG EVIDENCE] Yes. Excess caloric intake and high dietary fat produce visceral adipose accumulation and hepatic lipid deposition across many finfish species. High-fat diets disrupt lipid homeostasis by downregulating lipolytic genes and upregulating lipogenic genes, producing hepatic lipid accumulation. Most finfish studied exhibit growth inhibition, metabolic dysfunction, and greater disease susceptibility under prolonged high-fat exposure, with species-specific variation.

Zebrafish are an established laboratory model for diet-induced visceral obesity, which is itself strong confirmation that the phenomenon is real, reproducible, and dietary in origin. Excessive visceral adipose accumulation is associated with ectopic lipid deposition that impairs the function of major metabolic organs including the liver, and with systemic inflammatory responses.

What are the documented consequences?

Consequence	Evidence	Aquarium relevance
Hepatic steatosis (fatty liver)	[STRONG EVIDENCE] — well documented across cultured finfish; excessive lipid intake identified as the primary factor in the majority of cultured fish	The most likely serious consequence in a long-lived pet fish. Hepatic lipidosis is reported clinically in ornamental contexts, particularly in African cichlids and rainbowfish
Impaired immune function	[STRONG EVIDENCE] — immune suppression and greater disease susceptibility under prolonged high-fat diets	Plausibly contributes to the pattern of chronically overfed tanks with recurrent disease
Reduced growth	[STRONG EVIDENCE] — growth inhibition in most finfish under prolonged HFD exposure	Counterintuitive and worth stating: more fat does not mean faster growth
Altered lipid metabolism	[STRONG EVIDENCE]	Underlies the other effects
Reduced reproductive performance	[REASONABLE INFERENCE]	Widely reported in husbandry contexts; mechanistic basis is plausible
Reduced mobility	[ANECDOTAL OBSERVATION]	Commonly described by keepers; not systematically studied in ornamental fish
Reduced lifespan	[REASONABLE INFERENCE]	Follows from organ damage and immune effects; not directly measured in ornamental species

Which fish are most at risk?

[REASONABLE INFERENCE] Risk concentrates where three factors coincide: low activity level, enthusiastic and indiscriminate feeding behaviour, and an owner who enjoys feeding. Species-level risk assignments below are inferences from feeding behaviour and clinical reports rather than from controlled study in ornamentals.

Group	Why elevated risk	Contributing feeding practice
Fancy goldfish	Low activity, compressed body, voracious feeder, long-lived, often single-species tank with dedicated feeding	Frequent feeding of energy-dense pellets
Bettas	Very small body, highly responsive to the owner, kept singly so all food is theirs	Frequent feeding of high-fat treat foods, especially bloodworms

Group	Why elevated risk	Contributing feeding practice
African cichlids (particularly aufwuchs-grazing mbuna)	Hepatic lipidosis reported clinically in this group; natural diet is bulky and low-fat but captive diets often are not	Feeding high-protein, high-fat carnivore pellets to a grazer
Rainbowfish	Hepatic lipidosis reported clinically	Energy-dense diets with high feeding frequency
Large cichlids kept singly	Owner interaction focused on feeding; large discrete meals; long life	Feeder fish, high-fat foods, and reward feeding
Adult livebearers	Constant grazers with small capacity kept in well-fed communities	Continuous availability of food
Any single-specimen 'pet' fish	The whole ration goes to one animal and feeding is the relationship	Feeding as interaction rather than as nutrition

What about the food side of the equation?

Two dietary levers matter more than total volume for fat deposition specifically: **dietary lipid level** and **treat frequency**. Frozen bloodworms, freeze-dried invertebrates, and many “colour” and “conditioning” foods are energy-dense relative to a staple pellet. Because these are used as treats, they are usually added to the normal ration rather than substituted for it. Cross-reference Part XXIII on variety and Part XXXIV on bloodworms.

Part XIX — Signs of Overfeeding

Every sign in this section has at least one alternative explanation. The section is structured around that fact, because single-sign diagnosis is the most common analytical error in the hobby.

What are the candidate signs, and what else could cause them?

Possible sign of overfeeding	Other possible explanations	What makes it more likely to be feeding	Diagnostic value
Food remaining after feeding	Food form wrong for the fish present; fish are ill or stressed; water temperature too low; new fish not yet feeding; food is stale or unpalatable	Fish are healthy and actively feeding but stop while food remains	High for Type 1
Food collecting on substrate	Floating food fed to bottom feeders that never reach the surface; food particle too large; competitive exclusion at the surface	Food accumulates even though bottom dwellers are present and active	High — but distinguishes Type 1 from Type 3 poorly
Fish becoming excessively rounded	Gravidity in livebearers; egg development in females; normal body shape for a fancy variety; dropsy or organ enlargement; internal parasites; tumour	Multiple fish of both sexes rounding over months; no other clinical signs	Moderate — easily confounded
Increased visible faeces	Higher-fibre or less digestible food; larger fish; more fish	Faecal output rose after a feeding change with no stocking change	Moderate
Nitrate rising faster than before	Reduced water change frequency or volume; increased stocking; plants removed or in decline; nitrate in source water; substrate disturbance releasing accumulated nutrient	Trend began when feeding changed and no other variable moved	High, if tracked as a trend rather than a single reading
Cloudiness after feeding	New tank not yet cycled; filter media replaced; substrate disturbed; dosed supplement; dead animal	Reproducibly follows feeding events	High if the timing is consistent
Snail population increase	Decaying plant matter; dead animal; recently introduced snails; reduced maintenance; vegetables left in the tank	Boom follows a feeding increase and no other organic source is present	Low on its own; moderate in combination
Mechanical filter clogging faster	Filter media degraded; flow reduced; substrate stirred; plant debris; increased stocking	Cleaning interval shortened after a feeding change	Moderate — underused and quite informative
Organic buildup and detritus in low-flow zones	Insufficient circulation; inadequate substrate maintenance; heavily planted areas trapping debris	Buildup is food-coloured or recognisable as food particles	Moderate
Surface film or scum	Protein films from many organic sources; low surface agitation; oily foods	Follows feeding of oily or fatty foods	Low to moderate
Fish showing reduced interest at feeding time	Illness; stress; poor water quality; temperature; the food itself	Fish are otherwise healthy and simply satiated	Moderate — a genuinely useful early signal

Table XIX-1. Overfeeding signs and their differential. Note that no row in the third column says “this sign alone proves it.”

No single sign proves overfeeding

Overfeeding is diagnosed by **convergence**: several independent signs pointing the same direction, a plausible timeline connecting them to a feeding change, and the absence of a better explanation. A rising nitrate trend on its own means the nutrient budget is out of balance, which could equally be a water-change problem. A rising nitrate trend plus food on the substrate plus a shortened filter cleaning interval plus a snail boom, all beginning after a new food was introduced, is a diagnosis. [\[REASONABLE INFERENCE\]](#)

Is overfeeding really the number one cause of aquarium problems?

[CONTESTED CLAIM] This is asserted constantly and has no supporting data. It is plausible that overfeeding is a common contributing factor, particularly for beginners, and it is certainly a frequent cause of the specific problems in Parts XIV–XVII. But inadequate cycling, overstocking, insufficient water changes, unstable temperature, incompatible stocking, and inadequate acclimation are all also extremely common, and no one has ranked them.

The claim's practical danger is that it encourages single-variable diagnosis: a keeper with a struggling tank cuts feeding, sees no improvement because the real problem was stocking or maintenance, and either escalates the restriction or concludes the advice was worthless. Both outcomes are bad. The more defensible framing is that overfeeding is a common *contributor* that is unusually easy to correct, which is why it is worth checking first — not that it is the leading cause of anything.

Part XX — Underfeeding

Underfeeding receives a fraction of the attention overfeeding does, partly because its consequences are slower and less visible, and partly because the hobby has adopted “err on the side of less” as a near-universal default without examining where that default fails.

What does inadequate feeding look like?

Sign	What it indicates	Other possible explanations	Time to appear
Progressive weight loss	Sustained energy deficit	Internal parasites; mycobacteriosis; chronic disease; inability to compete; dental or jaw problems in older fish	Weeks to months
Hollow or concave belly	Depleted visceral fat and reduced gut fill	Species-normal body shape; recent spawning; internal parasites; wasting disease	Weeks
Sunken or 'pinched' area behind the head, above the operculum	Loss of dorsal musculature — a late and serious sign	Mycobacteriosis and other chronic wasting conditions	Weeks to months; indicates advanced depletion
Poor growth in juveniles	Inadequate ration or inadequate nutrient density during the growth phase	Genetics; temperature; water quality; stunting from other causes	Weeks; often permanent
Reduced activity and listlessness	Energy conservation	Illness; temperature; water quality; stress; normal species behaviour	Days to weeks
Increased aggression at feeding	Competition intensified by scarcity	Normal species behaviour; breeding condition; territorial disputes	Days
Fish picking at substrate, glass, and decor persistently	Active foraging driven by insufficient intake	Entirely normal behaviour for many benthic and grazing species	Days — low specificity
Loss of colour intensity	Depleted carotenoid and general condition	Stress; lighting; water quality; illness; normal variation	Weeks
Poor spawning or reduced fecundity	Insufficient energy for gamete production	Water parameters; age; pair compatibility; photoperiod	Weeks to months

Why is the “underfeeding is safer” default only half right?

[CONTESTED CLAIM] The advice to err toward underfeeding is well intentioned and usually harmless for healthy adult fish, where reserves buffer a short-term deficit. It becomes actively harmful in three situations: with fry and juveniles, where restriction can cause permanent stunting rather than recoverable weight loss; with small-bodied species that carry minimal reserves; and in any tank with a distribution problem, where reducing the total ration starves the subordinate fish first while the dominant fish remain well fed.

The most important distinction in this section

Not enough food in the tank and **the right amount of food distributed poorly** look similar from across the room and require opposite corrections.

If the total ration is inadequate, the correction is to feed more. If the total ration is adequate but distribution is poor, feeding more makes the tank dirtier *and* leaves the thin fish thin — because the food will continue to go where it was already going. The correction there is to change **where, when, and in what form** food is delivered, not how much. **[REASONABLE INFERENCE]**

How do you tell the two apart?

Observation	Suggests inadequate total ration	Suggests poor distribution
Which fish are thin?	All or most fish, across species and temperament	Specific individuals or specific species — typically the slow, shy, benthic, or nocturnal ones
Body condition of dominant fish	Dominant fish are also lean	Dominant fish are well conditioned or rounded
Food remaining after feeding	None; food is consumed instantly and fish continue searching	None at the top; none reaching the bottom — intercepted mid-column
Nitrate trend	Low and stable or declining	Normal or rising — the food is being eaten and metabolised, just by the wrong fish
Behaviour at feeding	All fish rush aggressively; competition is general	A subset of fish do not approach, or arrive after the food is gone
Effect of adding more food	Condition improves across the board	Dominant fish get rounder; thin fish stay thin; waste increases

Which fish go hungry in a well-fed tank?

A recurring set of profiles. Each is expanded in Parts LIII–LV.

- **Benthic species in a floating-food tank.** *Corydoras*, kuhli loaches, and many catfish simply never encounter food that is consumed at the surface.
- **Nocturnal species fed during the day.** Food offered at lights-on is gone long before the animal becomes active.
- **Slow and deliberate feeders.** Many anabantoids, some cichlids, and most loricariids feed methodically and lose to fast schooling fish every time.
- **Timid or newly introduced individuals.** Recent arrivals often will not compete for several days, which coincides with the period when their reserves are already depleted from transport.
- **Small individuals in a size-mixed group.** Where food particle size suits the larger fish, the smaller ones may be physically unable to take it. This is documented in aquaculture as a cause of size-graded growth divergence and mortality of smaller fish.
- **Subordinate individuals in hierarchical species.** Cichlids particularly.
- **Specialised feeders in a generalist community.** An *Otocinclus* in a tank with no algae and no supplementary vegetable is being asked to live on nothing.

Can a fish be malnourished while eating every day?

[STRONG EVIDENCE] Yes, by two distinct routes. First, quantity may be adequate while nutrient composition is not — the classic case being a fish maintained largely on a single high-fat invertebrate food that is not nutritionally complete. Second, water-soluble nutrients may leach before ingestion: a large percentage of vitamin C, vitamin B₁₂, choline, and pantothenic acid was lost into water within 30 seconds of feeding some commercial flake diets (Pannevis and Earle 1994a).

A third route is species mismatch: a strictly herbivorous or aufwuchs-grazing fish eating a high-protein carnivore pellet daily is consuming adequate calories in a form its digestive physiology handles poorly. Cross-reference Parts XXIV and L.

[STRONG EVIDENCE] Deficiency signs in ornamental species are documented. Oscars without ascorbic acid supplementation took 25 weeks to develop clinical deficiency signs (Fracalossi et al. 1998), and guppies fed phosphorus-deficient diets showed depressed appetite, scoliosis, and lordosis (Shim and Ho 1989). The 25-week latency in the oscar work is the important number for keepers: nutritional deficiency develops on a timescale of

months, long after the causal feeding pattern was established.

Part XXI — Body Condition Assessment

The most useful hobbyist-accessible measurement in this entire document, and the one least often taught.

Can aquarists evaluate body condition visually?

[REASONABLE INFERENCE] Yes, with practice and with species-specific calibration. Visual body condition assessment is standard in aquatic veterinary practice and in zoological husbandry, and the Merck Veterinary Manual explicitly identifies regularly checking whether fish are too fat or too thin as an important factor in proper feeding. Formal numerical body condition scoring systems exist for some fish species in fisheries and veterinary contexts but are not standardised across ornamental species.

What should you actually look at?

Feature	Underconditioned	Good condition	Overconditioned	Confounders
Dorsal musculature (viewed from above or head-on)	Narrow, knife-edged; spine palpable in profile	Smoothly rounded, symmetrical shoulders	Broad, thick, sometimes with a distinct fat pad	Species-normal shape; many characins are naturally slender
Belly profile (lateral view)	Concave or flat, tucking up sharply behind the pelvic fins	Gently convex, continuous curve	Distended, sometimes drooping below the pelvic line	Gravidity; recent feeding; egg development; bloat; dropsy
Head-to-body proportion	Head appears disproportionately large; the classic 'big head' of a wasted fish	Head flows smoothly into body	Head appears small relative to body mass	Juvenile proportions differ; some species are naturally large-headed
Area behind the operculum	Sunken, pinched	Smoothly filled	Full to bulging	One of the more reliable indicators — less confounded by gut fill
Caudal peduncle	Thin, with visible skeletal structure	Muscular and full	Thick	Reliable; little affected by feeding state
Overall symmetry and posture	Angular, with visible skeletal landmarks	Smooth contours	Rounded, sometimes with reduced manoeuvrability	Fancy varieties are deliberately deformed by selection

Table XXI-1. Visual body condition features. The features least affected by recent feeding — dorsal musculature, peduncle, and the area behind the operculum — are the most reliable for assessing chronic condition.

Why does species variation make this hard?

A hatchetfish, a discus, a kuhli loach, and a fancy oranda have almost nothing in common as body shapes, and three of the four are the product of deliberate morphological selection or extreme natural specialisation. Assessment therefore has to be **within-species and preferably within-individual over time**, not against a general ideal.

- **Fancy goldfish varieties** have compressed, globoid bodies by design. Some pearlscales are described as resembling golf balls, and some varieties normally swim slightly head-down. None of this is a condition problem.
- **Livebearers** are frequently gravid, and a gravid female looks much like an overfed one from a distance. The gravid spot, the shape of the distension (squared off rather than smoothly rounded), and the fish's reproductive history distinguish them.
- **Naturally deep-bodied species** — discus, angelfish, silver dollars — are assessed laterally, not from above.
- **Naturally slender species** — many tetras, danios, hatchetfish, and most loaches — will look thin to an eye calibrated on livebearers.

What is the practical protocol?

1. **Establish a baseline.** Photograph the fish in lateral view and, where possible, from directly above, shortly after acquisition and again once settled. This baseline is worth more than any generic condition chart.
2. **Observe before feeding, not after.** Gut fill confounds the belly profile substantially.
3. **Assess monthly, not daily.** Condition changes on a scale of weeks. Daily assessment produces noise and anxiety.
4. **Look at the peduncle and the area behind the operculum first.** These are the least confounded by recent feeding.
5. **Compare across the group.** If one fish of six is thin, the problem is distribution, competition, or that individual's health. If all six are thin, the problem is ration.
6. **Record it.** A dated note or photo in a tank journal converts a vague impression into a trend. This is precisely the use case the Daily Tank Journal is built around.

Body condition is the feedback signal the hobby is missing

Nitrate tells you about the system. Uneaten food tells you about one feeding event. **Body condition tells you whether the animal is actually getting what it needs**, which is the question the whole exercise is supposed to answer.

Every other measurement in this document is a proxy for this one. [\[REASONABLE INFERENCE\]](#)

Part XXII — Why Fish Always Act Hungry

The most common feeding question in the hobby, and one with a clean, well-evidenced answer that happens to be counterintuitive.

Is begging behaviour a hunger signal?

[STRONG EVIDENCE] Largely no. The behaviour keepers interpret as hunger is predominantly a conditioned anticipatory response with a circadian component, combined with an opportunistic feeding reflex. Food-anticipatory activity (FAA) — an increase in locomotor activity before a scheduled feeding time — is a documented and widely studied phenomenon in fish.

What is food-anticipatory activity?

FAA is treated in the literature as a genuine circadian rhythm rather than a simple stimulus-response. The criteria it meets are specific and instructive:

- **It persists when food is withheld.** In zebrafish shoals, when food was withheld but the light-dark cycle preserved, FAA persisted at the old clock time in 78 % of shoals. In cavefish, FAA free-ran under fasting conditions with an approximately 24-hour period. A fish continuing to “beg” at the usual time after a missed meal is running a clock, not reporting an empty stomach.
- **It is entrainable.** Scheduled feeding entrains peripheral circadian oscillators in zebrafish, including *per1* expression in the liver. Feeding time is a genuine zeitgeber.
- **It does not shift instantly.** After an abrupt change of mealtime, the rhythm adjusts over several transient cycles rather than immediately — one of the defining properties of a circadian process.
- **It is linked to reward expectation, not only energy demand.** FAA is described in the literature as linked not only to energy demands but also to food reward expectations.

What else drives the behaviour?

Driver	Mechanism	What it means for the keeper
Conditioned association with the keeper	Fish learn that human approach predicts food. The association is rapid and robust	The fish react to <i>you</i> , not to hunger. They will do it whether or not they were just fed
Circadian food-anticipatory activity	An entrained internal clock drives pre-meal activity	Behaviour at the usual time is scheduled, not requested
Opportunistic feeding reflex	Many fish evolved under unpredictable food availability. The suction feeding action triggers on food presence	The reflex is not gated by satiety in the way a mammal's appetite is
Social facilitation	One fish feeding triggers others	Group excitement amplifies the appearance of desperate hunger
Genuine hunger	Actual energy deficit	Real, and the reason this section does not say begging is <i>never</i> meaningful

When does begging actually mean something?

[REASONABLE INFERENCE] Begging becomes informative when it is accompanied by other evidence and when it deviates from the conditioned pattern.

- **Persistent foraging outside feeding times** — continuous substrate sifting, glass grazing, and searching behaviour hours away from any feeding — is more meaningful than excitement at the usual hour.

- **Begging plus declining body condition** is a real signal and should be acted on.
- **Begging in a fish that previously ignored feeding time** is a change worth noticing.
- **A specific individual begging harder than the group** may indicate it is not getting its share — a distribution problem rather than a ration problem.

How to explain this to an audience without sounding dismissive

The keeper's instinct is not foolish; it is a correct reading of a misleading signal. The fish genuinely are excited, genuinely will eat, and genuinely learned to associate the keeper with food. What is wrong is only the inference from “they want it” to “they need it.”

A useful reframe: *“Your fish aren't lying to you. They're telling you they've learned your schedule. That's actually a compliment — it means your routine is consistent enough for them to have learned it.”* **[STRONG EVIDENCE]**

Part XXIII — Variety in the Diet

One of the two secondary themes the eventual episode must handle. The evidence supports variety, but for reasons that differ from the ones usually given — and variety carries a specific, well-defined risk that is almost never mentioned.

Do freshwater aquarium fish benefit from dietary variety?

[REASONABLE INFERENCE] Probably yes, on balance, but the benefit is best characterised as nutritional insurance and behavioural enrichment rather than as correction of a deficiency that a good staple would leave. The Merck Veterinary Manual identifies feeding a wide variety of feeds as one way to address the risk of nutritional deficiency, and commercial ornamental producers in Singapore emphasise the importance of regular supplementation of formulated feeds with live feed because inclusion of live feed improves growth (Fernando et al. 1991, in Sales and Janssens 2003).

There is direct evidence of a functional benefit from supplementation in at least two ornamental contexts. Daily supplementation of *Daphnia* to swordtail broodstock maintained on an artificial flake diet produced a significant increase in fecundity, more rapid growth, a higher number of embryos, and improved feed conversion (Kruger et al. 2001a). Supplementation with *Artemia* increased growth of juvenile angelfish (Degani 1993).

What are the actual benefits of variety?

Proposed benefit	Evidence position	Assessment
Broader nutrient profile	Different food types genuinely differ in amino acid profile, fatty acid composition, and micronutrient content. Protein content of <i>Artemia</i> nauplii, chironomid larvae, and <i>Daphnia</i> was measured at 52.5%, 76%, and 39.7% respectively in one comparison	[STRONG EVIDENCE] that profiles differ; [REASONABLE INFERENCE] that the difference matters against a complete staple
Insurance against formulation gaps	Ornamental nutrient requirement data are sparse; Sales and Janssens concluded that ornamental fish nutrition rests on extrapolation from food fishes under intensive farming conditions	[REASONABLE INFERENCE] — and arguably the strongest argument for variety
Different textures and physical forms	Relevant to feeding competence, particle handling, and access for different mouth morphologies	[REASONABLE INFERENCE]
Behavioural enrichment	Live and moving prey elicits hunting, chasing, and searching behaviours that static pellets do not	[REASONABLE INFERENCE] ; see Part XCIII
Improved feeding response in reluctant feeders	Some species show consistent preference for one diet over another when offered simultaneously; ruby barb was reported to prefer live feed to artificial feeds	[STRONG EVIDENCE] for preference; useful clinically
Micronutrient diversity	Plausible but not demonstrated to be necessary where a vitamin-supplemented complete diet is used	[REASONABLE INFERENCE]
Improved reproductive performance	Demonstrated in swordtails with <i>Daphnia</i> supplementation and in angelfish with <i>Artemia</i>	[STRONG EVIDENCE] in those specific cases

What are the risks of variety?

[REASONABLE INFERENCE] The risks are real and under-discussed. The dominant one is additive overfeeding: because supplementary foods are framed as treats, they are normally *added* to the standard ration rather than *substituted* for it.

Risk	Mechanism	Mitigation
Additive overfeeding	Treats are added to, not substituted for, the staple ration. Two frozen cubes on top of normal feeding is a large ration increase	Substitute, do not add. A frozen-food day replaces that day's staple meal

Risk	Mechanism	Mitigation
Fat accumulation from energy-dense treats	Many popular supplementary foods are lipid-rich relative to a staple pellet; high-fat diets produce hepatic steatosis across many finfish	Limit high-fat items to occasional substitution; see Parts XVIII and XXXIV
Nutritional imbalance from over-reliance on one supplement	A fish maintained substantially on a single invertebrate food is not receiving a complete diet	Keep a nutritionally complete food as the dietary base
Pathogen and parasite introduction from live foods	Live feed can act as a transmitter of parasitic, bacterial, and viral disease if not stored or sourced correctly (Pannevis 1993; Earle 1995). Ornamental farms shifted from manure-cultured <i>Moina</i> to <i>Artemia</i> nauplii partly for this reason	Source carefully; prefer cultured to wild-collected; consider decapsulated cysts and frozen alternatives
Contaminant accumulation	Environmentally collected chironomid larvae have been reported with excess lead in some samples and low zinc in others	Prefer commercially produced over self-collected invertebrates from unknown water
Selective feeding and staple refusal	Fish offered a preferred rich food may subsequently refuse the complete staple	Keep supplements genuinely occasional; do not let treat foods become the expected default
Nutritional inconsistency of live foods	<i>Artemia</i> nutritional value is not constant but varies geographically and temporally; enrichment is standard practice precisely because of this	Do not assume live equals nutritious; treat live food as a component, not a complete diet

Is variety necessary if a high-quality complete food is used?

[CONTESTED CLAIM] This is the central question of this theme and it does not have a clean answer. The honest position has three parts.

1. **A good complete diet probably covers the known requirements.** Modern formulated diets are built against published nutrient requirement data and include stabilised vitamin and mineral premixes specifically to prevent the deficiencies documented in Part XXIV.
2. **But the requirement data for ornamental species are thin.** Sales and Janssens found that protein requirements had been studied for a handful of ornamental species, mineral requirements mainly in the guppy, and that few studies had concentrated on vitamin requirements of ornamental species at all. A formulation is therefore an educated approximation for most aquarium fish, not a precisely matched diet.
3. **And the tank contains several species with different requirements.** Sales and Janssens noted that it is impractical to feed very specific diets to individuals in an aquarium and proposed that a better approach than satisfying each species is to minimise deviations with an intermediate feed. That framing implies every single-food community aquarium is already feeding a compromise.

The defensible position on variety

Variety is insurance, not a cure. A fish on one good complete food is very unlikely to develop a clinical deficiency. A fish on two or three complementary good foods is more robust against the possibility that the formulation missed something for its particular species — and gets behavioural benefit besides.

But variety is also the most common route to accidental overfeeding in an otherwise careful tank. The keeper who feeds a measured pellet ration six days a week and then adds two cubes of bloodworms on Sunday *on top* has not improved the diet; they have increased weekly energy intake substantially and added the fattiest item in the cupboard.

The rule that resolves both: vary by substitution, not by addition. **[REASONABLE INFERENCE]**

Part XXIV — Does Variety Prevent Deficiencies?

What deficiency actually looks like in fish, and how modern formulated diets changed the risk landscape.

What nutritional deficiencies are documented in fish?

Nutrient	Documented deficiency signs	Evidence in ornamental species	Notes
Vitamin C (ascorbic acid)	Scoliosis and lordosis (spinal deformity, the 'broken back' condition), deformed opercula and jaws, haemorrhage in eyes and fins, impaired wound healing and immune function	Directly demonstrated. Oscars have a dietary requirement for ascorbic acid; unsupplemented fish took 25 weeks to show clinical signs (Fracalossi et al. 1998). A conservative requirement of 360 mg/kg diet was proposed for angelfish juveniles to maintain maximum tissue storage (Blom et al. 2000)	Fish cannot synthesise it. The most important single vitamin in fish feeds and the least stable in storage
Vitamin A	Impaired vision, epithelial changes, reduced growth	Not specifically studied in most ornamentals	Fat-soluble; heat- and oxidation-sensitive
Vitamin D	Impaired calcium and phosphorus metabolism	Minimal ornamental data	Fish obtain some minerals from water, complicating requirement studies
Vitamin E (tocopherol)	Muscular dystrophy, anaemia, reduced immune function; also functions as the primary antioxidant protecting dietary lipids	Vitamin E supplementation improved survival and spawning performance in <i>Betta splendens</i> in one study	Its antioxidant role in the feed itself matters as much as its nutritional role
B vitamins (including B ₁₂ , choline, pantothenic acid)	Varied — neurological signs, anaemia, poor growth, appetite loss	Leaching demonstrated: large proportions lost within 30 seconds of flake diets entering water	Water-soluble and therefore the most vulnerable to leaching
Thiamine (B ₁)	Neurological dysfunction, anaemia, reduced reproductive success, immunosuppression; early life stage mortality in offspring	Well documented in the context of thiaminase-containing prey; see Part XCI	Destroyed by thiaminase in certain prey species
Phosphorus	Depressed appetite, scoliosis, lordosis	Demonstrated in guppies (Shim and Ho 1989). Requirements reported at 0.52% for tiger barb and 0.53–1.23% for guppy	Low in natural water, so must come from the diet
Iron	Hypochromic microcytic anaemia	Requirement of 0.008% reported for guppy (Shim and Ong 1992)	Studied specifically in ornamentals
Magnesium / zinc	Reduced growth and feed conversion	Requirements of 0.054% (Mg) and 0.01% (Zn) reported for guppy	Among the few ornamental mineral requirement figures published
Essential fatty acids	Reduced growth and survival; impaired membrane function; reduced egg hatchability	Freshwater fish generally require dietary linoleic (18:2n-6) and/or linolenic (18:3n-3) acid. Dietary DHA levels significantly affected egg hatchability in common carp	See Part XXVI
Essential amino acids	Reduced growth, poor protein utilisation	Requirements published for goldfish larvae; markedly higher than juvenile tilapia values	See Part XXV

How much have complete commercial diets reduced deficiency risk?

[REASONABLE INFERENCE] Substantially. Sales and Janssens described the development of manufactured feed over the preceding fifty years as one of the contributing factors to the hobby's growth, and specifically noted that increased reliance on manufactured feed focused attention on nutritional requirements. Formulated diets incorporate stabilised vitamin forms precisely because the deficiency syndromes above were previously encountered.

The clinical implication is that classical deficiency disease is now uncommon in aquarium fish maintained on a modern complete diet, and that where it does appear the likely explanations are (a) a diet that was never complete — a fish maintained largely on bloodworms or on a single vegetable, (b) severely degraded food from prolonged or poor storage, or (c) nutrient leaching from food that sits in the water. All three are feeding practice problems rather than formulation problems.

Part XXV — Protein

The macronutrient the hobby argues about most and understands least, largely because “high protein” is a marketing phrase before it is a nutritional statement.

What are the published protein requirements?

Species	Life stage / size	Requirement	Source context
Guppy (<i>Poecilia reticulata</i>)	0.10 g	30–40%	Weight gain, feed conversion, gonadal development
Goldfish (<i>Carassius auratus</i>)	0.20 g juvenile	29%	Weight gain, feed conversion, protein efficiency ratio
Goldfish	0.008 g larvae	53%	Specific growth rate, feed efficiency, nutrient retention
Tin foil barb (<i>Barbodes altus</i>)	0.81 g	41.7%	Weight gain
Discus (<i>Symphysodon aequifasciata</i>)	4.45–4.65 g	44.9–50.1%	Specific growth rate
Redhead cichlid (<i>Cichlasoma synspilum</i>)	0.28 g fry	40.8%	Specific growth rate
Swordtail (<i>Xiphophorus helleri</i>)	6–8 weeks	At least 45% at 6% lipid	Best specific growth rate and feed conversion
Dwarf gourami (<i>Colisa lalia</i>)	Female broodstock	25–45% gave highest fecundity	Reproductive performance endpoint
<i>Betta splendens</i>	Broodstock	35% animal or plant protein gave highest egg number and hatching rate	Spawning trial

Table XXV-1. Published protein requirements for ornamental species (compiled in Sales and Janssens 2003 and related literature). Note the critical caveat the authors attach: these are dietary percentages needed for optimal **growth**, derived from 8–12 week dose-response studies, and are “only applicable to similar conditions under which it was evaluated.” They are not true requirements expressed as amount of protein per animal per day, and they are not maintenance figures for adult pet fish.

What drives protein requirement?

- **Species and trophic type.** Roughly 30 % for the omnivorous goldfish versus roughly 45–50 % for the carnivorous discus is the clearest illustration in the ornamental literature.
- **Life stage.** The goldfish larva-versus-juvenile comparison — 53 % versus 29 % — shows the effect is larger than most between-species differences.
- **Growth rate.** Protein requirement is largely a growth requirement. An adult at maintenance needs protein for turnover and repair, not for tissue accretion.
- **Temperature.** Higher temperature raises metabolic rate and therefore intake, which changes the protein delivered at a given dietary percentage.
- **Reproduction.** Gamete production is protein-expensive; dietary protein level affected ovary weight and fecundity in dwarf gouramis and in *Betta splendens*.
- **Dietary energy balance.** Fish eat to satisfy energy requirement, so protein and energy in the diet must be balanced. Excessive dietary energy can restrict protein consumption and subsequent growth.

Is more protein always better?

[UNSUPPORTED AQUARIUM FOLKLORE] No. Excess dietary protein above requirement is deaminated and used for energy, producing ammonia. Sales and Janssens make the point with unusual directness for the aquarium context: ornamental fish in captivity need to use dietary protein with the utmost efficiency because the breakdown products of protein metabolism — mainly ammonia — directly pollute their living environment.

They extend this to a specific warning against transplanting aquaculture protein levels into home aquaria: high dietary protein levels necessary for growth of juvenile fish kept in large ponds might present problems of nitrogen breakdown products with adult fish kept in small home aquaria, as well as ineffective use of excess protein in metabolism.

The protein claim rewritten

Instead of: “Higher protein is better quality food.”

Use: “Protein needs to match the species and the life stage. Protein above what the fish can use is converted to energy and excreted as ammonia — it becomes a water-quality problem instead of a nutrition benefit. A 50 % protein food is right for a growing discus and wrong for an adult goldfish.” **[STRONG EVIDENCE]**

Part XXVI — Fat and Lipids

The macronutrient most responsible for both the benefits and the harms of supplementary feeding.

What do dietary lipids do?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Dietary lipids are the most energy-dense macronutrient and the source of essential fatty acids. They also carry fat-soluble vitamins and sterols, form biological membranes at cellular and subcellular level, serve as components of hormones and precursors for functional metabolites such as prostaglandins, and contribute to feed flavour and texture (NRC 1983, in Sales and Janssens 2003).

Which fatty acids are essential?

[STRONG EVIDENCE] Fish in general require fatty acids of longer chain length and higher degree of unsaturation than mammals, because low melting points are needed at lower body temperatures to maintain cell membrane flexibility. Freshwater fish generally require dietary linoleic acid (18:2n-6), linolenic acid (18:3n-3), or both. Marine fish require the longer-chain EPA (20:5n-3) and/or DHA (22:6n-3), which they cannot biosynthesise *de novo* or from shorter-chain precursors.

The freshwater-versus-marine distinction matters for aquarium practice: it means most freshwater ornamental fish can elongate and desaturate shorter-chain precursors, so a diet adequate in 18-carbon essential fatty acids is generally sufficient. Marine-derived HUFA supplementation, which dominates the marine larviculture literature, is less critical for most freshwater species. It is not irrelevant, however — dietary DHA levels significantly affected egg hatchability in common carp (Shimma et al. 1977).

Why does energy density create a problem?

[STRONG EVIDENCE] Fish have a comparatively low energy demand and are therefore susceptible to deposition of excessive lipid (Earle 1995). Prolonged high-fat feeding produces hepatic steatosis, immune suppression, and reduced growth in most finfish studied, via downregulation of lipolytic genes and upregulation of lipogenic genes.

This is the mechanism connecting treat foods to Part XVIII. It is also why the commonly repeated advice to feed frozen bloodworms frequently is more consequential than it appears — the issue is not that the food is bad but that it is energy-dense and typically additive.

What about rancidity and oxidised fats?

[STRONG EVIDENCE] Fish feeds have high protein and oil content that can deteriorate rapidly. Lipid oxidation in products with high polyunsaturated fatty acid content leads to rancidity and off-flavour. The main antioxidant used in feeds is vitamin E, normally added as tocopherol acetate, specifically to stabilise lipids in the feed during storage — which is particularly important for fish feed with high PUFA content.

The practical consequence is direct: the same chemistry that makes fish feeds nutritionally valuable makes them perishable. A container of food that has been open for a year in a warm room has oxidising fats and degrading fat-soluble vitamins, and the vitamin E that was protecting the lipids is itself being consumed in the process. Cross-reference Part LXX.

Part XXVII — Carbohydrates

The macronutrient about which the hobby holds the most confidently wrong belief.

Can fish digest carbohydrates?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Yes, with very large variation between species. The blanket claim that fish cannot digest carbohydrates is false. Warmwater omnivorous fish have a substantially greater capacity to use dietary carbohydrate as an energy source, sparing protein, than coldwater and carnivorous species, which utilise it poorly or not at all.

Group	Carbohydrate handling	Underlying mechanism	Practical consequence
Warmwater herbivores and omnivores (goldfish, koi, tilapia, many cyprinids)	Good. Wilson recommended roughly 40% dietary carbohydrate for tilapia. Carbohydrate digestibility of about 70% has been reported in goldfish	Higher α -amylase activity; greater number and affinity of insulin receptors; hindgut microbial fermentation of complex carbohydrates in some species	Plant- and starch-containing foods are appropriate and protein-sparing
Warmwater omnivores with moderate capacity (many community fish)	Moderate. Carbohydrate digestibility of about 50% has been reported for moonlight gourami	Intermediate enzyme profile	Mixed diets suit them
Carnivores and coldwater species (trout, salmon, many predatory cichlids)	Poor. Carnivorous species generally tolerate no more than about 20% digestible carbohydrate; below 20% is suggested for marine and carnivorous species	Low intestinal glucose uptake rates; slow blood glucose clearance; poor regulation of hepatic metabolic pathways; persistent postprandial hyperglycaemia	High-starch foods are a genuine metabolic burden, not merely filler

Do fish require dietary carbohydrate?

[ESTABLISHED SCIENTIFIC PRINCIPLE] No dietary requirement for carbohydrate has been demonstrated in fish. Fish can synthesise glucose from non-carbohydrate precursors. However, if carbohydrates are not provided, other nutrients — protein and lipid — are catabolised for energy instead, which is both wasteful and, in the case of protein, ammonia-generating.

This is why “filler” is a misleading pejorative when applied to digestible carbohydrate in a food intended for an omnivorous or herbivorous fish. Carbohydrate that the species can actually use is a **protein-sparing energy source**, which reduces ammonia output. Carbohydrate that the species cannot use is genuinely a problem. The ingredient is the same; the species determines whether it is filler or function. Cross-reference Part LXXIV.

What about form and processing?

[STRONG EVIDENCE] Most species utilise cooked starch better than raw starch, and the relative use of dietary carbohydrates appears associated with the complexity of the carbohydrate. Some species use simple sugars as well as or better than complex carbohydrates; others do not use simple sugars as an energy source at all. Extrusion processing, which gelatinises starch, therefore improves carbohydrate availability in prepared foods.

Part XXVIII — Fibre and Plant Matter

Fibre is the dietary component most often invoked in aquarium advice and least often defined.

What does fibre actually do in a fish?

[CONTESTED CLAIM] Much less than the hobby assumes, and by a different mechanism. No fish species has been shown to efficiently digest cellulose, which is the principal structural fibre in plant material. Fibre therefore functions mainly as bulk and as a modulator of gut transit rather than as a nutrient source.

This has a specific and important consequence for the pea-for-constipation advice in Part XL: if a fish cannot digest cellulose, the mechanism by which fibre is supposed to relieve an impaction is not the mammalian mechanism, and the fibre content of the food being recommended is not actually unusual. Aquatic-veterinary commentary has noted that shelled green peas contain roughly 5 % fibre — the same as many prepared fish foods — and that a goldfish's metabolism is not robust enough to process large amounts of fibre in any case.

Where does plant matter genuinely matter?

Situation	Role of plant matter	Evidence
Herbivorous and aufwuchs-grazing species	Provides bulk, grazing behaviour, and a nutrient profile matched to their digestive physiology — including digestible carbohydrate they handle well	[STRONG EVIDENCE]
Omnivorous species	Contributes digestible carbohydrate and micronutrients; supports natural feeding behaviour	[REASONABLE INFERENCE]
Carnivorous species	Little demonstrated benefit; plant carbohydrate is poorly utilised and may be a metabolic burden at high levels	[STRONG EVIDENCE] for the poor utilisation
Gut transit in any species	Indigestible bulk plausibly affects transit rate, but this is asserted far more often than measured in aquarium fish	[INSUFFICIENT EVIDENCE]
Hindgut microbial function	Most herbivorous fish such as goldfish and koi carp use hindgut microflora to digest complex carbohydrates	[STRONG EVIDENCE]
Behavioural enrichment	Grazeable plant matter occupies grazing species for extended periods without adding concentrated energy	[REASONABLE INFERENCE] — and arguably the strongest practical argument

Part XXIX — Vitamins and Minerals

Both deficiency and excess matter, and the aquarium context adds a complication no terrestrial animal has: fish absorb some minerals directly from the water.

Why is mineral requirement hard to determine in fish?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Fish can absorb some water-soluble minerals directly from the surrounding water, which complicates the determination of dietary mineral requirements (Shim and Ho 1989). A requirement figure derived in one water chemistry may not transfer to another.

Phosphorus is the exception that proves the rule and the most important dietary mineral for fish. It is essential for growth, bone mineralisation, and lipid and carbohydrate metabolism, and it is needed in the diet precisely *because* its content in natural water is low. Calcium, by contrast, was found to be uncorrelated with growth in guppies, consistent with uptake from water. Cross-reference Part LXVIII on the water-quality consequences of dietary phosphorus.

Can excess matter as well as deficiency?

[REASONABLE INFERENCE] Yes, on two axes. Nutritionally, fat-soluble vitamins accumulate and excessive intake is a recognised risk in animal nutrition generally. Environmentally, excess dietary phosphorus is explicitly flagged as critical in the ornamental literature because the pollution of water by excess excreted phosphorus may lead to eutrophication — which in a closed aquarium means it stays in the box.

Which vitamins are most vulnerable in practice?

Vulnerability	Most affected nutrients	Cause	Mitigation
Leaching into water	Vitamin C, vitamin B ₁₂ , choline, pantothenic acid — large proportions lost within 30 seconds from some flake diets	Water solubility plus high surface-area-to-volume in flake form	Feed food that is consumed quickly; avoid extended soaking of flake; do not let food sit
Heat degradation during processing	Vitamin A and biotin are most sensitive to heat	Extrusion and drying temperatures	A manufacturing concern, addressed by overdosing premixes
Oxidation in storage	Vitamins A, D, and C are most prone to oxidation; vitamin E is consumed protecting dietary lipids	Oxygen exposure, especially with warm temperature and light	Cool, dark, dry, airtight storage; small containers; see Part LXX
Light degradation	Vitamin C particularly	Ascorbic acid stability ranked freezer > refrigerator > room temperature in black bags > room temperature in clear bags	Opaque containers; avoid windowsills and lit display shelves
Humidity	General degradation, plus mould growth and potential mycotoxin production	Moisture ingress into dry feed	Airtight containers; never return damp scoops to the container

The micronutrient point that changes behaviour

Vitamin content is a property of the food *at the moment the fish swallows it*, not a property printed on the label. Between manufacture and ingestion, a vitamin has to survive processing heat, months of storage, container opening, and then 30 seconds in water.

Two practical consequences follow: buy food in quantities that will be used within a few months, and feed in a way that gets the food into the fish quickly. Both are more consequential than choosing between brands. **[STRONG EVIDENCE]**

Food Types: Overview Comparison

The detailed sections follow. This table exists for retrieval: it answers the common comparative questions in one place before the nuance is added.

Food type	Complete staple?	Supplement?	Treat only?	Principal strength	Principal limitation
Quality flake	Yes, for suitable species	—	No	Surface presentation; small particle size; nutritionally complete formulations available	Rapid nutrient leaching; crumbles; poor for bottom feeders
Quality pellet / granule	Yes, for suitable species	—	No	Controlled sink rate; countable portions; less leaching than flake	Size matching is critical; floating types promote air ingestion in some species
Algae / herbivore wafer	Yes, for herbivorous grazers	For omnivores	No	Reaches the bottom; persists for slow grazers	Often fed to carnivorous bottom feeders that cannot use it
Sinking carnivore wafer / tablet	Yes, for benthic carnivores	Yes	No	Delivers protein to the substrate zone	Easily intercepted by mid-water fish
Frozen invertebrates (mysis, daphnia, cyclops)	No	Yes	—	Palatability; natural form; enrichment	Not nutritionally complete alone; variable composition
Frozen bloodworms	No	Limited	Best treated as a treat	Extremely high palatability; useful for reluctant feeders	Energy-dense, variable lipid content, commonly overfed
Freeze-dried invertebrates	No	Yes	—	Shelf-stable; convenient; no parasite risk from the process	Floats; lower palatability than frozen for some fish
Live foods	No	Yes	—	Behavioural enrichment; conditioning value; demonstrated reproductive benefit in some species	Pathogen and parasite risk; nutritionally inconsistent; culture effort
Vegetables	No	Yes for herbivores and omnivores	—	Grazing behaviour; bulk without concentrated energy	Mostly water; low nutrient density; must be removed
Gel diets	Yes, where formulated as complete	Yes	No	Low leaching; medication vehicle; texture control	Preparation required; shelf life once made

Table F-1. Food type role summary. Note that “complete staple” always carries the qualifier “for suitable species” — no single product serves every trophic type.

Part XXX — Flakes

The food form the hobby has decided to look down on, largely without evidence.

Are flakes nutritionally inferior?

[UNSUPPORTED AQUARIUM FOLKLORE] There is no evidence that the flake *form* is nutritionally inferior. Flake and pellet are manufacturing formats, and the nutritional content of either depends on its formulation. A well-formulated flake and a well-formulated pellet from the same manufacturer typically share a very similar ingredient and guaranteed-analysis profile.

The belief probably arises from three real observations that get generalised into a claim about quality: flakes leach faster, flakes are the default budget product at retail so cheap flake is very common, and flakes crumble in the container so the last portion of a canister is dust.

What is genuinely different about flake?

Property	Consequence	Evidence
Very high surface-area-to-volume ratio	Fastest nutrient leaching of any dry format. A large percentage of vitamin C, B ₁₂ , choline, and pantothenic acid was lost within 30 seconds from some commercial flake diets	[STRONG EVIDENCE]
Low bulk density	A given volume of flake weighs far less than the same volume of pellet. Various types of feed have dramatically different weights; flaked feeds tend to be much lighter than pelleted feeds	[AQUATIC-VETERINARY GUIDANCE]
Floats initially, then sinks slowly and unpredictably	Good for surface and upper-midwater feeders; poor for reliably delivering to the bottom	[COMMON AQUARIUM PRACTICE]
Crumbles readily	Produces fine dust that is unusable by larger fish and clouds the water	[COMMON AQUARIUM PRACTICE]
Small particle size at the crumb level	Genuinely useful for very small fish and for grow-out of juveniles	[COMMON AQUARIUM PRACTICE]
Large exposed surface in storage	Faster oxidation of lipids and fat-soluble vitamins once the container is opened	[REASONABLE INFERENCE]

What follows practically?

- **The leaching finding is the operative one.** If a substantial share of water-soluble vitamins leaves the flake within 30 seconds, then flake that is eaten immediately delivers meaningfully more nutrition than flake that drifts for two minutes. This is an argument for smaller portions eaten fast, and against the “let them pick at it” approach.
- **Do not pre-soak flake.** Soaking flake before feeding deliberately performs the leaching step outside the fish. Whatever case exists for soaking pellets (Part LXI) does not extend to flake.
- **Buy small containers.** The high surface area that causes leaching also accelerates oxidation in storage.
- **Flake is a poor delivery vehicle for benthic fish.** This is a form problem, not a quality problem, and it is solved by adding a sinking food rather than by abandoning flake.

Part XXXI — Pellets

The dominant format in both aquaculture and the premium aquarium market, for reasons that are mostly practical rather than nutritional.

What do the pellet subtypes actually do?

Pellet type	Behaviour in water	Best for	Watch for
Floating	Remains at surface until eaten or waterlogged	Surface feeders; visible portion control; large fish where uneaten pellets can be netted off	Air ingestion in physostomous species with vigorous surface feeding; see Part LI
Slow-sinking	Descends gradually through the water column	Midwater community fish; mixed communities; the most generally useful type for a community tank	May be intercepted entirely before reaching the bottom
Fast-sinking	Reaches the substrate quickly	Benthic species; target feeding a specific area	Can be lost into substrate; may go unnoticed and decay
Micro pellet / granule	Varies; usually slow-sinking	Small-mouthed species; nano fish; juvenile grow-out	Very easy to overfeed by volume because the particles are tiny
Wafer / tablet	Sinks immediately, holds together	Bottom feeders and slow grazers; persists long enough for methodical feeders	Dissolving remnants; mid-water interception
Stick / large pellet	Usually floating	Large cichlids, goldfish, koi	Size relative to gape; air ingestion

Pellets versus flakes: is there a winner?

[CONTESTED CLAIM] No universal winner. The choice is driven by feeding zone, mouth size, and portion-control preference rather than by nutritional superiority.

Criterion	Flake advantage	Pellet advantage
Nutrient retention in water	—	Lower surface area means slower leaching
Particle size for very small fish	Crumbs are naturally tiny	Micro pellets exist but have a floor
Portion control	—	Countable units; a keeper can literally count out six pellets
Surface presentation	Natural floating behaviour	Floating pellets exist but are larger
Reaching bottom feeders	—	Sinking types are designed for it
Water clarity	—	Less dust and clouding
Cost per feeding	Generally lower	—
Storage stability	—	Lower exposed surface area per unit mass
Suitability for automatic feeders	—	Flows and dispenses far more reliably

The honest comparative answer

“Pellets are better” is not supportable as a general claim. What is supportable: **pellets are easier to portion accurately, leach more slowly, and can be engineered to reach a specific feeding zone.** Those are meaningful practical advantages, and they are advantages of *control*, not of nutrition.

For most community tanks, the strongest position is that the *combination* — a quality staple in a form that suits the majority of fish, plus a sinking food that reliably reaches the substrate — outperforms either format alone, because it solves the distribution problem that quantity rules cannot. **[REASONABLE INFERENCE]**

Part XXXII — Wafers

The food form most often bought for the wrong animal.

Are algae wafers appropriate for all “algae eaters”?

[UNSUPPORTED AQUARIUM FOLKLORE] No. “Algae eater” is a commercial category, not a dietary one, and it includes animals with substantially different requirements. Feeding an algae wafer to a predominantly carnivorous bottom-dwelling catfish delivers a food it cannot use well, while feeding a protein-rich sinking pellet to an obligate aufwuchs grazer does the reverse.

Animal	Actual dietary requirement	Algae wafer appropriate?	What it also needs
<i>Otocinclus</i>	Genuine algivore — soft green algae, biofilm, diatoms	Yes, as a supplement	Established biofilm; blanched vegetables; cannot thrive in a spotless new tank
Bristlenose pleco (<i>Ancistrus</i>)	Herbivore-leaning omnivore; grazes wood surfaces	Yes	Wood surface to graze; vegetables; some protein
Common pleco (<i>Pterygoplichthys</i>)	Omnivore/detritivore, increasingly protein-inclined with size	Yes, but insufficient alone	Vegetables and a protein component; a large specimen needs substantial food
<i>Panaque</i> / <i>Panaqolus</i>	Xylivorous-detritivorous; consumes large quantities of wood but does not digest cellulose	Yes, as part of a mix	Wood; and specifically <i>not</i> the assumption that wood substitutes for food
Siamese algae eater (<i>Crossocheilus</i>)	Omnivore that eats algae opportunistically, and progressively less as it matures	Partly	General omnivore diet; will take prepared food readily
<i>Corydoras</i>	Benthic carnivore/omnivore — invertebrates and organic particles	No, not as a primary food	Sinking protein-containing food delivered deliberately; see Part XLIII
Kuhli loach	Benthic carnivore/detritivore, nocturnal	No, not as a primary food	Sinking protein food after lights-out
Hillstream loach	Aufwuchs grazer in fast water	Yes, as part of a mix	Biofilm-covered surfaces; high oxygen and flow
Mollies	Omnivore with a substantial plant fraction	Yes, as a supplement	General omnivore diet; they are not a cleanup solution
Nerite and other grazing snails	Algae and biofilm grazers	Yes	Supplemental feeding when tank algae is limited — they are livestock, not equipment

Practical wafer handling

- **Break wafers to portion them.** A whole wafer is often more than a small group needs, and half a wafer eaten is better than a whole wafer half-eaten and disintegrating.
- **Place, do not drop.** Using tongs or a feeding dish keeps the wafer where the target animal forages rather than wherever the current takes it.
- **Expect and plan for interception.** In a community tank, mid-water fish will often reach a sinking wafer first. Feeding after lights-out, or feeding the mid-water fish simultaneously at the surface as a decoy, both address this. See Part LV.
- **Remove remnants.** A partially dissolved wafer is a concentrated organic mass. The usual guidance is to remove uneaten portions the following morning at the latest.

Part XXXIII — Frozen Foods

The standard supplement category, with genuine benefits and one specific, systematic trap.

What are the common frozen foods and what do they offer?

Food	What it is	Nutritional character	Best role	Cautions
Bloodworms	Chironomid midge larvae	Consistently high protein (values around 76% protein reported in one comparison) but variable lipid content between batches	High-value treat; appetite stimulant for reluctant feeders	Energy-dense; variable composition; heavy-metal and zinc variability reported in environmentally collected samples; see Part XXXIV
Brine shrimp (adult <i>Artemia</i>)	Adult saltwater crustacean	Moderate protein; relatively low in essential fatty acids unless enriched; high water content	Palatable supplement; laxative reputation in the hobby	Nutritional value varies geographically and temporally; adult brine shrimp is not a complete food
Newly hatched <i>Artemia</i> nauplii	Freshly hatched brine shrimp	Protein around 52.5% reported; yolk reserves intact at hatch	Fry first food; conditioning	Nutritional value declines rapidly after hatch as yolk is consumed
Mysis shrimp	Small marine or freshwater opossum shrimp	Good protein with chitin content; generally better fatty acid profile than bloodworms	Regular supplement; good staple-adjacent food for many carnivores	Particle size too large for very small fish
<i>Daphnia</i>	Freshwater cladoceran	Protein around 39.7% reported — noticeably lower than <i>Artemia</i> or chironomids; high water and chitin content	Low-energy supplement; the food most often recommended for digestive complaints	Low nutrient density means it is a poor staple; see Part XXXVI
Cyclops	Small freshwater copepod	Small particle size; reasonable protein	Small fish and fry; juvenile grow-out	Very small particle may be ignored by larger fish
Frozen plankton / rotifers	Mixed microorganisms	Variable	Fry and filter-feeding species	Fouls quickly if uneaten
Frozen vegetable / spirulina preparations	Bound plant material	Plant matter with added binder	Herbivore supplement	Check for excessive binder content

Is frozen food healthier than dry food?

[CONTESTED CLAIM] There is no general evidence that frozen food is nutritionally superior to a quality prepared diet. Frozen invertebrates are typically *single ingredients*, and single ingredients are by definition not complete diets. A prepared food is formulated against a nutrient specification; a frozen bloodworm cube is whatever midge larvae happened to contain.

What frozen food reliably offers is palatability, natural form, and enrichment value — and, in the ornamental literature, there is direct evidence that supplementing a prepared diet with live or frozen invertebrates improves growth and reproductive performance in at least some species. The reasonable conclusion is that frozen foods are excellent *supplements* and poor *staples*, and that the hobby's framing of them as premium and of dry food as inferior inverts the relationship.

Practical frozen food handling

- **Thaw before feeding.** Feeding frozen blocks directly is commonly discouraged; a thawed portion disperses properly and is eaten rather than sinking as a solid mass.
- **Rinse and drain the pack fluid.** The liquid in a thawed portion carries dissolved organics and offers no nutritional benefit to the fish. Straining it off reduces the organic load.

- **Portion the cube.** A standard cube is frequently far more than a small community needs; cutting a cube into quarters before thawing is the simplest portion control available for frozen food.
- **Never refreeze.** Aquatic-veterinary guidance is explicit that thawed frozen feeds should not be refrozen.
- **Observe the shelf life.** Commercial frozen feeds such as brine shrimp, bloodworms, mysids, or glassworms should be discarded after one year.
- **Substitute, do not add.** A frozen feeding replaces that meal's staple ration.

Part XXXIV — Bloodworms

Given a dedicated section because they are the single most commonly overfed item in freshwater fishkeeping.

What are bloodworms, nutritionally?

[STRONG EVIDENCE] Bloodworms are chironomid midge larvae. Proximate analysis shows them to be consistently high in protein but notably more variable in lipid content, and variation in the lipid-to-protein ratio affects the amount and pathway of energy supply to the animal consuming them. One comparative study measured protein content of chironomid larvae at 76 %, against 52.5 % for *Artemia* nauplii and 39.7 % for *Daphnia*.

Are bloodworms a complete diet?

[STRONG EVIDENCE] No. They are a single invertebrate species with a variable composition. A fish maintained predominantly on bloodworms is receiving an unbalanced diet with no vitamin or mineral supplementation, and with a lipid-to-protein ratio that is not controlled.

Are bloodworms harmful?

[CONTESTED CLAIM] The hobby claim that bloodworms cause bloat is not supported by direct evidence, and the mechanism usually offered is vague. What is supportable is that bloodworms are energy-dense, highly palatable, and typically fed in addition to the normal ration — a combination that predicts overfeeding and fat deposition rather than an acute digestive event.

Bloodworm claim	Assessment
“Bloodworms cause bloat”	[CONTESTED CLAIM] — no direct evidence. The association is plausibly confounded by the fact that bloodworms are fed enthusiastically and in quantity. A fish that ate a very large meal of anything may look distended
“Bloodworms are too fatty”	[CONTESTED CLAIM] — lipid content is variable rather than uniformly high. The energy-density concern is real but the specific claim overstates the consistency of the data
“Bloodworms should be fed every day”	[UNSUPPORTED AQUARIUM FOLKLORE] — feeding a single unsupplemented invertebrate as a daily staple is not defensible for any species
“Bloodworms are the best food for bettas”	[CONTESTED CLAIM] — they are among the most readily accepted foods for bettas, which is not the same claim. A complete betta pellet with bloodworms as an occasional substitution is better supported
“Bloodworms can cause allergic reactions in people”	[REASONABLE INFERENCE] — chironomid haemoglobin is a documented occupational and hobbyist allergen in the human allergy literature. Reported reactions include respiratory and contact symptoms, typically with repeated exposure. Handling frozen or freeze-dried product with tongs rather than fingers and washing hands afterward is a reasonable precaution
“Wild-collected bloodworms are better”	[CONTESTED CLAIM] — environmentally collected chironomid samples have shown excess lead in some samples and low zinc in others, indicating potential for both contaminant accumulation and nutritional imbalance

The position on bloodworms

Bloodworms are a useful, highly palatable supplementary food and a genuinely valuable tool for getting a reluctant or recovering fish to eat. They are not a staple, they are not nutritionally complete, and their real risk in the hobby is not toxicity but frequency.

A defensible pattern: bloodworms as a substitution for one staple meal, once or twice a week at most, in species that take invertebrate prey. Not daily, and not on top of the normal ration. **[REASONABLE INFERENCE]**

Part XXXV — Brine Shrimp

Two quite different foods sold under one name.

How do adult and newly hatched brine shrimp differ?

	Newly hatched nauplii	Adult brine shrimp
Size	~400–500 µm; suitable for many fry	Several millimetres; suitable for adult fish
Protein	Around 52.5% reported	Lower on a dry-weight basis
Energy reserves	Yolk sac intact at hatch; energy content declines steadily thereafter as the nauplius consumes its own reserves	Dependent on what the shrimp was fed
Essential fatty acids	Highly variable by strain and batch. Some strains are deficient in n-3 HUFA; enrichment is standard commercial practice	Reflects the culture diet; can be enriched
Typical use	Fry first food; small fish; conditioning	Adult supplement; digestive-complaint remedy in hobby practice
Main caution	Feed soon after hatch; nutritional value falls as yolk is consumed	High water content means the nutritional delivery per volume fed is low

What is enrichment and why does it exist?

[STRONG EVIDENCE] *Artemia* nutritional value is not constant but varies both geographically and temporally. In its natural state *Artemia* contains suboptimal levels of highly unsaturated fatty acids, particularly DHA and EPA. Enrichment — holding nauplii in a suspension of oil emulsion, microalgae, yeast, or soy lecithin for a period before feeding — raises their content of these nutrients and demonstrably improves growth and survival in fish fed on them.

A classic demonstration: hybrid striped bass larvae fed Great Salt Lake *Artemia* known to be insufficient in n-3 HUFA grew poorly; the same strain enriched with menhaden oil for 3–9 hours produced growth and survival equal to a HUFA-sufficient strain. Enrichment is therefore not a refinement for specialists; it is a correction of a known deficiency in the base food.

The aquarium implication is modest but real: for freshwater fry, which can generally elongate shorter-chain fatty acids, unenriched nauplii are usually adequate. The point worth carrying into the episode is the broader one — **“live” does not mean “nutritionally complete,”** and the aquaculture industry spends real money correcting the nutritional shortcomings of live food.

What about decapsulated cysts?

[STRONG EVIDENCE] Decapsulated *Artemia* cysts were found to be a more hygienic and off-the-shelf alternative to hatched nauplii for growth of adult guppies and of fry from guppies, platies, swordtails, mollies, and black neon tetras (Lim et al. 2002a). This is a practical option for keepers unwilling to run a hatch cycle.

Part XXXVI — Daphnia

The food carrying the hobby's most persistent functional claim.

What is the nutritional value of Daphnia?

[STRONG EVIDENCE] Modest as a nutrient source. Measured protein content of *Daphnia* was 39.7 % in one comparative study, against 52.5 % for *Artemia* nauplii and 76 % for chironomid larvae. *Daphnia* also has high water content and a chitinous exoskeleton. It is a low-energy-density food.

That low density is not a criticism. In an aquarium context where the dominant error is excess energy delivery, a palatable low-energy food has genuine utility: it satisfies feeding behaviour and provides enrichment with far less caloric impact than an equal volume of bloodworms.

[STRONG EVIDENCE] There is also direct ornamental evidence of functional benefit. Daily supplementation of *Daphnia* to swordtail broodstock maintained on an artificial flake diet produced a significant increase in fecundity through more rapid growth, a higher number of embryos, and an improved feed conversion ratio (Kruger et al. 2001a).

Does Daphnia relieve constipation?

[CONTESTED CLAIM] The claim is very widely repeated in hobby and even in some clinical-adjacent sources, and no controlled evidence has been located to support it. The proposed mechanisms are the chitinous exoskeleton acting as indigestible bulk, and the high water content.

Two observations are worth carrying into the episode. First, the mechanism is at least *coherent* in a way the pea mechanism is not: chitin is genuinely poorly digested by many fish, so it does add indigestible bulk, and unlike a pea it is a food item that carnivorous and insectivorous fish will actually eat and are adapted to process. Second, a coherent mechanism is not evidence. The honest statement is that *Daphnia* is a safe, palatable, low-energy food that many keepers use for this purpose, that it is unlikely to do harm, and that the specific therapeutic claim has not been tested.

Part XXXVII — Freeze-Dried Foods

Convenient, shelf-stable, and attached to one of the hobby's most confidently stated procedural rules.

What does freeze-drying do?

Freeze-drying (lyophilisation) removes water by sublimation from the frozen state, leaving a porous, lightweight, shelf-stable product that retains structure and much of its nutrient content. Compared with frozen equivalents it requires no freezer, has a long ambient shelf life, and carries no risk of a freezer failure destroying the supply.

Property	Consequence
Very low density and high porosity	Floats persistently until saturated; may not reach benthic fish
Rehydrates on contact with water	Expands as it takes up water — the basis of the soaking debate
Shelf-stable at ambient temperature	Convenient; but still subject to lipid oxidation and vitamin degradation over time
Structure retained	Recognisable as prey; often well accepted
Process is not a sterilisation step	Freeze-drying reduces but does not reliably eliminate all pathogens; the main advantage over live food is the absence of living intermediate hosts
Lower palatability than frozen for some fish	Commonly reported; texture and leached attractants differ

Must freeze-dried food always be soaked first?

[CONTESTED CLAIM] This is stated as a rule across the hobby and no controlled study has been located testing it in aquarium fish. The underlying concern — that dried food rehydrating and expanding inside the gut causes distension or impaction — is the same claim examined for pellets in Part LXI, and it rests on the same absence of direct evidence.

What can be said with more confidence:

- **Soaking definitely changes buoyancy.** Saturated freeze-dried food sinks rather than floating, which is a genuine and useful effect if the target animal is not a surface feeder. This is a real reason to soak that does not depend on the expansion claim being true.
- **Soaking leaches water-soluble nutrients.** Given that a large proportion of some water-soluble vitamins can leave a dry food within 30 seconds of water contact, extended pre-soaking transfers nutrients into the soaking water rather than into the fish. Soaking is therefore not cost-free.
- **The expansion concern is plausible but unmeasured in fish.** Freeze-dried material does take up water and expand. Whether that occurs to a degree that matters inside a fish, and in which species, has not been demonstrated.
- **Feeding a smaller quantity addresses the concern regardless.** Any risk from expansion scales with how much was eaten. Portion control is the intervention that works whether or not the mechanism is real.

How to handle the soaking question honestly

“Soaking freeze-dried food is harmless, and it does make the food sink, which helps if you're feeding bottom dwellers. The reason people usually give — that it swells up inside the fish — hasn't actually been tested. And soaking does wash out some of the water-soluble vitamins. So: soak it if you want it to sink, don't soak it for long, and either way feed less than you think you need.” **[CONTESTED CLAIM]**

Part XXXVIII — Live Foods

The category with the highest enrichment value and the highest biosecurity risk.

What live foods are used in freshwater fishkeeping?

Live food	Culture difficulty	Nutritional character	Principal risk
Brine shrimp (hatched nauplii)	Low — hatch on demand	Good for fry; variable EFA content by strain	Minimal disease risk (marine organism, poor freshwater survival)
<i>Daphnia</i> / <i>Moina</i>	Moderate — needs a culture	Low energy density; good enrichment; demonstrated reproductive benefit in swordtails	Cultured in organic-rich media; <i>Moina</i> cultured in manure-enriched water was a specific concern that led ornamental farms to shift to <i>Artemia</i>
Blackworms / <i>Tubifex</i>	Purchased; require rinsing and holding	High protein; extremely palatable	Historically the highest-risk live food; collection environment matters enormously
Mosquito larvae	Free where legal and available	Excellent for surface-feeding insectivores; highly natural	Legality varies by jurisdiction; risk of introducing pathogens or chemicals from the collection site; risk of adults emerging indoors
Microworms / banana worms / walter worms	Low — simple grain cultures	Good fry food after <i>Artemia</i> size range	Culture contamination and odour
Grindal and white worms	Moderate soil cultures	High protein and fat; excellent conditioning food	Fat content; easy to overfeed
Fruit flies (wingless)	Low	Natural surface prey for insectivores	Escape; not suitable for all species
Vinegar eels	Low	Very small; useful for tiny fry	Limited nutritional profile
Feeder fish	Purchased	See Part XC — substantial concerns	Disease, parasites, thiaminase, welfare

What are the benefits of live food?

- **Demonstrated performance benefit in some ornamental species.** Large commercial ornamental producers emphasise the importance of regular supplementation of formulated feeds with live feed because inclusion improves growth (Fernando et al. 1991). Specific demonstrations exist for *Daphnia* with swordtail broodstock and *Artemia* with juvenile angelfish.
- **Feeding response in reluctant feeders.** Movement triggers predatory response in fish that will not take inert food. This is the standard clinical application.
- **Behavioural enrichment.** Hunting, chasing, and searching are behaviours a pellet cannot elicit. See Part XCIII.
- **Conditioning for breeding.** Widely used, with some supporting evidence in the fecundity studies cited above.
- **Reaches feeding zones inert food cannot.** Live blackworms burrow into substrate and remain available to benthic fish for extended periods.

What are the risks?

[STRONG EVIDENCE] Live feed can act as the transmitter of parasitic, bacterial, and viral disease if it is not sourced and stored correctly (Pannevis 1993; Earle 1995). Live feed has also been described as often nutritionally deficient — a point the *Artemia* enrichment literature confirms comprehensively.

The industry's own behaviour is informative here. An increasing number of ornamental fish farmers shifted away from *Moina*, which is cultured in water enriched with organic manure, toward *Artemia* nauplii specifically because of contamination concerns (Lim et al. 2001). Commercial producers who depend on live food for growth performance still moved away from the higher-risk source.

Risk	Highest for	Mitigation
Parasite introduction	Wild-collected invertebrates from water containing fish; <i>Tubifex</i> from polluted sources	Buy from reputable culture sources; avoid collection from water bodies containing fish
Bacterial pathogens	Organic-rich culture media; poorly maintained cultures	Rinse thoroughly; maintain cultures cleanly; discard cultures that crash
Chemical contamination	Wild collection near agricultural or urban runoff	Avoid unknown collection sites; prefer cultured sources
Heavy metal accumulation	Environmentally collected chironomids showed excess lead in some samples	Prefer commercially produced product
Nutritional inconsistency	All live foods	Treat as supplement, not staple; consider enrichment for fry applications
Introduction of unwanted organisms	Wild-collected anything — hydra, planaria, snails, dragonfly nymphs	Inspect; quarantine cultures; avoid wild collection for display tanks

Part XXXIX — Vegetables

Useful, widely recommended, and frequently the source of a hidden organic load.

Which vegetables are used and what do they offer?

Vegetable	Preparation	Nutritional note	Assessment
Zucchini / courgette	Blanched or raw slice, often weighted	Very high water content; modest nutrient density; soft texture accepted readily	The standard choice. Value is largely behavioural and bulk rather than nutritional
Cucumber	Raw or blanched slice	Even higher water content than zucchini; among the least nutritionally dense options	Accepted well; essentially a grazing substrate
Blanched spinach	Briefly blanched	Denser nutrient profile than zucchini or cucumber; contains oxalates	Better nutritional value; fouls faster
Shelled peas	Cooked, skin removed, often crushed	About 5% crude fibre — comparable to many prepared fish foods; contains starch	See Part XL. Nutritional value is real but the therapeutic claim is not
Blanched green beans	Blanched and split	Moderate fibre; firmer texture	Accepted by some herbivores; persists well
Lettuce (romaine)	Blanched	Very low nutrient density; largely water and cellulose	Grazing substrate only; low value
Blanched broccoli / cauliflower stem	Blanched	Firmer; persists longer	Useful for persistent grazers; fouls if left too long
Sweet potato / squash	Cooked	Starchy; higher energy than most vegetable options	Suits warmwater omnivores and herbivores that handle starch well; poor for carnivores
Nori / dried seaweed	Clipped, unseasoned only	Higher nutrient density than terrestrial vegetables; convenient	Good option for grazers; must be plain and unsalted

Which fish actually benefit?

- **Clear benefit:** herbivorous and aufwuchs-grazing species — many plecos, *Otocinclus*, silver dollars, mbuna, hillstream loaches, and mollies. These animals have the digestive physiology for plant material and the behavioural need to graze.
- **Moderate benefit:** omnivores such as barbs, goldfish, many livebearers, and some cichlids, which will take plant material and can digest its carbohydrate component.
- **Little or no benefit:** obligate carnivores and insectivores. A betta or an oscar has no demonstrated requirement for vegetable matter and limited capacity to use its carbohydrate.

How should vegetables be prepared and managed?

1. **Blanch to soften.** Brief boiling or steaming breaks down cell walls, making the material accessible to grazing mouths and reducing the time it spends in the tank.
2. **Use no salt, oil, or seasoning.** This applies especially to nori and prepared vegetables intended for human consumption.
3. **Weight or clip it.** A veggie clip, a stainless fork, or a plant weight keeps the item where the target animal will find it rather than drifting into the filter intake.
4. **Match the portion to the grazers present.** A full zucchini slice for two *Otocinclus* is a large organic mass that will mostly decompose.

5. **Remove the remnant.** The standard practice is removal within about 24 hours, or the following morning for an item fed overnight. A decaying vegetable is a substantial and frequently overlooked nutrient input — and a documented driver of snail population increases. See Part XVII.
6. **Count it as part of the ration.** Vegetable feeding is not free. It should partially displace prepared food for the species eating it.

Part XL — The “Pea for Constipation” Rule

One of the most widely repeated therapeutic recommendations in the hobby, and the one with the weakest supporting mechanism.

What is the claim?

That feeding a cooked, shelled, crushed green pea relieves constipation and associated buoyancy problems in aquarium fish, most commonly goldfish and bettas. It is typically presented with a mammalian-style fibre-as-laxative rationale.

What is the evidence?

[CONTESTED CLAIM] No controlled study has been located testing peas as a treatment for digestive impaction in aquarium fish. Aquatic-veterinary commentary has directly challenged the mechanism: shelled green peas contain about 5 % fibre, which is the same as many prepared fish foods, and a goldfish's metabolism is described as not robust enough to process large amounts of fibre in any case (Sanders, aquatic veterinary practice).

That observation is doing a lot of work and is worth restating plainly: **if the fibre content of a pea is the same as the fibre content of the food the fish already eats, then fibre cannot be the operative variable.** Whatever effect keepers observe, the proposed mechanism does not explain it.

Adding to the difficulty, no fish species has been shown to efficiently digest cellulose, so the plant structural fibre in a pea is not being processed in the way the mammalian analogy assumes.

Is constipation even the right diagnosis?

[CONTESTED CLAIM] Frequently not. Buoyancy disorders and abdominal distension in fish have multiple possible causes. Veterinary sources list polycystic kidneys, excess gas in the bowel, swim bladder filled with fluid, ruptured swim bladder, over- or under-inflation, infection, neoplasia, injury, and genetic and anatomical factors. X-ray is described as the main method of diagnosis, and many cases are untreatable because of the underlying cause. In fancy goldfish varieties, the globoid body shape itself predisposes to abnormal swim bladder function.

A keeper looking at a bloated, bottom-sitting, or floating fish is looking at a sign, not a diagnosis. Treating every such presentation as constipation and reaching for a pea means that cases of organ disease, infection, and structural abnormality receive a dietary intervention.

When is the advice actively inappropriate?

Situation	Why the pea advice fails
Carnivorous and insectivorous species (bettas, many cichlids, predatory catfish)	These species handle dietary carbohydrate poorly. A pea delivers starch to an animal with limited capacity to use it, and there is no demonstrated fibre benefit to offset that
Any fish with pineconing scales, reddened skin, or severe abdominal swelling	These signs point to systemic disease rather than digestive slowdown. Delay in seeking veterinary assessment is the real harm
Fancy goldfish with a chronic buoyancy tilt	Body shape and swim bladder anatomy may be the cause. A dietary intervention cannot correct a structural predisposition
A fish that has stopped eating	It will not eat the pea either, and the anorexia itself is the more diagnostically important sign. See Part LXXXI
Fish in cold water	Water temperature directly dictates metabolism in ectotherms; in cold water fish are not able to digest their diet properly. Adding food to a fish that cannot digest it is the opposite of helpful

Is there any defensible version of this advice?

[REASONABLE INFERENCE] A limited one. For an omnivorous, carbohydrate-tolerant species such as a goldfish, a cooked shelled pea is a safe food that the animal can digest, and its consumption in place of the normal high-energy ration amounts to a very light, low-energy meal. The plausible mechanism is therefore not fibre but **substitution for a heavy meal** — which is close to a short fast.

What the episode should say about peas

“Peas aren't magic, and the fibre explanation doesn't hold up — a pea has about the same fibre as the food your fish already eats. If a goldfish eats a pea instead of its normal meal, what's really happening is a very light meal, which is closer to a short fast than a treatment. That might help and it won't hurt an omnivore.

But for a betta or another carnivore, a pea is starch they're not built to use, and for a fish that's pineconing or bottom-sitting and not eating, the pea is a distraction from getting it looked at.” **[CONTESTED CLAIM]**

Part XLI — Algae Eaters

A commercial category that does enormous damage as a husbandry category. The phrase implies the animal is fed by the tank, which is rarely true and never guaranteed.

Why does “algae eater” not mean “needs no food”?

[REASONABLE INFERENCE] Three reasons, each sufficient on its own. First, aquarium algae growth is variable and often deliberately minimised by the keeper, so the food supply is unreliable and usually declining. Second, most animals sold as algae eaters are omnivores or detritivores whose natural diet includes a substantial animal fraction. Third, the standing crop of algae in a healthy aquarium is a small fraction of what a grazer encounters over a day of foraging in a river.

The category also creates a perverse incentive structure: the keeper buys the animal *because* there is algae, then succeeds at removing the algae, and the animal's food supply is the thing that was eliminated. An *Otocinclus* in a spotless tank is in a tank with no food in it.

Animal	Actual diet	Can it live on tank algae?	What it needs from the keeper
<i>Otocinclus</i>	Soft green algae, diatoms, biofilm — a genuine algivore with a narrow specialisation	Only in a mature, biofilm-rich tank with continuous soft algae growth — rarely sustained	Established biofilm; blanched vegetables; algae wafers; frequent small supplementation. Commonly starves slowly in clean new tanks
Bristlenose pleco (<i>Ancistrus</i>)	Herbivore-leaning omnivore; grazes wood and surfaces	No	Wood to graze; vegetables; sinking herbivore food; some protein
Common pleco (<i>Pterygoplichthys</i>)	Omnivore/detritivore, more protein-inclined with size	No — and adult size makes the deficit large	Substantial deliberate feeding; vegetables plus protein; wood
Siamese algae eater (<i>Crossocheilus oblongus</i>)	Omnivore that takes algae opportunistically, less so as it matures	No	General omnivore diet; readily takes prepared food
Chinese algae eater (<i>Gyrinocheilus</i>)	Juvenile grazer that becomes an aggressive omnivore	No — and behaviour changes make it a poor community choice	Omnivore diet; note the behavioural trajectory
Mollies	Omnivore with a substantial plant fraction; will graze algae	No	Full omnivore diet with plant matter
Hillstream loach	Aufwuchs grazer adapted to fast, oxygen-rich water	No — needs continuous grazing surface	Biofilm-covered hardscape; supplemental wafers; high flow and oxygen
Nerite snails	Algae and biofilm grazers	Sometimes, in an algae-rich tank	Supplemental feeding when algae is limited — they are livestock, not cleaning equipment
Amano shrimp	Omnivorous scavenger and algae grazer	No	Supplemental feeding; they compete poorly with fish for food

The framing this brand should use

Algae-grazing animals are **ecosystem participants with a dietary specialisation**, not maintenance equipment. Buying one to solve an algae problem is buying an animal whose food you are simultaneously trying to eliminate.

If the algae is gone and the grazer is still there, the keeper has acquired a feeding obligation, not a solution. **[REASONABLE INFERENCE]**

Part XLII — Plecos

The genus-level differences within Loricariidae are larger than the differences between many unrelated aquarium fish. Treating plecos as a single dietary group is the root of most pleco husbandry failure.

Do plecos digest wood?

[STRONG EVIDENCE] No, not in the sense usually meant. Gut content analysis shows wood constituting up to about 75 % of the digesta of wild *Panaque*, and roughly 70 % of daily intake has been described for the genus. But studies of gut transit time, digestive enzyme activity, and fermentative end-products concluded these fish are **detritivores that do not obtain energy from the digestion of wood**. *Panaque nigrolineatus* and *Pterygoplichthys disjunctivus* passed wood through their guts in under four hours, could not assimilate significant amounts of cellulose, and did not thrive on a woody diet in the laboratory (German and Bittong 2009).

The researcher's own summary is unusually blunt for a scientific source: species of *Panaque* do consume a large amount of decaying wood in the wild, but they cannot survive on wood alone, and no fish species has ever been shown to efficiently digest cellulose. Enteric microbial communities in these fish appear to lack the cellobiohydrolases needed for efficient cellulose hydrolysis, and comparative microbiome work concluded there is little role for gut bacteria in wood digestion.

What the fish appear to actually obtain is the **microbial and detrital fraction associated with the wood** — ingested microbes, microbial by-products, and the biofilm and fungal material growing on decaying surfaces.

The pleco statement that should replace the folklore

Instead of: “Plecos need wood because they digest it.”

Use: “Wood-grazing plecos rasp large amounts of wood in the wild, but the research says they can't digest the wood itself — no fish can digest cellulose efficiently. What they're actually getting is the biofilm, microbes, and decaying material on and in it. Wood is worth providing because it supports natural grazing behaviour and tooth wear, not because it is food.”

[STRONG EVIDENCE]

How different are pleco species from each other?

Group	Dietary character	Feeding requirement	Common failure
<i>Panaque</i> / <i>Panaqolus</i> (royal, clown)	Xylivorous-detritivorous; rasps wood heavily	Wood surface plus substantial supplementary food — vegetables, sinking herbivore-leaning wafers, some protein	Assuming wood substitutes for feeding; slow starvation
<i>Ancistrus</i> (bristlenose)	Herbivore-leaning omnivore; efficient grazer	Vegetables, algae wafers, wood, occasional protein	Relatively forgiving; usually fine if fed at all
<i>Pterygoplichthys</i> (common, sailfin)	Omnivore/detritivore; protein intake rises with size	Large volume of mixed food; substantial adult requirement	Massively underfed as adults on the assumption they scavenge
<i>Hypancistrus</i> (zebra, L-number types)	Predominantly carnivorous	Protein-rich sinking foods; frozen invertebrates	Fed algae wafers and vegetables they cannot use — the classic failure
<i>Baryancistrus</i> (gold nugget)	Aufwuchs grazer with a significant animal fraction	Mixed grazing and protein; high oxygen	Fed as a strict herbivore
<i>Otocinclus</i> (not a pleco sensu stricto but grouped commercially)	Specialist algivore	Continuous soft algae and biofilm; supplementation	Starvation in clean tanks; poor condition on import
<i>Farlowella</i> / <i>Sturisoma</i> (twig catfish)	Herbivore/aufwuchs grazer, slow and non-competitive	Deliberate target feeding; vegetables; wood	Outcompeted entirely in busy communities

Do plecos eat fish waste?

[UNSUPPORTED AQUARIUM FOLKLORE] No. This is one of the most persistent and least defensible claims in the hobby. Fish faeces are the residue of digestion — the fraction a fish could not extract nutrition from. A pleco rasping across a surface may incidentally ingest detritus containing faecal material along with biofilm and microorganisms, but it is grazing the microbial community, not consuming waste as a food source. No animal in an aquarium reduces the tank's nutrient load by eating waste; nutrients that pass through another animal are still in the system.

Part XLIII — *Corydoras*

The most commonly underfed fish in the freshwater hobby, and the clearest case study in why 'cleanup crew' framing causes harm.

Do *Corydoras* live on leftovers?

[UNSUPPORTED AQUARIUM FOLKLORE] No. *Corydoras* are benthic carnivores and omnivores that forage actively for invertebrates and organic particles in and on the substrate. They require deliberate feeding with sinking food that reaches the bottom in adequate quantity. The 'leftovers' model describes a tank that is simultaneously overfed at the top and underfed at the bottom.

The internal contradiction is worth stating explicitly for the episode: if the tank produces enough leftovers to sustain a group of *Corydoras*, then the tank is being substantially overfed at the surface. If the tank is fed correctly at the surface, there are no meaningful leftovers and the *Corydoras* are being starved. Both branches are husbandry failures, which is why 'they eat the leftovers' cannot be a plan.

How should *Corydoras* actually be fed?

1. **Use food designed to sink.** Sinking pellets, wafers, or tablets formulated with a protein component — not algae wafers as the sole item.
2. **Feed after the mid-water fish are fed, or after lights-out.** This reduces interception. Many *Corydoras* species are active at dusk and after dark.
3. **Deliver to a specific location.** A feeding dish or a consistent corner concentrates the food where the group forages and makes the remainder easy to see and remove.
4. **Watch them actually eat.** This is the single most useful verification available, and it is skipped constantly.
5. **Supplement with frozen or live invertebrates.** Frozen bloodworms, daphnia, and live blackworms are genuinely appropriate for this group as occasional substitutions.
6. **Check body condition monthly.** A thin *Corydoras* has a visibly narrow, pinched profile behind the head. See Part XXI.

Note also that *Corydoras* require a substrate they can forage in without barbel damage. A fish whose barbels are eroded by sharp gravel or by substrate bacterial load is a fish whose primary food-location sense is compromised — a feeding problem with a non-feeding cause.

Part XLIV — Loaches

A behaviourally diverse group united mostly by living near the bottom.

Group	Dietary character	Feeding notes
Kuhli loaches (<i>Pangio</i>)	Benthic carnivore/detritivore; nocturnal and secretive	Among the most frequently underfed aquarium fish. Feed sinking protein food after lights-out; they will not compete in daylight. See Part LIV
Clown loach (<i>Chromobotia</i>)	Omnivore with a strong invertebrate component; eats snails	Large adult size and substantial appetite; social feeder; will take most sinking foods readily
Yoyo / zebra loach (<i>Botia</i>)	Omnivore, invertebrate-inclined	Active daytime feeders; compete adequately; snail predation is real but should not be the feeding plan
Hillstream loaches (<i>Sewellia</i> , <i>Beaufortia</i>)	Aufwuchs grazers in fast current	Need biofilm-covered surfaces and high flow; supplement with wafers and blanched vegetables. Frequently starve in slow, clean tanks
Dojo / weather loach (<i>Misgurnus</i>)	Omnivore/detritivore, substrate sifter	Coolwater species; large; will take sinking pellets. Temperature strongly affects appetite
Horseface and other sand-sifting loaches	Benthic invertebrate feeders	Require fine sand to sift; food must reach the substrate intact

The common thread: loaches almost always feed at or below the substrate, and several are nocturnal or crepuscular. Food form and timing matter more for this group than food composition.

Part XLV — Bettas

The most singly-kept fish in the hobby, which means the entire ration goes to one animal and the owner's feeding instinct is unmoderated by competition.

What is a betta's natural diet?

[STRONG EVIDENCE] *Betta splendens* is a carnivore and specifically an insectivore, taking small invertebrate prey at and near the surface. Its requirements are protein-dominated; published spawning trial work found 35 % dietary protein optimal for reproductive performance, and commercial betta diets are commonly formulated around 40–48 % protein.

Is a betta's stomach really the size of its eye?

[UNSUPPORTED AQUARIUM FOLKLORE] As an anatomical claim, no reliable source supports the specific eye-size comparison. As a portion heuristic it is conservative and therefore relatively harmless — which is why it survives. The betta does have a genuinely small stomach and limited capacity, so the conclusion the rule points toward is correct even though the stated reason is not.

The more defensible framing is quantitative rather than anatomical: a betta is a small fish with small absolute energy requirements, kept alone, with an owner who enjoys feeding it. Two to four appropriately sized pellets once or twice daily is a commonly used and reasonable pattern, and it looks alarmingly small to a new keeper.

What does the feeding frequency evidence say for bettas?

[STRONG EVIDENCE] In juvenile *Betta splendens*, feeding, growth, and fecundity rates increased with feeding frequency up to two meals per day and then declined at three meals per day. Fish fed three times daily both consumed less and gained less than fish fed twice daily (James and Sampath 2004). Two feedings per day were also reported as optimal for growth and reproductive performance in this species.

For an adult betta at maintenance rather than a growing juvenile, once or twice daily is well within the supported range, and the practical difference between them is smaller than the difference made by portion size.

Betta-specific feeding issues

Issue	Assessment
Pellet size	Critical. Bettas have small mouths and many general tropical pellets are too large. Pellets designed for the species, or micro pellets, are appropriate. A fish that repeatedly takes and spits food is usually telling you the particle is too big. See Part LIX
Obesity	A genuine and common problem. Single housing, small body size, enthusiastic feeding response, and frequent high-fat treats combine badly. See Part XVIII
Bloodworms as a staple	[UNSUPPORTED AQUARIUM FOLKLORE] — extremely common practice, not defensible. A complete pellet as the base with bloodworms as an occasional substitution is better supported. See Part XXXIV
Fasting	Commonly practised, low risk for a healthy adult, and useful as portion discipline. Not a treatment for bloating of unknown cause. See Part X
Peas for bloating	Poorly supported for this species specifically. Bettas are carnivores with limited carbohydrate-handling capacity. See Part XL
Surface feeding and air ingestion	Bettas are labyrinth fish that routinely take atmospheric air. Surface feeding is natural for them, and the air-gulping concern raised for goldfish does not transfer directly

Issue	Assessment
Refusing food after a move or change	Common and usually temporary. See Part LXXXIII

Part XLVI — Gouramis

Slow, deliberate surface and mid-water feeders that lose consistently to faster tankmates.

Gouramis are anabantoids like bettas, generally omnivorous with an insectivorous lean, and they feed methodically rather than competitively. In the ornamental maintenance data, the moonlight gourami had the highest measured maintenance requirement of the five species tested — under 28.5 mg feed per day at 1.87 g body weight — roughly seven times that of a neon tetra, which illustrates how much body size dominates within a community.

Carbohydrate digestibility in moonlight gourami was reported at around 50 %, notably lower than the roughly 70 % reported for goldfish, which suggests starchy foods are less efficiently used by this group than by cyprinid omnivores.

- **Feeding speed is the practical problem.** In a tank with tetras, danios, or barbs, gouramis frequently arrive late. Feeding in more than one location, or feeding a slow-sinking food that persists in the water column, helps.
- **Surface orientation.** Floating and slow-sinking foods suit them; fast-sinking food is often missed entirely.
- **Dwarf species are small-mouthed.** Particle size matters as it does for bettas.
- **Protein level for reproduction.** Female dwarf gouramis receiving 25–45 % dietary protein had the highest fecundity (Shim et al. 1989).

Part XLVII — Tetras, Rasboras and Danios

The numerical majority of community fish, and the group the two-minute rule is implicitly calibrated on.

What characterises feeding in small schooling fish?

- **Small particle requirement.** Neon tetras, ember tetras, chili rasboras and similar species have very small mouths. Crumbled flake, micro pellets, and small frozen items such as cyclops and daphnia are appropriate; standard community pellets are often too large.
- **Midwater and upper-column feeding.** Most will not forage effectively on the substrate. Food that sinks past them is effectively lost from their perspective.
- **Fast and competitive.** Danios in particular are the fastest feeders in most community tanks and will strip the surface before anything else arrives.
- **Very low absolute requirement.** The neon tetra maintenance figure — about 3.8 mg of feed per day — is the most useful single number in this document for calibrating how little a small schooling fish actually needs.
- **Multiple small portions distribute better than one large one.** Feeding a small amount, waiting, and feeding a second small amount gives slower fish a second opportunity without increasing the total.

[STRONG EVIDENCE] Group size affects feeding confidence. Schooling species held in inadequate numbers show elevated stress behaviours and reduced feeding activity in general husbandry experience; this is a welfare recommendation with wide support in the aquarium literature though it has been quantified mainly in aquaculture and behavioural contexts rather than in ornamental trials. Cross-reference Part XCV.

Part XLVIII — Barbs

Robust omnivores with strong carbohydrate-handling capacity and assertive feeding behaviour.

Barbs are cyprinids and therefore agastric — no stomach, no acid digestion phase. Combined with generally good amylase activity, this makes them well suited to frequent small meals containing a meaningful plant and starch fraction, and poorly suited to infrequent large meals.

Published ornamental requirement data exist for two species: tin foil barb (*Barbodes altus*) at about 41.7 % dietary protein for weight gain in juveniles, and tiger barb (*Barbus tetrazona*) with a phosphorus requirement of about 0.52 %. Fish meal was successfully substituted by soybean meal up to 37 % of the diet, replacing 33 % of the fish meal protein, for normal growth in juvenile tin foil barb (Elangovan and Shim 2000) — relevant to Part LXXV on ingredient interpretation.

- **Assertive feeders.** Barbs commonly outcompete slower tankmates, which makes them a frequent contributor to Type 3 misdirected feeding in community tanks.
- **Omnivorous balance matters.** A purely high-protein diet is not appropriate; plant matter and digestible carbohydrate are genuinely usable by this group.
- **Some species are noted for preferring live food.** The ruby barb (*Puntius nigrofasciatus*) was described as a voracious and indiscriminate feeder that prefers live feed to artificial feeds.

Part XLIX — Livebearers

Continuous grazers with small gut capacity, high reproductive output, and a persistent presence in beginner tanks.

Species	Dietary lean	Feeding notes
Guppy (<i>Poecilia reticulata</i>)	Omnivore; the best-studied ornamental species for nutrient requirements	Protein requirement 30–40% for growth and gonadal development. Mineral requirements published: phosphorus 0.53–1.23%, magnesium 0.054%, zinc 0.01%, iron 0.008%. Vitamin C supplementation at 1000–2000 mg/kg improved stress resistance to osmotic shock (Lim et al. 2002b)
Molly (<i>Poecilia sphenops</i> , <i>P. latipinna</i>)	Omnivore with the strongest plant fraction of the group	Genuinely benefits from algae, vegetables, and spirulina-based foods. Frequently sold as an algae eater, which it is not in the sense buyers expect
Platy (<i>Xiphophorus maculatus</i>)	Omnivore	Undemanding; continuous grazer; easily overfed
Swordtail (<i>Xiphophorus helleri</i>)	Omnivore	A diet with at least 45% protein at 6% lipid gave best specific growth rate and feed conversion in growing juveniles (Kruger et al. 2001b). Daily <i>Daphnia</i> supplementation to broodstock on flake significantly increased fecundity

Do livebearers need frequent feeding?

[REASONABLE INFERENCE] They are adapted to near-continuous grazing with limited gut capacity, which supports dividing the daily ration into smaller portions rather than increasing it. The common hobby advice to feed livebearers more often is defensible only in that form — more meals of the same total, not more food.

Two livebearer-specific complications:

- **Gravid females confound body condition assessment.** A distended abdomen in a female livebearer is usually pregnancy, not overfeeding. The gravid spot and the squared-off shape of the distension distinguish them. See Part XXI.
- **Fry appear continuously.** A community with livebearers acquires a population of fry that requires frequent small feeding while the adults do not. The usual resolution is that fry either survive on fine particles and biofilm or are predated; deliberately feeding for the fry in a display tank raises total input substantially. See Part VII.

Part L — Cichlids

The family with the widest dietary range in the hobby and the most misleading single food category on the shelf.

Why can one “cichlid food” not suit every cichlid?

[STRONG EVIDENCE] Cichlid dietary strategies span obligate piscivory, invertebrate carnivory, generalist omnivory, aufwuchs grazing, and specialised herbivory. Published protein requirements within the ornamental cichlid literature alone range from about 40.8 % for redhead cichlid fry to 44.9–50.1 % for discus. Carbohydrate handling differs correspondingly, and excessive dietary carbohydrate is a genuine metabolic burden for the carnivorous members of the family.

Dietary group	Representative species	Diet character	Feeding consequence	Common error
Aufwuchs grazers	Most mbuna (<i>Pseudotropheus</i> , <i>Labidochromis</i> , <i>Metriaclicma</i>)	Graze the algal and microfaunal surface community; near-continuous intake of bulky, low-fat material	Frequent small meals of a plant-inclined, moderate-protein, low-fat food; grazeable surfaces	Fed high-protein high-fat carnivore pellets. Hepatic lipidosis is reported clinically in African cichlids
Micropredators / invertebrate feeders	Many <i>Apistogramma</i> , rams, kribensis, shell dwellers	Small invertebrates taken continuously	Small particulate protein food; frozen invertebrates suit them well	Particle size too large; outcompeted by faster fish
Generalist omnivores	Severum, firemouth, convict, many <i>Cichlasoma</i> types	Mixed animal and plant	Balanced omnivore diet; handle a range of foods	Overfeeding; these are the easiest cichlids to make obese
Piscivores and large-prey predators	Oscar, many <i>Crenicichla</i> , peacock bass	Large infrequent prey	Larger meals at longer intervals; veterinary guidance permits once or twice weekly for large-meal carnivores	Feeder fish (see Part XC); daily large feeding leading to obesity
Specialised herbivores	<i>Tropheus</i> , some <i>Petrochromis</i>	Highly specialised algal grazing with long gut	Low-protein, low-fat, high-fibre grazing diet; very sensitive to dietary error	High-protein feeding, associated in husbandry practice with serious digestive disease
Discus and related	<i>Symphysodon</i>	Carnivorous with high protein requirement (44.9–50.1%)	High-protein diet, frequent small feedings, warm and very clean water	Underfeeding protein; high-protein feeding without the corresponding water change regime

A specific note on discus: because their protein requirement is near the top of the ornamental range and they are typically kept in high densities, they generate exceptional nitrogen loading. Sales and Janssens noted that soybean meal inclusion in discus diets reduced food intake owing to low palatability and decreased digestibility, and that longer acclimatisation periods are recommended when changing diet. Large-scale discus breeders rely heavily on live food such as *Tubifex*, bloodworms, and *Artemia* nauplii for growing fry.

Part LI — Goldfish

Agastric, omnivorous, physostomous, long-lived, frequently kept in poorly sized systems, and attached to more feeding folklore than any other aquarium fish.

What is a goldfish's actual dietary profile?

[STRONG EVIDENCE] Goldfish are omnivorous cyprinids with no stomach. Published protein requirements are around 29 % for juveniles (Lochmann and Phillips 1994), with one study concluding 40–45 % significantly enhanced growth and dietary utilisation in young goldfish, and 53 % for larvae. Carbohydrate digestibility of around 70 % has been reported — among the highest in the ornamental data — and most herbivorous fish such as goldfish and koi use hindgut microflora to digest complex carbohydrates.

The practical reading is that goldfish are genuinely well equipped to use plant matter and digestible carbohydrate, which distinguishes them from most of the carnivorous and omnivorous species discussed elsewhere in this document, and which makes goldfish-specific diets meaningfully different from general tropical foods rather than a marketing distinction.

Does floating food cause swim bladder problems?

[CONTESTED CLAIM] Partly supported, with important qualifications, and it is not the whole story. Goldfish are physostomous — the swim bladder retains a pneumatic duct connecting it to the oesophagus. Aquatic-veterinary sources state that this allows air swallowed during surface feeding to enter the swim bladder and that switching to a sinking or neutrally buoyant diet may reduce excess air intake in mild cases. Canadian veterinary guidance phrases it more cautiously still: some report that fish take in excess air when they access food at the surface.

Against that, the same veterinary sources are clear that many buoyancy disorders are not diet-related at all. Documented causes include polycystic kidneys, excess gas in the bowel, swim bladder filled with fluid, rupture, over- or under-inflation, bacterial or viral infection thickening the bladder lining, neoplasia, injury, and genetic and anatomical predisposition. Fancy varieties bred for globoid bodies have compressed coelomic cavities and correspondingly shortened, rounded swim bladders, which predisposes them from birth. Buoyancy disorders are described as affecting long-bodied and short-bodied fancy fish roughly equally, X-ray is the main diagnostic method, and many cases are untreatable because of the underlying cause.

The honest goldfish buoyancy statement

“Goldfish have a direct connection between their gut and their swim bladder, so gulping air at the surface can contribute to buoyancy problems — and switching to sinking food is a reasonable, low-cost thing to try. But floating food is not *the* cause of swim bladder disease. Fancy goldfish are predisposed by the body shape they were bred for, and a lot of buoyancy cases turn out to be infection, organ disease, or structural problems that no change of food will fix. If a fish is persistently floating or sinking, that's a case for an aquatic vet, not a case for a pea.” **[CONTESTED CLAIM]**

Goldfish feeding practicalities

- **Sinking food is a reasonable default** for fancy varieties, for the reason above and because it slows a notoriously fast feeder.
- **Temperature governs everything.** Goldfish are ectotherms and commonly kept cool or in ponds. In cold water they cannot properly digest their diet; standard pond practice reduces feeding and switches to wheat-germ-based food below about 15 °C and stops feeding altogether below about 10 °C, because digestive processes effectively cease. See Part LXXXV.

- **Body shape confounds condition assessment.** Pearlscales and similar varieties are naturally globoid, and some individuals normally swim slightly head-down. Neither is a condition problem.
- **Carotenoid nutrition is real.** Goldfish metabolise zeaxanthin to astaxanthin readily but metabolise very little beta-carotene and no lutein to astaxanthin (Hata and Hata 1972). See Part LXXIX.
- **Longevity raises the stakes on chronic overfeeding.** A goldfish may live for a decade or more, which gives hepatic and metabolic consequences ample time to develop.

Part LII — Labyrinth Fish and Surface Feeding

A short section because the practical points are narrow, but they recur across bettas, gouramis, paradise fish, and relatives.

Labyrinth fish possess a suprabranchial accessory breathing organ and routinely take atmospheric air at the surface. This has three consequences for feeding:

- **Surface access is non-negotiable.** Anything that obstructs the surface — dense floating plant cover with no gaps, a tightly fitted lid with minimal air space, heavy surface film — interferes with respiration independent of feeding.
- **Surface feeding is natural, and the air-ingestion concern raised for goldfish does not transfer.** These fish take air deliberately and routinely; the physostomous goldfish situation is a different mechanism.
- **Feeding pace is slow and deliberate.** Labyrinth fish inspect food before taking it and lose consistently to fast competitors. Floating or very slow-sinking food that remains available for a minute or two suits them; fast-sinking food is frequently missed.
- **Mouth size is small relative to body size** in most of the group, which makes particle size a recurring issue. Spitting food out is usually a size complaint. See Part LIX.

Part LIII — Feeding Bottom Dwellers

One of the two most practically important sections in this document, because it addresses the failure mode that quantity rules cannot detect: a tank that is correctly fed in aggregate and starving at the bottom.

Why do bottom dwellers go underfed in well-fed tanks?

[REASONABLE INFERENCE] Because food delivery is vertical and consumption is competitive. Food entering at the surface passes through every other fish's feeding zone before it reaches the substrate. In a stocked community, the probability that a given particle survives that gauntlet is low, and it falls further with every additional mid-water fish.

The keeper's feedback loop makes this worse rather than better. The visible signal — no food left on the substrate — reads as correct feeding. It is equally consistent with the bottom dwellers receiving nothing. There is no observation at the surface that distinguishes the two, which is why this problem persists in otherwise carefully managed tanks.

What are the working methods?

Method	How it works	Effectiveness	Limitations
Sinking food	Food formulated to reach the substrate quickly rather than drifting	Necessary but not sufficient — fast-sinking food is still intercepted in busy tanks	Mid-water fish learn to chase sinking pellets within days
Feed after lights-out	Delivered when diurnal competitors have settled and nocturnal bottom dwellers become active	High — probably the single most effective method for nocturnal species	The keeper cannot observe consumption; remnants must be checked in the morning
Decoy feeding	Feed the mid-water fish at the surface first, then add sinking food while they are occupied	High and immediately practical	Requires the keeper to be present for both steps; increases total input unless portions are split
Target feeding	Deliver food directly to the animal or its location using a pipette, turkey baster, tube, or tongs	Very high for individual animals	Labour intensive; not scalable to a large group; some fish are disturbed by the approach
Feeding dish or station	A shallow dish, terracotta saucer, or defined area receives sinking food	High — concentrates food, makes remnants visible and easy to siphon	Fish must learn the location; dominant fish may claim it
Multiple simultaneous locations	Food added at two or more points in the tank at once	Moderate to high — splits the competitive response	Requires more portions to be prepared; can increase total input
Weighted or clipped food	Vegetables or wafers held at the substrate	High for grazers	Requires removal; a persistent organic mass if ignored
Live burrowing food	Blackworms and similar organisms enter the substrate and remain available	High for substrate foragers; strong enrichment value	Sourcing and biosecurity; see Part XXXVIII

How do you verify a bottom dweller is being fed?

1. **Watch the animal eat.** Not the tank — the animal. Use a dim light or a red light after dark if necessary.
2. **Check body condition monthly.** Corydoras, loaches, and small plecos show underconditioning clearly in the profile behind the head and along the abdomen. See Part XXI.
3. **Count the remnant.** If you place a known quantity in a dish and it is gone by morning, something ate it — and if it is still there, nothing did.
4. **Look for foraging effort.** An animal that forages constantly and visibly across every surface for hours is working harder than an adequately fed animal would.

5. **Distrust 'they'll find something.'** This is the assumption this entire section exists to remove.

Part LIV — Nocturnal Fish

A specific case of the bottom-dweller problem, with an additional temporal dimension.

Which aquarium fish are genuinely nocturnal or crepuscular?

Group	Activity pattern	Feeding consequence
Kuhli loaches (<i>Pangio</i>)	Strongly nocturnal; often entirely hidden during daylight	Daytime feeding frequently reaches them not at all
Many <i>Corydoras</i> species	Crepuscular to nocturnal, with considerable species and individual variation	Evening or after-dark feeding improves access substantially
<i>Ancistrus</i> and most plecos	Nocturnal grazers	Wafers and vegetables offered at lights-out are far better used
<i>Synodontis</i> and many catfish	Nocturnal	Will learn to feed in daylight in captivity, but competitive access is poor
Some knife-fish and elephant-nose	Nocturnal, often electroreceptive foragers	Almost always require deliberate after-dark or target feeding
Certain eels and spiny eels	Nocturnal, substrate-oriented	Target feeding is usually necessary
Hillstream loaches	Diurnal grazers — the exception in this list	Daytime biofilm grazing is normal for them

Is feeding before or after lights-out beneficial?

[REASONABLE INFERENCE] Yes for genuinely nocturnal species, and the mechanism is straightforward: feeding at the time the animal is behaviourally prepared to forage removes competition from diurnal tankmates and coincides with the animal's natural activity peak. Fish entrain readily to scheduled feeding times regardless of whether those times fall in the light or dark phase — zebrafish entrained to feeding in the mid-dark phase showed a seven-hour phase advance in hepatic *per1* expression compared with mid-light feeding, confirming that dark-phase feeding is a fully effective zeitgeber.

The practical protocol: add sinking food a few minutes after the lights go off, when diurnal fish have settled but before the keeper leaves the room, then check for remnants the next morning and siphon anything left. Where this is done consistently, nocturnal species learn the schedule and emerge promptly — which also gives the keeper an observation opportunity that otherwise does not exist.

A caution: after-dark feeding removes the keeper's ability to observe consumption in real time, which makes portion discipline and morning remnant checks more important, not less.

Part LV — Community Tank Feeding

The second of the two most practically important sections. Everything about quantity assumes food reaches the fish it was intended for. In a community tank, that assumption does most of the failing.

What is the structural problem?

[STRONG EVIDENCE] Sales and Janssens identified this directly as one of the main problems in ornamental fish nutrition: the diversity of species kept in home and public aquaria and how to provide them with adequate diets. A single diet must be suitable for herbivores, omnivores, and carnivores simultaneously, whose nutrient requirements differ and whose digestibility of various dietary components differs depending on natural diet and intestinal morphology. The physical characteristics of the diet and the feeding regime must additionally satisfy surface, middle, and bottom feeders and their diurnal variation in feeding. Food particles must be small enough for the smaller species to ingest but large enough to be identified and eaten by the larger ones.

That is a formal statement, from a peer-reviewed review, that the community aquarium is a genuinely hard nutritional problem rather than a matter of choosing a good food. The authors' own proposed solution is notably modest: rather than satisfying the needs of individual species, develop an intermediate feed that minimises deviation from the crude requirements of all of them.

Who loses in a community tank, and why?

Profile	Why they lose	Countermeasure
Fast surface feeders (danios, some barbs, hatchetfish)	They win — they are the reason others lose	Feed them deliberately at the surface first as a decoy
Slow deliberate feeders (gouramis, many cichlids, loricariids)	Inspect food before taking it; arrive after the rush	Slow-sinking food that persists; a second small portion after the first is gone
Timid and newly introduced fish	Will not compete for days after introduction	Feed in a location away from the dominant group; feed a second portion once the rush disperses
Bottom dwellers	Food is intercepted above them	Sinking food, decoy feeding, feeding station, after-dark feeding
Nocturnal fish	Not active when food is available	After-dark feeding
Small individuals in a size-mixed group	Cannot physically take the particle size that suits larger fish; documented in aquaculture as a cause of divergent growth and mortality of smaller fish	Feed a mixture of particle sizes, or the largest size the <i>smallest</i> fish can take
Specialised feeders	The community diet does not match their requirement	Species-specific supplementation, target fed
Subordinate individuals of hierarchical species	Actively excluded from food by dominants	Multiple feeding locations simultaneously; visual barriers; reassess stocking

What is the community feeding strategy that follows?

1. **Inventory by feeding zone and time, not by species list.** Write down which fish feed at the surface, in midwater, and on the substrate, and which are active after dark. This is a five-minute exercise that reframes the whole problem.
2. **Split the daily ration rather than increasing it.** Two or three deliveries of the same total, staged across zones, outperforms one large delivery.

3. **Sequence deliberately.** Surface first, then midwater, then sinking food while the upper-column fish are occupied.
4. **Use more than one food form.** A floating or slow-sinking staple plus a genuine sinking food is not variety for its own sake; it is a distribution mechanism.
5. **Use physical separation.** Feeding rings, dishes, and separated drop points split the competitive response spatially.
6. **Add an after-dark delivery if nocturnal species are present.**
7. **Verify individually.** Body condition of the species you *expect* to lose is the measurement that tells you whether the strategy works. See Part XXI.

The reframe that resolves the central question

Most keepers ask “how much should I feed?” when the operative question in a community tank is “how do I get the right amount to each of these animals?”

Reducing the quantity in a tank with a distribution problem makes the distribution problem worse. Increasing it makes the water quality worse without helping the fish that are missing out. Neither adjustment to the quantity dial can solve a problem that is not about quantity. [\[REASONABLE INFERENCE\]](#)

Part LVI — Multiple Feeding Zones

A short structural section that the previous two depend on.

Zone	Typical residents	Appropriate food form	How to deliver	Failure signal
Surface	Hatchetfish, bettas, gouramis, guppies, killifish, some danios	Floating flake, floating pellets, floating freeze-dried, live insects	Broadcast or into a feeding ring	Food waterlogs and sinks before being taken
Upper midwater	Danios, many tetras, rasboras	Slow-sinking flake and micro pellets	Broadcast; these fish intercept most things	Rarely fails — this zone is usually over-served
Lower midwater	Many cichlids, larger tetras, barbs	Slow-sinking pellets	Broadcast; allow time	Food passes through too fast for slow feeders
Substrate surface	<i>Corydoras</i> , loaches, many catfish, plecos	Fast-sinking pellets, wafers, tablets, frozen invertebrates	Feeding dish, target feeding, after-dark, decoy method	Nothing reaches the bottom; bottom dwellers thin
Within substrate	Sand-sifting species, kuhli loaches	Live blackworms; fine sinking food that settles into sand	Direct placement; live food burrows naturally	Rarely verifiable without observation
Surfaces and hardscape (aufwuchs)	Otocinclus, hillstream loaches, mbuna, grazing snails	Biofilm, algae, clipped vegetables, wafers left in place	Clip, weight, or place on hardscape	Grazers thin despite a 'clean' tank

Part LVII — Target Feeding

The most precise delivery method available to a hobbyist, and the least used.

Tool	Best for	Notes
Pipette or turkey baster	Frozen and live foods delivered to a specific animal or location	Allows delivery under decor, into caves, and directly to a shy fish. The standard tool for target feeding invertebrates and benthic fish
Rigid feeding tube	Sinking pellets and wafers delivered to a precise substrate location in a deeper tank	Prevents food dispersing on the way down; effectively a chute
Long tongs or forceps	Wafers, vegetables, larger frozen items, whole prey	Also useful for retrieving remnants without a full siphon
Dedicated feeding dish or saucer	Any sinking food for a benthic group	Concentrates food, makes remnants visible, and turns removal into a single lift rather than a substrate vacuum
Veggie clip or weight	Vegetables for grazers	Keeps material in place and off the filter intake
Feeding ring	Floating foods	See Part LVIII

When is target feeding actually warranted?

- A specific individual is visibly underconditioned while others are not.
- A shy, newly introduced, or recovering fish is not competing.
- A specialised feeder requires a food the rest of the community will otherwise take.
- A nocturnal or cave-dwelling animal cannot be reached by broadcast feeding.
- The keeper needs to confirm that a particular animal is eating — a diagnostic use, not only a nutritional one.
- Medicated food must reach a specific patient. See Part LXXX.

Part LVIII — Feeding Rings

A simple device with two real functions and one common misunderstanding.

A feeding ring is a floating collar that confines floating food to a defined area of the surface. Its genuine benefits:

- **Prevents food from being drawn into the filter intake or overflow** before it is eaten — a direct reduction in wasted food and in mechanical filter loading.
- **Keeps food away from the surface film's edges and corners** where it accumulates and decays out of sight.
- **Creates a predictable feeding location**, which fish learn rapidly. This is useful for reducing the scatter of a broadcast feeding and for concentrating a shy fish's attention.
- **Makes portioning visible**. A defined area gives the keeper a consistent visual reference for portion size across feedings.

The misunderstanding: a feeding ring concentrates competition rather than relieving it. In a tank where a dominant fish monopolises feeding, a ring makes monopolisation easier by putting all the food in one defensible place. Where the problem is competition rather than food loss, multiple drop points are the correct intervention and a single ring is counterproductive.

Part LIX — Food Size

A constraint that is absolute rather than gradual: a particle the fish cannot take is not food.

Why does particle size matter so much?

[STRONG EVIDENCE] Aquaculture practice treats particle size as a primary variable, not a detail. Standard guidance is to feed the largest crumble or pellet the fish can consume and not to exceed it, with an explicit warning that differential growth occurs if only some fish are able to feed — which can result in the death of the smaller fish, because larger fish can eat the larger feed while smaller fish cannot. A mixture of sizes is recommended in some stages.

The angelfish larval work makes the same point at the other end of the scale: albino angelfish larvae with a smaller mouth gape than common angelfish achieved about 59 % survival on rotifers or a microdiet versus about 2 % on *Artemia* alone. A 57-percentage-point survival difference driven by particle size.

Problem	Cause	Signs	Fix
Pellet too large	Particle exceeds comfortable gape	Fish takes food and spits it out repeatedly; food ignored; fish mouths it and abandons it	Smaller pellet, crushed pellet, or crumbled flake
Food too fine	Particle below the size the fish registers or can capture efficiently	Larger fish ignore it; it clouds the water and is drawn into filtration	Larger particle for the larger fish; keep fine food for the small ones
Single size in a size-mixed community	One particle size cannot serve all mouths	Small fish thin while large fish thrive, or vice versa	Feed two particle sizes, or size the particle to the smallest mouth
Wafer too large for the group	One wafer exceeds what a small group consumes before it disintegrates	Persistent wafer remnants; snail aggregation; fouling	Break wafers; portion to the group
Frozen item too large	Whole bloodworms or mysis exceed small fish gape	Food taken and dropped; partial consumption; remnants	Chop the thawed portion, or choose a smaller organism such as cyclops or daphnia

Is spitting food out always a size problem?

[REASONABLE INFERENCE] Most often, but not always. Alternative explanations include unpalatability (stale or oxidised food), unfamiliarity with a new food, texture, illness, and normal manipulation behaviour — many cichlids and loricariids routinely mouth, reposition, and re-take food as part of feeding, which is not rejection.

Part LX — Food Hardness and Texture

Less studied than particle size and occasionally decisive.

Situation	Texture consideration	Evidence
Very small fish and fry	Soft, readily fragmented particles; hard extruded pellets may be unmanageable	[REASONABLE INFERENCE]
Older fish with reduced feeding competence	Softer food or pre-moistened pellets may be taken more reliably	[INSUFFICIENT EVIDENCE] — clinical reasoning only
Rasping grazers (loricariids, <i>Otocinclus</i>)	Require a firm substrate to rasp against; a wafer that disintegrates instantly does not serve the feeding behaviour	[REASONABLE INFERENCE]
Species with pharyngeal crushing dentition (many cyprinids, some cichlids, loaches)	Handle firm foods well; snail-eating species need hard prey to exercise the apparatus	[REASONABLE INFERENCE]
Slow inspecting feeders	Food that holds together long enough to be assessed and taken; food that dissolves in seconds is lost to them	[COMMON AQUARIUM PRACTICE]
Gel diets	Texture is adjustable, leaching is low, and medication or supplements can be incorporated	[AQUATIC-VETERINARY GUIDANCE] — gel diets are recognised as a medication vehicle in aquatic veterinary practice

Part LXI — Pre-Soaking Pellets

A near-universal piece of hobby procedure resting on an unverified mechanism.

What is the claim?

That dry pellets absorb water and expand inside the fish, causing distension, digestive discomfort, constipation, or buoyancy problems, and that soaking the pellet beforehand allows the expansion to occur outside the animal.

What is the evidence?

[CONTESTED CLAIM] No controlled study has been located testing whether dry pellets cause gastric or intestinal distension in aquarium fish at feeding-relevant quantities. The mechanism is physically plausible — dry extruded feed does absorb water and expand — but the step from 'expands' to 'causes clinical harm at normal feeding quantities' has not been demonstrated.

Consideration	Assessment
Dry pellets do absorb water and expand	[ESTABLISHED SCIENTIFIC PRINCIPLE] — not in dispute
Fish swallow pellets whole and they rehydrate internally	[REASONABLE INFERENCE] — follows directly
This expansion causes clinical problems at normal feeding quantities	[CONTESTED CLAIM] — not demonstrated
Soaking prevents those problems	[CONTESTED CLAIM] — cannot be established while the underlying harm is undemonstrated
Soaking leaches water-soluble nutrients	[STRONG EVIDENCE] — a large proportion of some water-soluble vitamins was lost from commercial flake within 30 seconds of water contact. Pellets leach more slowly than flake, but they do leach
Soaking changes buoyancy and sink rate	[STRONG EVIDENCE] — a real, useful, and often the actual reason worth soaking
Aquatic-veterinary guidance on food dissolution	[AQUATIC-VETERINARY GUIDANCE] — Merck advises that pellets fed in water should not be allowed to dissolve before being eaten, to prevent pollution of the water. That is an argument <i>against</i> extended soaking

The position on soaking

Soaking is not harmful in itself, and there are two good reasons to do it briefly: it makes floating food sink, and it softens food for a fish that struggles with hard particles. There is one good reason not to soak it for long: nutrients leach into the soaking water.

The expansion argument is unproven, and in any case the intervention it implies — feeding less — is the same intervention that solves the problem regardless of whether the mechanism is real. A keeper feeding an appropriate small portion does not need to litigate the question. **[CONTESTED CLAIM]**

Part LXII — Floating Versus Sinking Food

A delivery decision, not a quality decision.

	Floating food	Sinking food
Serves	Surface and upper-column feeders	Benthic species and slow mid-water feeders
Keeper visibility	High — uneaten food is obvious and removable	Low — remnants settle and disappear into substrate
Portion control	Easy to observe and correct mid-feeding	Harder; benefits from a feeding dish
Waste risk	Low if removed promptly; can be drawn into overflows and intakes	Higher — unnoticed remnants decompose in substrate
Air ingestion concern	Raised for physostomous species with vigorous surface feeding, notably goldfish and koi; contested as a primary cause of buoyancy disorder	Avoids the concern entirely
Competition effects	Concentrates competition at the surface	Distributes it vertically
Suitability for auto-feeders	Good	Good; both dispense similarly, though flake is the poor performer regardless

What is the actual air-swallowing claim?

[CONTESTED CLAIM] See Part LI for the full treatment. In summary: goldfish and koi are physostomous, with a pneumatic duct connecting the oesophagus to the swim bladder, and veterinary sources state that air swallowed during surface feeding can enter the swim bladder, with a switch to sinking or neutrally buoyant food sometimes helping in mild cases. The claim is species-specific and should not be generalised to aquarium fish at large, particularly not to labyrinth fish for which taking surface air is normal respiratory behaviour.

Part LXIII — How Long Should Food Stay in the Tank?

The evidence does not support one number, but it does support different numbers for different food forms — which is more useful anyway.

What governs the answer?

- **Nutrient leaching rate.** The faster a food gives up its soluble content, the less value there is in leaving it. Flake leaches fastest; a large proportion of several water-soluble vitamins left some commercial flakes within 30 seconds.
- **Structural integrity.** A wafer that holds together for an hour is still being eaten; a flake that has disintegrated is no longer available as food, only as organic load.
- **Which animal it is for.** A slow grazer needs the food to persist; a fast schooling fish does not.
- **Decomposition rate.** Wet organic material in warm water begins microbial decomposition quickly.
- **Aquatic-veterinary guidance** is explicit that pellets fed in water should not be allowed to dissolve before being eaten, in order to prevent pollution of the water.

Food form	Reasonable dwell time	Reasoning	Removal method
Flake	Minutes — should be consumed essentially at once	Fastest leaching and fastest disintegration of any format	Usually not recoverable; correct by feeding less next time
Floating pellets	Minutes; remove what remains	Visible and easily netted; leach more slowly than flake but still dissolve	Net off the surface
Sinking pellets and granules	Up to roughly an hour for slow feeders; remove remnants the same session	Persist longer than flake; risk is loss into substrate	Feeding dish, or targeted siphon
Wafers and tablets	Several hours is normal for slow grazers; overnight is common practice, with removal the following morning	Designed to persist; grazers feed methodically over extended periods	Tongs or siphon; a feeding dish makes this trivial
Vegetables	Up to about 24 hours; remove the remnant	Grazers work on them over hours, but a decaying vegetable is a large nutrient input and a documented driver of snail increases	Clip removal; discard the remnant entirely
Frozen foods	Minutes — treat as immediately consumable	High water content and already-degraded tissue decompose fast	Siphon; strain before feeding to reduce the fluid load
Freeze-dried	Minutes once rehydrated	Floats until saturated then behaves like a soft food	Net or siphon
Live food	Indefinite while the organisms survive — this is a genuine advantage	Living prey does not decompose; blackworms remain available in substrate	None needed while alive; remove dead individuals

The general principle behind the table

Food that is being eaten is food. Food that is no longer being eaten is waste, and the transition happens at a different time for every format and every tank.

The operationally useful version: *remove anything that is no longer being actively worked on, and note that you fed too much of it.* The removal is maintenance; the note is the calibration signal that matters. [\[REASONABLE INFERENCE\]](#)

Part LXIV — Feeding and Tank Cleanliness

The secondary theme the episode is built around: how to feed well while keeping the aquarium cleaner. The answer is never “feed less than the animals need.”

The constraint that governs this entire section

Animal nutrition and welfare come first. Every technique below reduces waste while holding nutrition constant, or improves nutrition at constant waste. None of them works by underfeeding.

A clean tank achieved by restricting an animal's food is not a husbandry success. It is a welfare problem with good water parameters.

What actually reduces waste without reducing nutrition?

Technique	Mechanism	Waste reduction	Nutrition effect	Effort
Portion calibration	Feed the amount the fish actually require rather than the amount they will accept	Large — addresses both uneaten food and surplus intake	Neutral or positive; removes chronic excess	Low, once established
Split the ration across feedings	Smaller portions are consumed more completely and distribute better	Moderate — less uneaten food per event	Positive — better distribution	Low
Match food form to feeding zone	Food reaches the intended animal instead of settling uneaten	Large	Strongly positive for underserved species	Low — a purchasing decision
Match particle size to mouth size	Eliminates food that is taken, rejected, and abandoned	Moderate	Positive	Low
Use a feeding dish or station for sinking food	Concentrates food; remnants are visible and removable in one action	Large for benthic feeding	Positive — verification becomes possible	Low
Remove uneaten food promptly	Prevents decomposition of the highest-nutrient organic material in the tank	Moderate to large	Neutral	Moderate — requires attention at feeding time
Strain the fluid from thawed frozen food	Removes dissolved organics that provide no nutrition to the fish	Small but free	Neutral	Very low
Feed food that is consumed quickly	Reduces nutrient leaching into the water	Moderate	Positive — more of the vitamin content reaches the fish	Low
Substitute rather than add supplementary foods	Prevents treat feeding from becoming a ration increase	Moderate to large in tanks that use supplements	Neutral or positive	Low — purely a discipline change
Adjust portions to body condition over time	Corrects the slow drift as fish mature and requirements fall	Large over months	Positive — this is the feedback loop	Low; monthly
Use a fasting day	Removes roughly 14% of weekly input; demonstrated reduction in nitrogen discharge in trial data	Moderate	Neutral for healthy adults; harmful for fry and sick fish	None
Prune and export plant biomass	Removes nutrient that food imported, rather than reducing the import	Moderate — this is the export side of the budget	Neutral	Moderate

Table LXIV-1. Waste-reduction techniques that do not reduce nutrition. Note that the highest-impact items are all about precision and delivery rather than restriction.

What is the target state?

Not a sterile tank. Detritus, biofilm, and microfauna are functional parts of an established aquarium: they support grazers, host bacterial communities, and provide continuous low-level food for fry and invertebrates. The objective is to avoid **excessive and accelerating accumulation** — a nutrient budget that trends upward faster than the export mechanisms can clear it.

Stated as a budget, which is the framing this brand's system-level approach favours:

Nutrient inputs	Nutrient exports
Food (usually dominant)	Water changes
Source water (nitrate and phosphate vary widely by supply)	Plant biomass removed by pruning
Fertiliser and supplements, where dosed	Removed snails, shrimp, and fish
Decomposing plant matter and dead organisms	Siphoned detritus and cleaned mechanical media
Introduced organics on new decor or substrate	Gaseous nitrogen loss where denitrification occurs

When the left column exceeds the right consistently, parameters trend upward regardless of how carefully any individual feeding is managed. The keeper has two levers, and feeding is only one of them.

Part LXV — Does Feeding Less Mean Fewer Water Changes?

A question that must be answered carefully because the technically correct answer invites a genuinely harmful practice.

Is the mechanism real?

[REASONABLE INFERENCE] Yes. Reducing food input reduces the nitrogen and phosphorus entering a closed system, which reduces the rate at which nitrate and phosphate accumulate, which reduces the water exchange required to hold them at a given level. The arithmetic is not in doubt.

Why is the conclusion still wrong as usually stated?

[CONTESTED CLAIM] Because the relationship is being run backwards. Feeding is determined by what the animals need; maintenance is then set to handle the resulting waste. Reducing feeding in order to reduce maintenance reverses the dependency and uses the animals' nutrition as the adjustable variable.

Framing	Logic	Assessment
"I feed my fish what they need, and I change water often enough to handle the resulting waste."	Nutrition is fixed; maintenance adjusts	Correct ordering
"I reduced overfeeding, and my nitrate now rises more slowly so I change water less often."	Correcting an error produced a maintenance benefit	Legitimate — this is the good version
"I feed less so I don't have to do water changes."	Maintenance is fixed; nutrition adjusts	Inverted priority. This is the framing to argue against
"My tank is clean because I barely feed."	Cleanliness as the goal, nutrition as the cost	A welfare problem presented as a technique

What else does a water change do?

The argument for maintaining water changes is not only about nitrate. Partial water changes also dilute dissolved organic compounds, replenish depleted minerals and buffering capacity, reset accumulated substances that hobbyists do not test for, and provide the occasion for substrate cleaning and equipment inspection. A feeding reduction addresses one of these and none of the others.

The statement for the episode

"Yes, feeding less does mean less waste going into the tank, and if you were overfeeding, fixing that will slow down how fast your nitrate climbs. That's a real benefit.

But feeding is not the dial you use to manage maintenance. You feed your fish what they need, and then you do whatever maintenance that amount of food requires. If the only way to keep your parameters stable is to keep your fish hungry, the problem isn't the feeding — it's stocking, filtration, or the maintenance schedule itself." [REASONABLE INFERENCE]

Part LXVI — Feeding and Filter Maintenance

An underused diagnostic channel: the mechanical filter is quietly measuring the particulate output of the feeding regime.

How does feeding load the filter?

Pathway	What it produces	Where it shows up
Uneaten food drawn into the intake	Whole or partial food particles	Mechanical media, rapidly
Fine food dust and crumbled flake	Fine particulate that passes coarse media	Fine mechanical media and the biological media surface
Faecal solids	Fine to medium particulate; higher output at higher intake	Mechanical media and low-flow zones
Dissolved organics and leached nutrients	Bacterial substrate rather than particulate	Biological media as increased heterotrophic growth; not visible as debris
Decomposing remnants in substrate	Slow release of fine particulate and dissolved organics	Substrate first; filter secondarily

Why is filter cleaning interval a useful signal?

[REASONABLE INFERENCE] Because it integrates particulate production over time in a way that spot observation of a single feeding cannot. A mechanical media cleaning interval that shortens noticeably without a stocking change is evidence that particulate input has increased — and food is the most common reason it would.

This is worth teaching because it is one of the few overfeeding signals that does not require the keeper to catch a feeding in progress. The keeper who used to rinse sponge media monthly and is now doing it every ten days has a measurement, even if they have never seen a leftover flake.

Two cautions. First, the signal is confounded by anything else that raises particulate load — substrate disturbance, plant debris, media degradation, reduced flow. Second, increased mechanical filter loading is *not* the same as increased biological loading, and a clogged sponge reduces flow and therefore oxygenation, which compounds the oxygen demand the organic load already created.

Part LXVII — Feeding and Nitrate

The most-watched parameter in the hobby and the one most directly connected to feeding — but not the only thing feeding it.

How does dietary nitrogen become nitrate?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Dietary protein contains nitrogen. Protein consumed above requirement, and protein catabolised for energy, is deaminated, producing ammonia, which teleost fish excrete primarily across the gills. Uneaten food and faecal solids release additional ammonia during microbial decomposition. Nitrifying bacteria oxidise ammonia to nitrite and nitrite to nitrate. Nitrate accumulates unless it is exported.

The quantitative links are covered in Part XIV: only about 25–40 % of feed nitrogen is retained in fish tissue in growing production fish, most nitrogen waste (60–90 %) is in dissolved form as ammonia, and in adult fish at maintenance essentially all dietary nitrogen eventually leaves the animal.

Is all aquarium nitrate from feeding?

[REASONABLE INFERENCE] No. Feeding is usually the dominant source in a stocked aquarium, but several others contribute meaningfully and are routinely overlooked when a keeper attributes a rising nitrate trend to overfeeding.

Nitrate source	Typical contribution	How to check
Fish food (via consumed and uneaten fractions)	Usually dominant in a stocked tank	Correlate the trend with feeding changes
Tap or source water	Can be substantial; municipal supplies in agricultural regions sometimes carry appreciable nitrate	Test the source water directly — the single most overlooked diagnostic step
Decomposing plant matter	Moderate after a trim, a melt, or a plant die-off	Correlate with plant events
Dead or dying animals	Potentially large and acute	Headcount, including snails and shrimp
Accumulated substrate detritus being disturbed	Pulses on disturbance	Correlate with maintenance events
Fertiliser containing nitrogen	Deliberate and controlled in planted tanks	Review dosing regime
Reduced export (fewer or smaller water changes; plants removed or declining)	Frequently the actual cause of a 'sudden' rise	Review the maintenance record rather than the feeding

How should nitrate be used as a feeding signal?

[REASONABLE INFERENCE] As a **trend**, not a reading. A single nitrate value describes the current balance of input and export. The rate at which nitrate climbs between water changes is what carries information about input, and that requires two or more measurements at known intervals under a known maintenance schedule.

A practical protocol: measure nitrate immediately after a water change and again immediately before the next one, keeping the interval consistent. The difference is the accumulation over that period. If that difference grows while stocking and maintenance are unchanged, input has risen. This is the kind of measurement a tank journal makes trivial and memory makes impossible.

Part LXVIII — Feeding and Phosphate

The nutrient the hobby tests for least and food delivers most inefficiently.

How much phosphorus does food contribute?

[STRONG EVIDENCE] A great deal, because retention is poor. Phosphorus excretion has been reported at roughly 69–86 % of dietary phosphorus in aquaculture contexts, with retention in fish biomass typically in the 17–40 % range. Commercially fed Nile tilapia was estimated to retain about 28 % of feed phosphorus. Unlike nitrogen, a larger proportion of phosphorus leaves the animal as particulate matter in faeces — reported in the range of 25–85 % depending on species — though some species, including tilapia hybrids, release a large share (60–62 %) in dissolved form.

Phosphorus is also the mineral fish most need from the diet, precisely because natural water contains little of it. That creates the tension: a diet adequate in phosphorus for the fish necessarily delivers phosphorus to a system that has no natural phosphorus sink. Sales and Janssens flagged this explicitly, noting that pollution of water by excess excreted phosphorus appeared highly critical because it may lead to eutrophication.

Why does the particulate fraction matter for aquarium practice?

[REASONABLE INFERENCE] Because it changes which maintenance action removes it. Nitrogen, being predominantly dissolved, is exported mainly by water change. Phosphorus, with a larger particulate fraction bound in faeces and uneaten food, accumulates in substrate and detritus and is exported substantially by **substrate cleaning and mechanical media removal**, not only by dilution.

This explains a pattern keepers sometimes report: a tank with diligent water changes but minimal substrate maintenance that nonetheless develops persistent nutrient-driven problems. The phosphorus reservoir is in the substrate, not the water column, and a water change does not reach it.

Phosphate source in an aquarium	Notes
Fish food — both consumed and uneaten fractions	Usually the dominant input in a stocked tank
Source water	Some municipal supplies add phosphate as a corrosion inhibitor; worth testing
Decomposing organic matter in the substrate	A reservoir rather than an input; releases over time
Phosphate-containing fertilisers	Deliberate in planted tanks
Some buffers and additives	Check product composition
Certain substrates and decor	Occasionally leach phosphate

Part LXIX — Feeding and Dissolved Organics

The invisible fraction. Hobbyists have no standard test for it, which is exactly why it accumulates unnoticed.

Does feeding contribute to dissolved organic compounds?

[REASONABLE INFERENCE] Yes, through several documented routes. Nutrient leaching from food into water is directly established — a large proportion of several water-soluble vitamins left commercial flake within 30 seconds of water contact, and leaching of soluble components is a recognised issue in feed formulation. Uneaten food undergoing decomposition releases dissolved organic material continuously. Thawed frozen food carries dissolved organics in its pack fluid. Fish mucus and metabolic products add further.

Route	Character	Mitigation
Leaching from food before ingestion	Immediate; fastest for flake and finely divided food	Feed food that is eaten quickly; avoid extended soaking; avoid over-fine particles
Decomposition of uneaten food	Continuous over hours to days	Remove remnants; portion correctly; use a feeding dish
Pack fluid from thawed frozen food	A single dose per feeding, entirely avoidable	Strain and discard the fluid before feeding
Fish metabolic products and mucus	Continuous and proportional to stocking and intake	Water changes; adequate biological filtration
Decaying plant and vegetable matter	Continuous while present	Remove vegetable remnants within about 24 hours; prune and export plant material

What are the observable consequences?

- **Yellowing or tea-tinted water** that is clear rather than cloudy is a common visual signature of accumulated dissolved organics, though tannins from wood and leaf litter produce the same appearance benignly.
- **Surface film or protein scum** forms where dissolved organics accumulate at the air-water interface, particularly with low surface agitation and after feeding oily foods.
- **Increased heterotrophic bacterial activity**, which raises oxygen demand and can express as cloudiness. See Part XVI.
- **Increased biofilm growth** on surfaces — which is a food source for grazers as well as an aesthetic issue, and is therefore not purely negative.
- **Reduced efficacy of some treatments and additives**, since dissolved organics bind a range of compounds.

Management is straightforward in principle: reduce input (precise feeding, prompt remnant removal, straining frozen food), increase export (water changes, mechanical media servicing, activated carbon where appropriate), and maintain adequate surface agitation and circulation so that organics do not stratify and accumulate in dead zones.

Part LXX — Food Storage

The most neglected variable in aquarium nutrition. A correctly chosen food stored badly for a year is a different food by the time it is eaten.

Why does storage matter so much for fish food?

[STRONG EVIDENCE] Fish feeds usually have high protein and oil content which can deteriorate rapidly. High humidity and high temperature result in degradation of the diet, promotion of mould growth and potential mycotoxin production, and rancidity (Merck Veterinary Manual). Lipid oxidation is the central chemical problem: feeds with high polyunsaturated fatty acid content are particularly vulnerable, which is why vitamin E is added as tocopherol acetate specifically to stabilise lipids in the feed during storage.

Agent	What it degrades	Evidence	Control
Heat	Vitamin C, vitamin A and biotin (most heat-sensitive), general lipid stability	[STRONG EVIDENCE] — vitamin C sustainability is low and most functionality is lost through exposure to high temperature, oxygen, and light	Store cool; never above a heater, near a light fixture, or on a warm hood
Oxygen	Vitamins A, D and C (most prone to oxidation); unsaturated fatty acids	[STRONG EVIDENCE]	Airtight container; minimise headspace; small containers
Light	Vitamin C particularly	[STRONG EVIDENCE] — ascorbic acid stability ranked freezer > refrigerator > room temperature in black bags > room temperature in clear bags (Soliman et al. 1987)	Opaque container; dark cupboard
Moisture	General degradation; enables mould growth and mycotoxin production	[AQUATIC-VETERINARY GUIDANCE]	Airtight container; never return a wet scoop or wet fingers to the container; do not store above a running aquarium
Time	Cumulative loss of vitamins and progressive lipid oxidation	[STRONG EVIDENCE]	Buy quantities that will be used within months
Pests	Contamination and consumption	[AQUATIC-VETERINARY GUIDANCE] — dry feeds should be kept in a pest-proof container	Sealed container

Does buying a giant container reduce nutritional quality?

[REASONABLE INFERENCE] Yes, and the effect is larger than most keepers assume. Once opened, a container is repeatedly exposed to air and ambient humidity and warms toward room temperature with every use. Vitamin C is heat-, oxygen-, and light-labile; approximately 75 % of supplemented vitamin C was lost during processing alone in one shrimp feed study, before any storage. A container that takes two years to finish is delivering progressively degraded food for most of its life.

The arithmetic that should change buying behaviour

A small community tank might use 15–30 g of dry food per year. A 250 g canister — a common and apparently economical size — would take the better part of a decade to consume.

Buying the large container is not thrift. It is buying eight years of progressively oxidising food at a discount. The correct purchase for most home aquariums is the smallest container available, replaced often. **[REASONABLE INFERENCE]**

Practical storage protocol

1. **Buy small.** Size the container to two to four months of use.

2. **Keep it sealed.** Close immediately after use; do not leave lids open on the tank hood.
3. **Keep it dry.** Never use a wet scoop or wet fingers. Humidity is the agent most under the keeper's direct control and the one most often ignored.
4. **Keep it cool and dark.** A closed cupboard away from the aquarium's heat and lighting, not a shelf above the tank.
5. **Consider decanting.** Splitting a container into a small working portion and a sealed reserve limits how often the bulk is exposed to air and humidity.
6. **Date it on opening.** Write the opening date on the container. This converts a vague impression into a decision rule.
7. **Discard on schedule, not on appearance.** Degraded food does not necessarily look or smell obviously wrong, particularly for vitamin loss.

Part LXXI — Expiration Dates and Shelf Life After Opening

The printed date and the useful life of an opened container are different quantities, and the second is much shorter.

What does a printed expiration date actually mean?

[MANUFACTURER RECOMMENDATION] Printed dates on dry fish food are manufacturer best-before dates, typically established for an unopened container under specified storage conditions. They are not independently verified statements about nutritional adequacy, and they do not account for how the container has actually been stored or how long it has been open.

What does aquatic-veterinary guidance say about opened containers?

[AQUATIC-VETERINARY GUIDANCE] The Merck Veterinary Manual gives specific and notably conservative figures: a feed container kept at room temperature should be discarded two months after opening; commercial feeds stored in the freezer should be discarded after six months; commercial frozen feeds such as brine shrimp, bloodworms, mysids, or glassworms should be discarded after one year, and thawed feeds should not be refrozen.

Storage situation	Recommended limit	Source
Dry feed container at room temperature, after opening	Discard after 2 months	[AQUATIC-VETERINARY GUIDANCE]
Commercial feed stored in the freezer	Discard after 6 months	[AQUATIC-VETERINARY GUIDANCE]
Commercial frozen feeds (brine shrimp, bloodworms, mysids, glassworms)	Discard after 1 year; never refreeze once thawed	[AQUATIC-VETERINARY GUIDANCE]
Unopened dry feed	Manufacturer best-before date, under specified conditions	[MANUFACTURER RECOMMENDATION]
Freeze-dried foods	No specific published limit located; treat as dry feed and apply the same reasoning	[REASONABLE INFERENCE]

The two-month figure is substantially shorter than common hobby practice, and it is worth flagging as a point of divergence rather than smoothing over. It is a conservative clinical recommendation, likely calibrated for facilities where nutritional deficiency would be a diagnosable problem. Whether a healthy adult community fish experiences measurable harm from four-month-old flake has not been tested. The direction of the advice, however, is clearly supported by the vitamin stability literature.

Part LXXII — Refrigerating and Freezing Dry Food

A practice with real supporting evidence and one specific execution risk.

Does cold storage help?

[STRONG EVIDENCE] Yes, for vitamin retention. Ascorbic acid stability under different storage conditions ranked, in descending order: freezer (−20 °C), refrigerator (5–8 °C), room temperature in black bags (22–24 °C), and room temperature in clear bags (Soliman et al. 1987). Merck notes that many feeds can be frozen, which will prolong their shelf life.

What is the risk?

[REASONABLE INFERENCE] Condensation. A cold container opened in a warm room accumulates moisture on the food and on the container's interior surfaces. Moisture in dry feed promotes mould growth and potential mycotoxin production — a worse outcome than the vitamin loss the cold storage was intended to prevent.

Approach	Vitamin benefit	Moisture risk	Assessment
Room temperature, small opaque airtight container, used within 2 months	Adequate given the short window	Low	The simplest defensible default for most keepers
Refrigerated working container opened daily	Moderate	High — daily cold-to-warm cycling	Not recommended; the condensation risk outweighs the benefit
Frozen bulk reserve plus small room-temperature working portion	High for the reserve	Low if the reserve is allowed to reach room temperature fully before opening, and is opened rarely	The best-supported arrangement for a keeper who must buy a large container
Frozen container opened while cold	High in principle	High — condensation on every use	Counterproductive

The operating rule if freezing is used: **divide first, then freeze**. Portion the food into several small sealed containers before freezing, remove one at a time, allow it to reach room temperature fully before opening, and keep it at room temperature until finished. This avoids repeated cold-to-warm cycling of the bulk supply.

Part LXXIII — Reading a Fish Food Label

What the label tells you, what it cannot tell you, and where hobby interpretation goes wrong.

What is on a typical label?

Element	What it tells you	What it does not tell you
Ingredient list (descending order by weight)	Which materials are present and, approximately, their rank by inclusion weight	Digestibility, amino acid profile, nutrient availability, processing quality, or whether the ranking reflects nutritional contribution
Crude protein (minimum)	Total nitrogen $\times 6.25$. A ceiling on protein content	How much of that protein is digestible or biologically available to that species; amino acid balance
Crude fat (minimum)	Total lipid content; the main energy-density indicator	Fatty acid profile, including whether essential fatty acids are present; oxidation state
Crude fibre (maximum)	Indigestible plant structural material	Whether it has any functional role for that species — recall that no fish digests cellulose efficiently
Moisture (maximum)	Water content; important for comparing dry with frozen or gel foods on a like-for-like basis	Anything about nutritional quality
Ash (maximum, where listed)	Total mineral content	Which minerals, in what proportions, or their availability
Added vitamins and minerals	That supplementation is present, and sometimes the form used	How much survives processing and storage, or how much leaches before ingestion
Feeding directions	The manufacturer's recommendation	Anything species-, size-, or life-stage-specific to your tank; treat as an upper bound

What are the limitations of guaranteed analysis?

[STRONG EVIDENCE] Guaranteed analysis describes gross composition, not nutritional value. Crude protein measures nitrogen, not usable protein: non-protein nitrogen and poorly digestible protein both contribute to the figure. Sales and Janssens noted directly that digestibility values of nutrients in feed ingredients and diets were not established in ornamental species, and that their availability would be a valuable tool both for diet formulation and for reducing pollution of the living environment.

The practical consequence for a keeper comparing two products: a difference of a few percentage points in crude protein carries very little information, while the species the food is intended for, the particle size, the sink behaviour, and the container size all carry a great deal.

Part LXXIV — Ingredient Order

The most confidently applied and least reliable label heuristic in the hobby.

Must the first ingredient be whole fish?

[UNSUPPORTED AQUARIUM FOLKLORE] As a rule, no. Ingredient order is by inclusion weight *as added*, which means it is heavily influenced by moisture content, and it says nothing about digestibility, amino acid profile, or nutrient availability.

The specific failure mode is moisture. A wet ingredient such as whole fish is largely water; a dried meal of the same material has the water removed and therefore contributes far more nutrient per unit of listed weight. Listing whole fish first can, depending on formulation, mean that the finished product contains *less* fish-derived protein than a product listing fish meal first. The heuristic can point the wrong way.

Ingredient-list belief	Assessment
“The first ingredient must be whole fish”	[UNSUPPORTED AQUARIUM FOLKLORE] — moisture content makes the comparison unreliable
“Meal means low quality”	[UNSUPPORTED AQUARIUM FOLKLORE] — see Part LXXV
“Plant ingredients are filler”	[CONTESTED CLAIM] — digestible carbohydrate is a protein-sparing energy source for omnivorous and herbivorous species, which reduces ammonia output. It is genuinely poorly used by carnivores. The species determines the answer, not the ingredient
“Fewer ingredients is better”	[UNSUPPORTED AQUARIUM FOLKLORE] — meeting a complete nutrient specification generally requires more ingredients, not fewer. A short list may indicate a supplement rather than a complete diet
“Ingredient splitting hides the real composition”	[REASONABLE INFERENCE] — listing several fractions of one material separately can lower each fraction's rank. This is a genuine limitation of ingredient-order reasoning and a further argument against relying on it
“The ingredient list tells you the nutritional quality”	[UNSUPPORTED AQUARIUM FOLKLORE] — it tells you what is present and roughly in what order by as-added weight. Digestibility and availability are not on the label

What to actually look at on a label

- 1. Is it intended for this kind of fish?** A herbivore food for a herbivore, a carnivore food for a carnivore. This is the decision that matters most and is the easiest to get right.
- 2. Is the particle size right?** The most common practical failure.
- 3. Does it float, sink, or sink slowly?** Delivery determines whether the food reaches the intended animal at all.
- 4. Is the container small enough to be used within a few months?**
- 5. Does the crude protein level roughly suit the species?** Roughly — a few percentage points is noise.

Ingredient order is well below all five of these in practical importance. **[REASONABLE INFERENCE]**

Part LXXV — Fish Meal

A processing term that the hobby reads as a quality judgment.

What is fish meal?

Fish meal is a dried, ground product made from whole fish, fish trimmings, or by-catch, with most of the water and often much of the oil removed. It is the historical backbone of aquaculture feed formulation because it provides a highly digestible protein with an amino acid profile well matched to fish requirements.

[STRONG EVIDENCE] Fish meal quality varies with raw material and processing but the category is not inherently low quality. The entire aquaculture research programme on fish meal *replacement* exists because fish meal is the benchmark against which alternatives are measured — and because of cost and sustainability pressures, not because of nutritional inadequacy. Sales and Janssens framed the search for alternatives explicitly in terms of the increasing cost of high quality fish meal and the need to reduce fishing pressure on declining stocks.

Replacement studies also show where the limits lie. Fish meal was substituted by soybean meal up to 37 % of the diet, replacing 33 % of the fish meal protein, to achieve normal growth in juvenile tin foil barb (Elangovan and Shim 2000) — partial replacement worked, complete replacement was not demonstrated. In discus, high levels of soybean incorporation reduced food intake owing to low palatability and decreased digestibility, and longer acclimatisation periods were recommended.

The fish meal statement

“Meal” describes a processing step — drying and grinding — not a quality tier. Removing water concentrates nutrients, which is why meals rank high on ingredient lists and why a product listing fish meal first may contain more fish-derived protein than one listing whole fish first. **[STRONG EVIDENCE]**

Part LXXVI — Insect-Based Foods

The fastest-moving ingredient category in aquafeed, with a real body of evidence and one species-dependent limitation.

What does the black soldier fly larvae literature show?

[STRONG EVIDENCE] Black soldier fly larvae meal (*Hermetia illucens*, BSFLM) is a viable partial fish meal replacement across a range of species. On a dry matter basis it contains roughly 42–60 % crude protein and 10–35 % crude lipid, varying with the rearing substrate. In general it meets amino acid requirements for Atlantic salmon except for methionine and lysine, which suggests it is more suitable than soy protein concentrate as a fish meal replacement.

Species / context	Finding
Zebrafish (<i>Danio rerio</i>)	Defatted BSF larvae and prepupae meals totally replaced fishmeal without negatively affecting growth or wellness over a 60-day trial, with prepupae meal performing particularly well
Goldfish (<i>Carassius auratus</i>)	BSFLM incorporated up to 15% in practical diets partially replaced fish meal without negative effects on feed intake, growth, feed utilisation, or plasma biochemistry — an ornamental-species result directly relevant to the hobby
Juvenile Jian carp	Good growth, carcass composition, antioxidant and digestive enzyme activity with defatted BSFLM up to 50% of fish meal; 75% replacement caused dietary stress and intestinal histopathological damage
Atlantic salmon (commercial-scale field trial)	Suitable alternative protein ingredient at inclusions up to at least 8% under industrial conditions, with no attributable histopathological changes
Rainbow trout	Chitin up to 3% of the diet did not influence growth performance, blood biochemistry, nutrient digestibility, or intestinal activity
Turbot	Diets containing 1.47% chitin from 16.5% BSFL prepupae meal significantly reduced growth performance

What is the chitin issue?

[STRONG EVIDENCE] Chitin is the limiting factor and its effect is species-dependent. BSFL contain roughly 3–10 % chitin depending on developmental stage and processing. Fish can digest chitin, but digestibility decreases as dietary chitin rises. Chitin may reduce feed intake and the availability and digestibility of nutrients. A review concluded that the role of chitin in aquafeeds is highly variable and depends on both inclusion level and the species considered.

For the hobby, the relevant summary is that insect-based foods are a legitimate and well-evidenced ingredient category with demonstrated performance in at least one ornamental species (goldfish), that they carry genuine sustainability advantages because larvae can be reared efficiently on organic waste streams, and that the appropriate inclusion level varies enough between species that a blanket endorsement would overstate the evidence.

Part LXXVII — Spirulina

A genuine functional ingredient surrounded by unusually heavy marketing.

What is spirulina and what does it actually provide?

Spirulina is dried biomass of the cyanobacterium *Arthrospira* (formerly *Spirulina*). In fish feed it serves three roles with differing levels of support:

Claimed role	Assessment
Carotenoid and pigment source	[STRONG EVIDENCE] — the best-supported role. Under certain culture conditions (nitrogen depletion, high salinity, high light intensity) <i>Arthrospira maxima</i> accumulates secondary carotenoids and has been evaluated as a replacement for costly synthetic colourings in ornamental fish feed (Gouveia et al. 2003). Dietary inclusion of 5–10% spirulina powder significantly changed measured skin colour parameters in fancy carp
Digestible plant protein source	[REASONABLE INFERENCE] — high protein content and generally good digestibility; a legitimate feed ingredient
Appropriate plant component for herbivores and grazers	[REASONABLE INFERENCE] — a sensible ingredient for herbivore-directed formulations, and one reason spirulina-based flakes and wafers are appropriate for mbuna, mollies, and algae grazers
Immune enhancement	[CONTESTED CLAIM] — asserted heavily in marketing; the aquaculture evidence for immunostimulant effects of microalgal biomass is mixed and dose-dependent
General health tonic or superfood	[UNSUPPORTED AQUARIUM FOLKLORE] — marketing language rather than a nutritional claim

The honest summary: spirulina is a good ingredient in a herbivore or grazer diet and a well-evidenced carotenoid source. It is not a supplement that improves an already adequate carnivore diet, and the product category has accumulated claims well beyond what the ingredient supports.

Part LXXVIII — Garlic

An additive with a surprisingly substantial research literature and a large gap between what that literature shows and what the hobby claims.

What has actually been studied?

[STRONG EVIDENCE] Garlic (*Allium sativum*) has a real and growing aquaculture research base. Reviews describe antimicrobial, antiviral, antiparasitic, and immunostimulatory properties, with effective dosages reported across a wide range of roughly 0.05–40 g per kg of feed. Antiparasitic activity is among the most reported effects in the literature.

Hobby claim	What the research shows	Assessment
“Garlic boosts immunity”	Immunostimulatory effects are reported across multiple studies in aquaculture species, with dose- and species-dependent results	[CONTESTED CLAIM] — supported in principle in production species at specified doses; not validated for a drop of garlic juice on a pinch of flake
“Garlic treats parasites”	Antiparasitic activity is well reported, but predominantly using immersion baths rather than dietary administration. In vivo treatment of carp with garlic and onion extracts reduced skin <i>Gyrodactylus</i> by 14.4% and 19.8% respectively. Dietary studies exist — one reported that a 30-day supplementation did not influence infection success while 60 days of conditioning reduced parasite prevalence and intensity	[CONTESTED CLAIM] — the route matters enormously, the effect sizes are partial, and the timescales are long. Garlic is not a treatment for an active parasite outbreak
“Garlic improves appetite”	Appetite stimulation is described in review literature; palatability enhancement is the likely mechanism	[REASONABLE INFERENCE] — the most defensible hobby use. Garlic-soaked food to tempt a reluctant feeder is low-risk and plausibly helpful
“Garlic is a safe natural alternative to medication”	Garlic extract has a measurable toxicity: a 96-hour LC50 of 0.4 g/L was determined for garlic extract in Nile tilapia in one bath study	[CONTESTED CLAIM] — ‘natural’ does not mean ‘without a dose-response curve’. Bath applications in particular have a real toxicity threshold
“Allicin survives in the water long enough to work”	Incorporation of garlic into a pressure-extruded pellet was found effective as a delivery method, with minimal leaching — less than 3% of allicin detected in the water during the interval between delivery and consumption	[STRONG EVIDENCE] for extruded delivery; the implication is that garlic works best incorporated into food, not dosed into water

The garlic position

Garlic is not snake oil — there is a legitimate research literature and the compounds are biologically active. But the hobby's use of it (a few drops of garlic extract on food, as a parasite treatment) does not match the conditions under which effects were demonstrated (specified dietary inclusion rates over weeks, or immersion baths at controlled concentrations).

Reasonable use: as a palatability enhancer for a fish that is reluctant to eat. **Not a reasonable use:** as a substitute for diagnosing and treating a parasite problem. **[CONTESTED CLAIM]**

Part LXXIX — Colour-Enhancing Foods

One of the few marketing claims in this document that the evidence broadly supports — with an important limit on what it can achieve.

How does dietary colour enhancement work?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Fish cannot synthesise carotenoid pigments and rely entirely on dietary supply to achieve their natural skin pigmentation. Carotenoids commonly occurring in freshwater include beta-carotene, lutein, taraxanthin, astaxanthin, tunaxanthin, doradexanthins, and zeaxanthin. Red colouration in goldfish and common carp is imparted by astaxanthin, which is readily metabolised from the yellow pigment zeaxanthin.

Critically, the conversion pathways are specific rather than general. Goldfish metabolised very little beta-carotene and no lutein to astaxanthin (Hata and Hata 1972). The source of the carotenoid therefore matters, not merely its presence.

What do the feeding trials show?

Study context	Finding
Goldfish, five months, five carotenoid sources	Best red colour was obtained with astaxanthin (75 mg/kg), canthaxanthin (75 mg/kg), and paprika oleoresin (180 mg/kg) diets. Growth and feed efficiency did not differ significantly among groups
Fancy carp (koi), 8 weeks, nine diets	Skin colour parameters were significantly changed by carotenoid source but not by carotenoid level , across 5–10% spirulina, 5–10% paprika, 50–100 ppm astaxanthin and 50–100 ppm canthaxanthin. Survival, growth and feed efficiency were unaffected
Koi carp, carotenoid deposition	Around 250 mg/kg astaxanthin gave the highest carotenoid deposition; in adult dwarf cichlids a saturation point was reached at a low concentration around 60 mg/kg
Microalgal biomass as a synthetic replacement	Under defined culture conditions <i>Chlorella vulgaris</i> , <i>Haematococcus pluvialis</i> , and <i>Arthrospira maxima</i> accumulate secondary carotenoids and may replace costly synthetic colourings in ornamental fish feed

What does colour enhancement actually mean?

[STRONG EVIDENCE] It means supplying the dietary pigment precursors a fish needs to express the colouration it is genetically capable of. It does not mean adding colour the fish does not have, and the trials consistently show saturation — beyond a certain dietary level, further carotenoid produces no further deposition.

Two practical consequences. First, a fish showing poor colour for reasons unrelated to diet — stress, poor water quality, unsuitable lighting, illness, or simply genetics — will not respond to a colour food. Second, colour-enhancing food is not a category of superior nutrition; in the trials above, growth and feed efficiency were consistently unaffected by carotenoid supplementation. It affects appearance, and that is a legitimate but narrow benefit.

Part LXXX — Medicated Food

Covered briefly and deliberately. This is not a treatment guide and must not become one.

When is oral medication used?

[AQUATIC-VETERINARY GUIDANCE] Medicated feed is one of several routes of drug administration in aquatic veterinary practice, alongside immersion baths, injection, and topical application. Some medicated feeds are available commercially for aquaculture or pet fish, and custom medicated feeds can be prepared using flake, pellet, or gel diets as a base. Cooking oil spray is described as an effective binder for pelleted or flake foods; medication is easily incorporated into commercially available gel diets as the gel cools, though it should not be added while the gelatin is hot because some medications, particularly oxytetracycline, are heat labile.

Why should medicated food not be used casually?

[AQUATIC-VETERINARY GUIDANCE] Use of prophylactic medication in the absence of diagnostic testing is strongly discouraged and may contribute to resistant bacterial infection and other complications (Merck Veterinary Manual). That statement is about medication generally and applies directly to the hobby practice of medicating food “just in case.”

Three further complications specific to oral delivery:

- **A sick fish often will not eat.** Anorexia is one of the earliest signs of illness, which means the fish most in need of oral medication is frequently the least able to receive it. Dose delivery is therefore unreliable exactly when it matters.
- **Dose control in a community is effectively impossible.** The dominant feeder receives a large dose, the target fish may receive none, and every other animal in the tank — including invertebrates — is exposed.
- **Uneaten medicated food enters the water.** This exposes the biological filter and non-target organisms to antimicrobials, with consequences for both the nitrogen cycle and resistance development.

The appropriate framing for this brand: medicated food is a veterinary tool, it requires a diagnosis to be useful, and the practical questions — which drug, what dose, how long, delivered to which animal — are outside what a general aquarium episode should attempt to answer. Keepers with a sick fish should be directed toward diagnosis, whether through an aquatic veterinarian or a structured differential, rather than toward the medication shelf.

Part LXXXI — Sick Fish That Stop Eating

Appetite loss is among the most diagnostically valuable observations available to an aquarist, and among the most commonly misattributed.

Why is anorexia diagnostically important?

[REASONABLE INFERENCE] Because it is an early, sensitive, and non-specific sign. Feeding behaviour is energetically expensive and behaviourally elaborate, and it is one of the first things a fish down-regulates under stress, disease, or adverse environmental conditions. Its non-specificity is precisely what makes it useful: it does not tell you what is wrong, but it reliably tells you that something is.

Category	Specific causes	How to check
Water quality	Ammonia or nitrite present; extreme nitrate; pH shift; chlorine/chloramine; dissolved oxygen deficit	Test first, always. This is the cheapest and highest-yield check available and it rules out the most rapidly lethal causes
Temperature	Too low for digestion; too high causing stress; unstable or swinging	Thermometer — and verify against the heater setting rather than trusting it
Stress and environment	Recent move or introduction; aggression; inadequate cover; inappropriate flow; excessive activity outside the tank; inadequate group size for a schooling species	Observation; consider what changed in the preceding week
Competition	The fish is being excluded from food rather than refusing it	Watch individual feeding; try target feeding. See Part XX
Disease	Bacterial, parasitic, fungal, viral; internal parasites are particularly associated with reduced condition despite apparent feeding	Look for other clinical signs: respiration rate, skin and fin condition, faeces character, buoyancy, posture
The food itself	New food not recognised; stale or oxidised food; particle size wrong; food inappropriate for the species	Offer a previously accepted food as a control
Recent transport or acclimation	Normal transient anorexia after shipping or introduction	Time and stable conditions. See Part LXXXIII
Reproductive state	Some species reduce or stop feeding while spawning, guarding, or mouthbrooding	Species behaviour; check for eggs, nest, or brooding posture
Age and physical capability	Reduced feeding competence, jaw or dentition problems, barbel damage in benthic species	Observe the strike; check barbels and mouth

The sequence that should be taught

- 1. Test the water.** Always first. It is the most common cause, the most rapidly dangerous, and the easiest to check.
- 2. Check temperature.** Including stability, not just the current reading.
- 3. Observe the individual at feeding.** Is it refusing food, or being prevented from reaching it?
- 4. Look for other clinical signs.** Anorexia alone is non-specific; anorexia plus a second sign narrows the differential.
- 5. Consider what changed.** New fish, new food, new decor, water change, temperature change, maintenance event.

This document does not diagnose. It establishes that appetite loss warrants investigation rather than a feeding adjustment.

[REASONABLE INFERENCE]

Part LXXXII — Feeding During Illness

A genuinely difficult judgment with limited direct evidence and two competing considerations.

Should feeding be adjusted when fish are ill?

[CONTESTED CLAIM] The considerations pull in opposite directions and the balance is case-specific.

Consideration	Argues for	Notes
A sick fish has reduced reserves and an immune system with a real energy cost	Continuing to offer food	Nutritional deficiency compounds disease; a fish that stops eating for an extended period deteriorates from that alone
Uneaten food in a tank with a compromised animal degrades water quality	Reducing the offered quantity	Water quality is frequently either the cause or an aggravating factor
Some diseases and treatments impair digestion	Reducing quantity; offering highly digestible food	Little direct evidence in ornamental species
Vitamin C and other micronutrients support immune function and wound healing	Maintaining nutritional quality rather than restricting	Individual micronutrients including vitamins C, B6, E, and A and minerals such as iron are known to affect disease resistance in fish
Appetite itself is the recovery indicator	Continuing to offer food as a diagnostic	A fish resuming feeding is one of the clearest positive signs available
Treatment chemicals may affect the biological filter	Reducing input during treatment	A common and legitimate reason to feed less during medication

What is the defensible general position?

[REASONABLE INFERENCE] Continue to **offer** small quantities of a high-quality, readily accepted food, remove anything uneaten promptly, and do not treat fasting as therapy. A fish that will not eat is telling you something; a fish that is not offered food tells you nothing.

Two specific cautions. First, do not withhold food as a treatment in itself — there is no evidence base for it and it removes the single most useful recovery indicator. Second, during medication that may affect the biological filter, reducing input while increasing monitoring is reasonable, but this is a water-quality measure rather than a clinical one and should be described as such.

Part LXXXIII — Feeding After Transport and New Arrivals

Covered narrowly to avoid duplicating acclimation content.

Should new fish be fed immediately?

[COMMON AQUARIUM PRACTICE] Standard practice is to delay feeding for a period after introduction — commonly the remainder of the day or up to 24 hours — and then to offer a small quantity. The reasoning is that transported fish are stressed, often will not eat, and that uneaten food adds an avoidable water-quality burden at the moment the fish is least able to tolerate it. This is well-established practice rather than a tested protocol.

Two countervailing points deserve mention. First, shipping depletes reserves: fish are often fasted before transport to reduce waste in the bag, then spend hours or days in transit. The animal arriving in the tank has typically not eaten for some time already, which shortens the window in which further delay is harmless. Second, a fish that eats readily on arrival is giving a strong positive signal, and offering a very small quantity to test that is informative.

Timeframe	Reasonable approach	What to watch
Day of introduction	Do not feed, or offer a very small test quantity once the fish has settled and is exploring	Whether the fish orients to food at all
Day 1–3	Small portions; remove anything uneaten; do not increase to compensate for refusals	First acceptance; whether the fish can compete; whether it is hiding persistently
Day 3–7	Build toward normal ration if feeding normally	Body condition; competitive access; interactions with established fish
Beyond a week without eating	Investigate rather than wait. See Part LXXXI	Water parameters; aggression; species-specific dietary needs; signs of disease

A note on species: some wild-caught and specialised fish — certain loricariids, some anabantoids, many wild-type cichlids — are notably slower to begin feeding in captivity and may require live or frozen food to initiate feeding before transitioning to prepared diets. This is a known application for live food (Part XXXVIII) and for target feeding (Part LVII).

Part LXXXIV — Feeding During Heat, Low Oxygen, and Power Outages

A specific situation where reducing feeding is the correct intervention, for reasons that have nothing to do with nutrition.

Why reduce feeding during a heat event?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Two independent mechanisms converge. First, warm water holds less dissolved oxygen. Second, higher temperature raises ectotherm metabolic rate and therefore oxygen demand. The result is reduced supply meeting increased demand. Feeding adds a third load: digestion and assimilation themselves consume oxygen, and any uneaten or excreted organic material supports heterotrophic bacterial respiration, which consumes more.

Situation	Recommended feeding adjustment	Reasoning
Heat wave with elevated tank temperature	Reduce quantity substantially; consider skipping feedings until temperature stabilises	Oxygen supply is reduced while demand is elevated; adding digestive and bacterial oxygen demand compounds the deficit
Power outage with filtration and aeration stopped	Stop feeding entirely for the duration	No biological filtration, no circulation, no gas exchange. Any food added becomes an uncontrolled ammonia and oxygen-demand source
Visible signs of oxygen deficit (fish at the surface, rapid gill movement, lethargy)	Stop feeding; address aeration and temperature first	The immediate problem is respiratory, not nutritional
Following a large water change to reduce temperature	Resume cautiously and at reduced quantity	Allow the system to stabilise before adding load
Sustained high temperature in a heavily stocked tank	Reduce and split into smaller portions	Smaller meals reduce the peak oxygen demand associated with any single feeding
Cold conditions (pond or unheated tank)	Reduce sharply and eventually stop entirely	In cold water fish cannot properly digest their diet. Standard pond practice reduces feeding and switches to wheat-germ-based food below about 15 °C, and stops feeding altogether below about 10 °C, because digestive bacterial processes effectively cease

The principle behind all of these

Feeding is an oxygen-consuming and waste-generating event. When the system's capacity to supply oxygen or process waste is compromised, reducing food input is a legitimate emergency measure — and one of the few situations in this entire document where restricting feeding is straightforwardly correct.

Healthy adult fish tolerate short periods without food without difficulty (Part XI). They tolerate oxygen deficit much less well. **[ESTABLISHED SCIENTIFIC PRINCIPLE]**

Part LXXXV — Temperature and Appetite

The environmental variable with the largest effect on feeding, and the one most often left out of feeding advice entirely.

How does temperature change feeding physiology?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Fish are ectotherms; their metabolic rate is set by water temperature. Within a species' tolerance range, rising temperature increases metabolic rate, digestion rate, gastric evacuation rate, appetite, and oxygen demand. Falling temperature does the reverse, and low temperature causes sluggishness by retarding digestion speed. Each species has an optimum temperature range for growth, and if growth is optimised, feed intake will be at its maximum to support that growth.

Variable	Effect of rising temperature (within tolerance)	Effect of falling temperature
Metabolic rate	Increases	Decreases
Appetite and feed intake	Increases	Decreases, eventually to zero
Gastric evacuation rate	Faster — appetite returns sooner	Slower — longer intervals between meals are appropriate
Appropriate feeding frequency	Higher frequency becomes appropriate	Lower frequency; eventually cessation
Appropriate % body weight ration	Higher	Lower
Dissolved oxygen availability	Decreases	Increases
Oxygen demand	Increases	Decreases
Bacterial decomposition of waste	Faster	Slower — uneaten food persists longer
Tolerance of a period without food	Reduced — reserves are consumed faster	Increased — metabolic down-regulation conserves reserves

What does this mean practically?

- **The same tank at 24 °C and 28 °C is a different feeding problem.** A ration calibrated at one temperature is not correct at the other, and keepers who raise temperature for treatment or lower it seasonally should expect feeding behaviour to change.
- **Coolwater species kept warm are being pushed.** Goldfish, white cloud mountain minnows, and hillstream loaches kept at tropical temperatures have elevated metabolic demand, which accelerates ageing and raises feeding requirements.
- **Ponds and unheated tanks require seasonal feeding protocols.** The standard progression — reduce feeding, switch to a more digestible food below about 15 °C, stop entirely below about 10 °C — exists because digestion effectively ceases at low temperature and food in the gut of a fish that cannot digest it is a problem rather than a benefit.
- **A fish that stops eating after a temperature drop is behaving normally.** This is a frequent false alarm, particularly in winter and after heater failure.

Part LXXXVI — Feeding at Lights-On Versus Lights-Off

A question with less evidence behind it than the confidence of the advice suggests, and with a genuinely species-specific answer.

Does timing relative to the photoperiod matter?

[REASONABLE INFERENCE] For diurnal species the evidence suggests timing matters less than consistency, because fish entrain readily to whatever schedule is imposed. For nocturnal species it matters a great deal, because the animal's activity window and the food availability window must overlap at all.

The entrainment literature is clear that scheduled feeding is a potent synchroniser in fish. Feeding at mid-dark advanced the phase of hepatic *per1* expression by about seven hours compared with mid-light feeding in zebrafish, and scheduled feeding entrained peripheral oscillators under constant light. Food anticipatory activity began before lights-on or lights-off even in shoals fed an hour after those times. In practical terms, fish will adapt to the schedule you set.

Timing	Arguments for	Arguments against	Best for
Shortly after lights-on	Matches the diurnal activity onset; the keeper is present and can observe; food has the full day to be found and consumed	Some hobby sources suggest allowing the lights on for a period first so fish are fully active; a fish that has just emerged may feed less efficiently	Diurnal community fish; the common default
Mid-photoperiod	Fish are fully active; observation is easy	Often inconvenient for the keeper's schedule	Any diurnal species; a second feeding in a split ration
Shortly before lights-off	Gives slower and more cautious fish a calmer feeding opportunity; leaves food available as nocturnal species emerge	Remnants sit overnight if overfed; keeper cannot observe long	Mixed communities; the second of two daily feedings
After lights-off	The only reliable approach for genuinely nocturnal species; eliminates diurnal competition	No observation of consumption; requires a morning remnant check	Kuhli loaches, plecos, nocturnal catfish
Variable / random times	Some keepers argue it prevents over-conditioning and reduces begging	Undermines entrainment; random feeding disrupted energy homeostasis in goldfish in chronodisruption work	Not recommended as a default

One specific hobby recommendation worth noting: some sources advise ensuring the aquarium light has been on for at least 30 minutes before the morning feeding and remains on for at least 30 minutes after the evening feeding. No supporting study has been located, but the reasoning is sensible — fish are fully active and able to locate food, and the keeper retains visibility to observe consumption and remove remnants. This is a **[USEFUL RULE OF THUMB]** rather than an evidence-based protocol.

Part LXXXVII — Routine and Conditioning

Fish learn feeding schedules and locations well, and this is a tool rather than merely a curiosity.

What do fish actually learn?

[STRONG EVIDENCE] Fish entrain both to feeding *time* and to feeding *location*. Food anticipatory activity is a circadian phenomenon that persists under constant conditions, is entrainable by scheduled feeding, and adjusts its phase over several transient cycles after a change of mealtime rather than instantly. Scheduled feeding entrains peripheral oscillators including hepatic clock gene expression. Increases in activity near the feeder generally did not appear at times unrelated to feeding.

How can routine be used deliberately?

Application	How it works	Value
Consistent feeding time	Fish are behaviourally prepared to feed when food arrives, reducing the interval during which food sits uneaten	Less waste; better consumption; observable feeding response as a health indicator
Consistent feeding location	Fish learn where food appears; a feeding ring or station becomes a reliable rendezvous	Concentrates feeding; reduces scatter; simplifies remnant removal
Separate locations for separate groups	Different fish learn different stations, reducing competitive overlap	Improves distribution in mixed communities
A distinct pre-feeding cue	Tapping a lid, a light change, or simply the keeper's approach becomes a conditioned signal	Brings shy fish out before food is added; useful for target feeding
Feeding response as a daily health check	A conditioned response that fails to appear is a deviation worth noticing	One of the best early-warning signals available to a keeper
After-dark routine for nocturnal fish	Nocturnal species learn the schedule and emerge promptly	Enables observation of animals that are otherwise never seen feeding

A caution against over-reading: conditioned begging is not hunger (Part XXII), and the strength of a fish's learned response is a measure of the consistency of the keeper's routine, not of the animal's nutritional state.

Part LXXXVIII — Automatic Feeders

Useful equipment with a specific and consequential failure mode.

Aspect	Assessment
Reliability of timing	Generally good in modern units; timing is the part that works
Reliability of quantity	The weak point. Dispensed quantity varies with food form, humidity, hopper fill level, and mechanism wear. Calibration over several days is essential rather than optional
Food form compatibility	Pellets and granules dispense far more consistently than flake, which bridges and clumps in most mechanisms
Humidity	The principal cause of failure. Moisture from the tank surface enters the hopper, causing clumping, jamming, or release of a single mass. Mounting away from the water surface and using a desiccant where the design allows both help
Power and battery	Battery-powered units are preferable where outages are plausible: a feeder that stops is far safer than a filter that stops
Portion sizing	Set to underfeed. A week of slight underfeeding in healthy adults is recoverable; a hopper dump into an unattended aquarium may not be
Distribution	A single dispense point concentrates food in one place, which disadvantages bottom dwellers and nocturnal species. Automatic feeding does not solve the distribution problem and may worsen it
Long-term routine use	Legitimate for keepers with irregular schedules, and supports entrainment through consistent timing. The distribution limitation still applies

Note that automatic feeders were used as the food delivery mechanism in circadian entrainment research, with rations calculated at 1 % of fish body weight — a reminder that the equipment is capable of precision when the quantity is actually calculated rather than estimated by turns of a dial.

Part LXXXIX — Human Food

A short section with a clear answer for most items and a narrow exception for plain vegetables.

Item	Assessment	Reasoning
Bread and baked goods	Avoid	Refined starch, salt, yeast, preservatives, and fats fish are not adapted to. Swells substantially in water and in the gut. No nutritional case
Mammalian meat (beef heart, liver, mince)	Generally avoid for community fish	Mammalian fat is poorly suited to fish lipid metabolism. Beef heart appears historically in discus and large cichlid conditioning practice, and was included in some ornamental feeding regimes, but modern formulated diets have largely displaced it and the water-quality cost is substantial
Cooked rice, pasta, potato	Avoid	Starch with little other nutritional content; poorly used by carnivores; adds organic load
Fruit	Avoid as a routine practice	Sugars are poorly utilised by most species; acidity and rapid decomposition; no demonstrated benefit. Some large herbivorous species in specialist keeping receive fruit, which is a narrow exception rather than general advice
Plain blanched vegetables	Appropriate for herbivores and omnivores	The legitimate exception. See Part XXXIX. Must be unseasoned, unsalted, and removed after grazing
Plain unsalted nori / seaweed	Appropriate for grazers	Higher nutrient density than terrestrial vegetables; must be plain
Raw seafood (shrimp, fish, mussel)	Caution	Used in specialist carnivore feeding but carries pathogen risk and a thiaminase issue for certain items — mussels, crustaceans, and cyprinid fishes are the groups identified as thiaminase-rich. See Part XCI
Egg yolk	Limited, specialist use	Used historically as a fry food; largely superseded by rotifers and microdiets, which gave better growth and survival in comparative work. Fouls water severely
Dairy	Avoid	No basis; fish have no lactose-handling capacity of relevance
Anything seasoned, salted, oiled, or sauced	Avoid absolutely	Salt, oils, and seasonings have no place in an aquarium

The general principle: prepared aquarium foods exist because fish nutrient requirements differ substantially from human ones, and because food entering a closed aquatic system has water-quality consequences that food on a plate does not. The exception for plain blanched vegetables holds because those are close analogues of what herbivorous species eat naturally.

Part XC — Feeder Fish

Covered as a feeding-practice question, not as a predator care guide.

What are the documented problems?

Problem	Detail	Evidence
Disease and parasite transmission	Feeder fish are typically produced and held at very high density with minimal health management. Historically, aquarists noted that predatory fish kept on live feeder diets frequently showed poor health and died prematurely, with parasites and bacterial infections among the reasons	[STRONG EVIDENCE]
Thiaminase	Goldfish and fathead (rosy red) minnows are cyprinids and contain thiaminase, an enzyme that degrades thiamine. Cyprinid fishes, mussels, and crustaceans are identified as the three groups of foods dangerously rich in thiaminase. Sustained feeding leads to vitamin B ₁ deficiency	[STRONG EVIDENCE] — see Part XCI
Nutritional inadequacy	Feeder goldfish are high in fat and are a single unsupplemented food item. A diet of one prey species is not a balanced diet	[REASONABLE INFERENCE]
Welfare	The welfare of the feeder animals themselves is a legitimate consideration, as is the stress of live prey in a confined space with no escape	[REASONABLE INFERENCE]
Availability of better alternatives	Most predatory aquarium fish can be maintained and will thrive on prepared and frozen diets, which are nutritionally controlled and carry no pathogen load	[COMMON AQUARIUM PRACTICE]

The brand-appropriate position is straightforward: feeder fish of the cyprinid types most commonly sold carry a documented thiaminase problem, a documented disease risk, and no nutritional advantage over frozen and prepared alternatives. Where live prey is genuinely required to initiate feeding in a specialist fish, that is a narrow clinical application rather than a routine diet.

Part XCI — Thiaminase

A specific, well-documented nutritional hazard with clear practical implications.

What is thiaminase and why does it matter?

[ESTABLISHED SCIENTIFIC PRINCIPLE] Thiaminase is an enzyme that degrades thiamine (vitamin B₁) into its pyrimidine and thiazole components. Thiamine deficiency in predatory fishes is often attributed to a diet of prey fishes with high thiaminase levels. Consequences include neurological dysfunction, anaemia, reduced reproductive success, immunosuppression, and early life stage mortality in offspring resulting from low egg thiamine concentrations.

The scale of the phenomenon in the wild is substantial: thiamine deficiency has been implicated in population declines in the Laurentian Great Lakes, the Baltic Sea, and Pacific coast salmon fisheries. A laboratory feeding study confirmed that lake trout fed an experimental diet of 100 % alewife for two years produced fry with early life stage mortality resulting from thiamine deficiency. Field observations showed an inverse relationship between alewife populations and lake trout egg thiamine concentrations.

Which foods contain thiaminase?

Category	Examples	Aquarium relevance
Cyprinid fishes	Goldfish, carp, minnows including fathead/rosy red, danios and relatives	Directly relevant — these are the standard feeder fish species
Mussels and bivalves	Mussel meat	Used as a fresh or frozen food for large carnivores
Crustaceans	Prawns and shrimp used as fresh or frozen food	Commonly fed to large predatory fish
Certain forage fishes	Alewife, rainbow smelt, anchovies	Primarily a fisheries and wild-population issue, but relevant where marine whole fish is used as food
Reported thiaminase-free	Cockles, tilapia, coley, cod, haddock, smelt sold as lancefish; terrestrial foods generally including earthworms, bloodworms, and crickets	Useful alternatives where whole-prey feeding is required

Two clarifications that circulate incorrectly in the hobby. First, thiaminase content is a property of the species, not of whether the animal is alive, dead, fresh, or frozen — the claim that the enzyme forms after death appears in several sources and is not supported. Second, it does not sort by freshwater versus marine origin: there are freshwater fish with and without thiaminase, and marine fish with and without it.

It is also worth noting that thiaminase activity depends on co-substrates and can be modified by other dietary components: ascorbic acid was identified as an efficient inhibitor of thiaminase I, while pyridoxine (vitamin B₆) acted as an efficient co-substrate promoting degradation. This is mechanistically interesting and does not change the practical advice, which is to avoid making thiaminase-rich species a dietary staple.

Part XCII — Food-Borne Pathogens

Where the risk is real, where it is overstated, and what actually reduces it.

Food type	Pathogen risk	Basis	Mitigation
Live wild-collected invertebrates from water containing fish	Highest	Can act as intermediate hosts or mechanical vectors for parasites and pathogens of fish	Avoid collection from fish-bearing water; prefer cultured sources
Live feeder fish	High	Live fish from high-density production carry fish-specific pathogens directly	Avoid; use frozen and prepared alternatives
Live cultured invertebrates (<i>Artemia</i> , microworms, <i>Daphnia</i> from a clean culture)	Low to moderate	<i>Artemia</i> is a marine organism that does not host freshwater fish pathogens. Risk in other cultures relates to the culture medium; <i>Moina</i> cultured in manure-enriched water prompted ornamental farms to shift to <i>Artemia</i>	Maintain cultures cleanly; rinse before feeding; discard crashed cultures
Frozen invertebrates	Low	Freezing reduces but does not reliably eliminate all pathogens; commercial production and the absence of living intermediate hosts substantially reduce risk relative to live	Buy from reputable producers; observe shelf life; never refreeze
Freeze-dried	Low	Processing is not a validated sterilisation step but removes living organisms	Standard storage practice
Prepared dry foods	Very low for fish pathogens; the relevant hazard is mould and mycotoxin from damp storage	High humidity and temperature promote mould growth and potential mycotoxin production	Dry, cool, airtight storage; discard damp or musty food
Raw seafood from human food supply	Moderate	Not produced or handled for aquatic animal biosecurity	Use sparingly; freeze; consider thiaminase content (Part XCI)

The practical hierarchy is consistent across sources: the pathogen risk rises with how closely the food resembles, and how recently it shared water with, live fish. Prepared and frozen foods sit at the low end; wild-collected live organisms from fish-bearing water sit at the high end; and the commercial ornamental industry's own migration away from manure-cultured live food toward *Artemia* is a useful indicator of how seriously producers take the distinction.

Part XCIII — Enrichment Through Feeding

Feeding is the most frequent and most controllable interaction an aquarium fish has with its environment, which makes it the largest available enrichment lever.

Is feeding enrichment a real concept for fish?

[REASONABLE INFERENCE] Yes, though the formal literature on environmental enrichment in ornamental fish is thin. Sales and Janssens identified the underlying gap: measurement of nutrient requirements has to be correlated with animal welfare, a field they described as still unexploited with aquatic species. The behavioural case rests on the observation that natural feeding involves searching, pursuing, manipulating, and processing food, and that a pellet delivered to a fixed point removes all four.

Technique	Behaviour elicited	Practical note
Scatter feeding rather than a single drop point	Searching and foraging across the tank	Increases the chance food is lost to substrate; best combined with careful portioning
Live food	Hunting, pursuit, capture	The strongest enrichment available; see Part XXXVIII for the risk profile
Grazeable material (vegetables, biofilm-covered hardscape, algae)	Extended grazing over hours rather than a meal in seconds	Low caloric cost for the behavioural benefit; ideal for grazers
Food placed in or under decor	Searching, probing, working for access	Suits benthic and secretive species; use a location you can siphon
Varying feeding locations between sessions	Exploration; prevents a single dominant fish from monopolising one station	Also improves distribution; see Part XCIV
Whole or larger food items for species adapted to process them	Manipulation, tearing, repositioning	Only for species whose feeding morphology suits it
Live blackworms in sand	Substrate sifting and probing over an extended period	Excellent for <i>Corydoras</i> and sand-sifting species
Varying food type between sessions	Investigation and discrimination	Overlaps with dietary variety; substitute rather than add (Part XXIII)

A caveat worth stating: enrichment techniques that spread food more widely also make remnants harder to see and remove. The enrichment benefit is real and so is the waste cost, and the resolution is smaller portions rather than abandoning either.

Part XCIV — Overcompetition and Feeding Aggression

The behavioural half of the distribution problem introduced in Parts XX and LV.

What drives feeding aggression?

- **Resource concentration.** A single food source in a single location is a defensible resource. The more concentrated the delivery, the more worthwhile defence becomes.
- **Scarcity or perceived scarcity.** Infrequent feeding and restricted rations intensify competition. This is one of the underappreciated costs of scheduled fasting in tanks with existing hierarchy problems (Part X).
- **Established hierarchy.** In cichlids and other hierarchical species, feeding is an occasion on which rank is enforced.
- **Territorial context.** A fish defending a territory that happens to contain the feeding station will defend the food incidentally.
- **Size disparity.** Larger individuals both outcompete and intimidate smaller ones.
- **Inadequate group size in schooling species.** Reduced confidence in under-sized shoals expresses partly as reduced feeding activity.

What strategies work?

Strategy	Mechanism	Effectiveness	Caveat
Multiple simultaneous feeding locations	A dominant fish cannot defend two places at once	High — the single most effective intervention	Requires the keeper to divide the portion rather than double it
Smaller portions delivered more frequently	Reduces the value of any single defence event and gives subordinates repeated opportunities	Moderate to high	Portion discipline must hold across all deliveries
Broadcast rather than point feeding	Disperses the resource so it cannot be monopolised	Moderate	Increases the chance of food reaching substrate uneaten
Visual barriers and broken sightlines	A dominant fish cannot police what it cannot see	High in aggressive species; standard practice in cichlid keeping	Requires aquascaping change rather than feeding change
Target feeding the subordinate	Bypasses competition entirely	Very high for individuals	Labour intensive; not scalable
Feeding shy fish separately in time	Feed the dominant group first, then deliver a second portion once they disperse	High	Requires the keeper to stay for both steps
Increasing group size in schooling species	Restores normal shoaling confidence and dilutes aggression	High where under-stocking is the cause	A stocking change, not a feeding change
Reassessing stocking compatibility	Some combinations cannot be fixed by feeding technique	Definitive when it applies	The honest answer in a subset of cases

When feeding technique is not the answer

Some competition problems are stocking problems wearing a feeding costume. A persistently harassed fish that cannot reach food in any configuration, a species kept in a group too small to behave normally, or a size mismatch that no distribution technique resolves — these require a change to the community, not a change to the feeding routine.

It is worth saying this plainly in the episode, because feeding technique is the more comfortable answer and the less honest one. [\[REASONABLE INFERENCE\]](#)

Part XCV — Feeding Schooling Fish

Group size affects feeding behaviour through confidence rather than through competition alone.

Does group size affect feeding?

[REASONABLE INFERENCE] Yes, in two directions that partly offset. Larger groups increase intraspecific competition for a fixed ration, but they also substantially increase feeding confidence in obligate shoaling species, which typically spend less time hiding and more time foraging. In practice the confidence effect usually dominates for species kept at or above their behavioural minimum group size, because a shoal kept in inadequate numbers often does not feed properly at all.

Social facilitation also operates: one fish beginning to feed triggers others, which is why a shoal of adequate size typically converges on food faster and more completely than a pair of the same species. The corollary is that an isolated or under-shoaled individual of a schooling species may simply not initiate feeding.

Situation	Feeding consequence	Response
Schooling species kept in inadequate numbers	Reduced feeding activity, more hiding, poorer condition despite adequate food in the tank	Increase group size; this is a welfare fix, not a feeding fix
Adequately sized shoal	Rapid convergence on food; confident feeding; competition is distributed rather than hierarchical	Standard broadcast feeding usually works well
Very large shoal in a modest tank	Food consumed extremely rapidly; slower tankmates disadvantaged	Split the ration; feed in more than one location; use slow-sinking food
Mixed shoal of different sizes	Smaller individuals may be excluded by both speed and particle size	Match particle size to the smallest mouth; feed a second small portion
Newly introduced individuals joining an established shoal	Often slow to feed for several days	Patience; a second delayed portion; observe rather than increase the ration

Part XCVI — Feeding in Heavily Planted Tanks

Where natural food production is genuinely higher — and where its contribution is most often overestimated.

Do planted tanks reduce feeding requirements?

[CONTESTED CLAIM] Partially and species-specifically. A mature, heavily planted aquarium supports substantially more biofilm surface area, microfauna, and grazing substrate than a bare tank. For species adapted to graze that material continuously, this is a real nutritional contribution. For midwater particulate feeders and carnivores, it contributes very little.

Species type	Benefit from a planted tank's natural production	Practical consequence
Aufwuchs grazers and algivores (otocinclus, hillstream loaches, some plecos)	Substantial — this is their natural food, continuously renewed on plant and hardscape surfaces	Genuinely reduces supplementary feeding requirement, though rarely to zero
Shrimp and grazing snails	Substantial — biofilm is a primary food source	Established planted tanks support small populations with minimal supplementation
Benthic omnivores (<i>Corydoras</i> , some loaches)	Modest — microfauna in substrate and leaf litter contributes	Reduces but does not replace deliberate feeding
Small omnivorous fish and fry	Modest to moderate — microfauna and infusoria are a real food source for very small mouths	Meaningfully improves fry survival in established planted tanks
Midwater particulate feeders (most tetras, rasboras, danios)	Minimal — they are not adapted to graze surfaces	Feeding requirement essentially unchanged
Carnivores and piscivores	Negligible	No change
Plant-eating herbivores	Variable — they may eat the plants, which is a different problem	Consider plant selection alongside feeding

Two further considerations specific to planted tanks. First, plants compete with algae for the nutrients that food introduces, so a healthy growing plant mass buffers the nutrient consequences of feeding — but only while the plants are actually growing, and only if the biomass is periodically harvested and removed. Nutrient uptake without biomass export is temporary storage, not removal. Second, dense planting traps uneaten food and detritus in low-flow zones where it is difficult to see and to siphon, which makes portion discipline more important in a planted tank rather than less.

Part XCVII — Natural Foods in Mature Aquariums

Quantifying, as far as the evidence allows, what an established tank actually produces.

Natural food source	What it is	Who eats it	Realistic contribution
Biofilm	Bacterial and microalgal layer on every wetted surface	Otocinclus, hillstream loaches, shrimp, snails, fry, some loricariids	Genuinely significant for specialised grazers and for very small mouths; negligible for everything else
Algae	Filamentous, film, and diatom growth	Algivores and grazing snails	Variable and usually declining, because the keeper is actively removing it
Microfauna	Copepods, ostracods, rotifers, <i>Daphnia</i> , seed shrimp, nematodes	Fry, very small fish, benthic foragers	Real and self-renewing in mature tanks; supports fry survival notably. Populations are themselves food-limited, so they respond to feeding rather than replacing it
Detritus and associated microorganisms	Decomposing organic matter plus its bacterial and fungal load	Detritivores, benthic foragers, shrimp, snails	The microbial component is the nutritional value, not the dead matter. Modest
Infusoria	Mixed protozoan and micro-organism assemblage, typically in established or green water	Very small fry	Meaningful specifically for first-feeding fry
Aquatic plant tissue	Soft plant growth	Herbivores	Substantial where the fish will eat the plants — which is usually treated as damage rather than as feeding
Leaf litter and its decomposer community	Botanical material plus the organisms it hosts	Shrimp, benthic fish, fry	A genuine and often-overlooked continuous low-level food source
Pest snail populations	Self-reproducing invertebrates	Snail-eating fish (loaches, puffers, some cichlids)	Real for those species, though rarely sufficient alone and unpredictable in supply

How meaningful is this overall?

[REASONABLE INFERENCE] Meaningful for a specific and narrow set of animals — specialised grazers, invertebrates, and fry — and negligible for the majority of fish in a typical community aquarium. The common error is to generalise from the genuine case (an *Otocinclus* grazing biofilm in a mature planted tank) to the general case (a tank feeding its own fish).

There is also a circularity worth naming. Microfauna and biofilm populations in a closed aquarium are themselves sustained largely by the organic input the keeper provides — principally food. A tank that produces abundant microfauna is usually a tank receiving abundant organic input. Natural food production is therefore better understood as a partial recycling of what was already added than as an independent source.

Part XCVIII — Can a Tank “Feed Itself”?

The ecosystem-tank question, handled without romanticising scarcity.

What is actually being claimed?

A family of related claims circulates around Walstad-method tanks, low-tech planted systems, dirtied tanks, and “balanced ecosystem” aquariums: that a sufficiently mature, sufficiently planted system produces enough natural food that supplementary feeding can be reduced or eliminated. Some versions present minimal or no feeding as a superior, more natural way of keeping fish.

What is defensible?

Claim	Assessment
A mature planted tank produces genuine natural food	[STRONG EVIDENCE] — biofilm, microfauna, algae, and detrital communities are real and self-renewing. See Part XCVII
Some animals can meet a substantial share of their needs from it	[REASONABLE INFERENCE] — specifically shrimp, grazing snails, biofilm-grazing fish, and fry. This is a genuinely narrow set
Feeding can be reduced in a mature, lightly stocked, heavily planted tank containing appropriate species	[REASONABLE INFERENCE] — with all four qualifiers doing real work
A planted tank can support a normally stocked community with no supplementary feeding	[UNSUPPORTED AQUARIUM FOLKLORE] — no evidence supports this for midwater particulate feeders or carnivores, which is most community fish
Reduced feeding is more natural and therefore better	[UNSUPPORTED AQUARIUM FOLKLORE] — wild fish forage across areas orders of magnitude larger than an aquarium. Restricting food in a closed system is not a reproduction of wild conditions
A no-feed tank demonstrates good husbandry	[UNSUPPORTED AQUARIUM FOLKLORE] — it demonstrates that the animals present have not yet visibly declined, which is a different statement

Under what specific circumstances might feeding be reduced?

All of the following conditions would need to hold simultaneously, and the reduction would still be partial:

1. The system is **mature** — established biofilm, stable microfauna populations, and a functioning detrital community.
2. The planting is **dense and actively growing**, providing extensive grazing surface.
3. Stocking is **light**, so that per-animal natural food availability is meaningful.
4. The species present are **grazers, detritivores, shrimp, or snails** — animals adapted to extract nutrition from surface and substrate communities.
5. The keeper is **monitoring body condition** regularly and is prepared to resume feeding on the first sign of decline.
6. No animal in the system is **growing, breeding, or recovering**, since all three raise requirements substantially.

The position on ecosystem tanks

A mature, heavily planted, lightly stocked tank containing grazing species genuinely does supply some of its animals' food, and reducing supplementary feeding in that specific case is defensible — with body condition monitored.

What is not defensible is treating minimal feeding as an achievement, or a low-feeding tank as evidence of superior husbandry. A fish that is slowly losing condition in a beautiful planted tank is still a fish that is slowly losing condition. Nutrition and welfare come first; the aesthetics and the philosophy come after. [\[REASONABLE INFERENCE\]](#)

Part XCIX — Feeding and Aquarium Maintenance: The Whole Relationship

Pulling Parts XIII–XIX and LXIV–LXIX into a single causal picture. This is likely to be one of the episode's central themes.

The chain, stated once, end to end

Stage	What happens	Keeper's lever	Observable signal
1. Food input	Nitrogen, phosphorus, and organic carbon enter the closed system	Portion size; feeding frequency; food type and energy density; supplement discipline	What you put in — the only stage you fully control
2. Consumption	Some food is eaten; some is not	Food form matched to feeding zone; particle size; delivery method; timing	Uneaten food remaining after feeding
3. Assimilation	Fish retain a fraction of nutrients; the rest is excreted or egested	Diet quality and species match; feeding frequency affects retention efficiency	Body condition over weeks; faecal output
4. Waste generation	Ammonia from gills; faecal solids; decomposition of uneaten food	Indirect — determined by stages 1–3	Rarely directly observable
5. Biological processing	Nitrification converts ammonia to nitrite to nitrate; heterotrophs consume organics and oxygen	Filtration capacity; circulation; oxygenation	Ammonia and nitrite at zero in a functioning system; cloudiness when heterotrophs bloom
6. Accumulation	Nitrate, phosphate, and dissolved organics build up; detritus settles	Nothing at this stage — accumulation is the default	Rising nitrate trend; detritus in low-flow zones; filter clogging interval; yellowing water
7. Export	Water changes, substrate cleaning, mechanical media servicing, plant biomass removal	Maintenance frequency and thoroughness	Post-change parameters; volume of detritus removed
8. Consequences if export lags input	Elevated nitrate and phosphate; organic loading; algae pressure; snail population growth; oxygen demand	Return to stage 1 or increase stage 7	The symptoms most keepers notice first, and the furthest downstream

Table XCIX-1. The complete feeding-to-maintenance chain. Note that the signals keepers act on are concentrated at stages 6 and 8, while the levers with the most effect are at stages 1 and 2.

Where does the keeper actually have control?

Two places, and they are the first two stages. Everything downstream is either automatic (stages 3–6) or corrective (stage 7). The practical implication is that **feeding precision is upstream leverage and maintenance is downstream compensation**. Both are necessary; they are not interchangeable.

The theme, stated for the episode

Feeding is not just a nutrition decision. Every time you feed, you are importing nitrogen and phosphorus into a sealed box, and the only way any of it leaves is when you take it out.

That does not mean feeding less is the answer — your fish still need what they need. It means feeding *precisely* is the answer: the right amount, in a form that reaches the right animal, eaten promptly, with the leftovers removed. Everything else about aquarium maintenance gets easier when that upstream step is right, and nothing downstream fully compensates when it is wrong. [\[REASONABLE INFERENCE\]](#)

Part C — A Practical Feeding Framework

Everything in the preceding ninety-nine parts converges here. The framework below is a decision process, not a formula, and the research supports it in that form and only in that form.

Why this is a process and not a number

The research does not support a universal quantity, and it does not support replacing one universal quantity with another. What it does support is that the variables governing the correct amount are knowable, that most of them are observable by an ordinary keeper, and that a structured sequence of observations converges on a workable answer for a specific tank.

The framework is therefore best understood as **a set of questions that produce a starting point, plus a feedback loop that corrects it.** [\[REASONABLE INFERENCE\]](#)

The ten questions

#	Question	Why it matters	How to answer it	Covered in
1	What species am I feeding, and what is each one's natural dietary strategy?	Determines food type, protein level, plant fraction, and carbohydrate tolerance	Identify each species' trophic category and, more importantly, its feeding zone and activity period	Parts I, XLI–LII
2	Where and when does each species feed?	The single most common cause of an animal going underfed in a well-fed tank	Inventory by zone (surface / midwater / substrate) and by time (diurnal / crepuscular / nocturnal)	Parts LIII–LVI
3	How large and how old are they?	Ration as a percentage of body weight falls steeply with size; fry and juveniles need proportionally far more	Estimate adult versus growing status; note that a tank stocked with juveniles needs downward adjustment as they mature	Parts IV, VII–IX
4	What is the total biomass, not the fish count?	Ten tetras and one large cichlid are not comparable feeding problems	Rough estimate of total living fish mass, weighted for the fact that many small fish need more than one large fish of equal mass	Part V
5	What is the water temperature?	Governs metabolic rate, digestion speed, appetite, and oxygen demand	Thermometer; and note whether the tank runs warm or cool for the species present	Part LXXXV
6	Which fish actually get access to the food?	A correct total distributed badly produces simultaneous over- and underfeeding	Watch a full feeding. Identify who eats first, who eats last, and who does not appear	Parts XX, LV, XCIV
7	How quickly is the food consumed, and does any remain?	The immediate Type 1 signal	Observe the feeding through to completion; check the substrate afterward	Parts XIII, XIX, LXIII
8	What does body condition look like, per species and per individual?	The only measurement that directly answers whether the animal is getting what it needs	Monthly visual assessment of dorsal musculature, peduncle, and the area behind the operculum. Photograph a baseline	Part XXI
9	Is waste accumulating faster than before?	Integrates input over time in a way single observations cannot	Nitrate accumulation rate between water changes; mechanical filter cleaning interval; detritus in low-flow zones; snail population trend	Parts XIV, LXVI–LXIX
10	Does the amount need adjusting, and in which direction?	The decision the other nine questions exist to inform	Change one variable at a time and wait long enough to see the effect	Parts CI–CII

Does the evidence support this specific model?

[REASONABLE INFERENCE] The individual components are each supported: body-condition assessment is standard veterinary practice, nutrient accumulation rate is a direct measure of input-export balance, feeding-zone mismatch is a documented cause of underfeeding in mixed systems, and incremental adjustment with observation is standard aquaculture feeding practice. The specific ten-question sequence is a synthesis rather than a validated protocol, and should be presented as a structured way of thinking rather than as an established method.

The three-signal check

For an audience that will not run a ten-question audit, the framework compresses into three signals that between them cover the three types of overfeeding and the underfeeding case:

Signal	What it catches	How often to check
Is food left over after feeding?	Type 1 overfeeding — uneaten food. Also flags food-form mismatch	Every feeding — it costs nothing
How does each species' body condition look?	Type 2 overfeeding (excess consumed) and Type 3 (misdirected food), plus underfeeding	Monthly, with a photographic baseline
How fast is nitrate climbing between water changes?	Total system input versus export; catches chronic excess with no visible leftovers	Every water change, recorded

These three signals operate on different timescales — immediate, monthly, and per-maintenance-cycle — which is why all three are needed. No single one of them detects all four failure modes.

Part CI — “Start Small and Observe”

Testing whether this is genuinely a better beginner approach than giving one exact quantity.

Is it better than a fixed prescription?

[REASONABLE INFERENCE] Yes, for four independent reasons, and it is worth being explicit about them because “start small and observe” can otherwise sound like an evasion.

1. **It errs in the recoverable direction.** A healthy adult fish tolerates a period of modest underfeeding without lasting harm (Part XI). A tank tolerates chronic overfeeding less well, and the consequences — accumulated nutrient, organic loading, fat deposition — develop slowly and are slow to reverse. The asymmetry favours starting low. *With the important exception of fry and juveniles, where restriction can cause permanent stunting — see Part VII.*
2. **It converges on a tank-specific answer.** A prescribed quantity is correct for a hypothetical average tank and wrong for every real one. An iterative approach produces an answer calibrated to the actual species, biomass, temperature, and competitive structure present.
3. **It builds the observational habit the rest of fishkeeping requires.** A keeper who watches a feeding closely enough to calibrate a portion is also watching closely enough to notice a fish that is not eating, a fish that is being harassed, and a fish whose condition is declining. The feeding calibration is a training exercise for everything else.
4. **It is honest about what is knowable.** The alternative — giving a confident number — requires asserting something the evidence does not support, and it fails the first time the viewer's tank does not match the assumption behind it.

What does it need to work?

The approach has one genuine failure mode: a keeper who starts small and never adjusts. “Start small” without “and observe” is simply underfeeding. Three things convert it into a working method:

- **A defined review point.** “Check body condition in four weeks” is a method; “keep an eye on them” is not.
- **A written record.** Portion, food type, and a dated condition note. Memory is not adequate for a variable that changes over months.
- **An explicit permission to increase.** The advice is overwhelmingly framed around not overfeeding, which means keepers read it as one-directional. The correction loop has to run both ways.

Part CII — Portion Calibration

The operational procedure. This is the section that most directly answers the episode's title question.

The calibration procedure

Step	Action	What you are measuring	Decision rule
1. Prepare	Choose a food form appropriate to the dominant feeding zone. If the tank has both surface and benthic fish, plan two food forms from the outset	Whether delivery is even possible	Do not proceed with a calibration you know cannot reach the bottom dwellers
2. Start deliberately small	Offer noticeably less than instinct suggests — for a small community, a quantity that looks inadequate	Baseline	The measured maintenance figures in Part III suggest instinct is roughly an order of magnitude high
3. Watch it through	Observe the entire feeding rather than leaving the room	Consumption speed; who eats; who does not; where food goes	Note anything that does not reach an intended animal
4. Add a small increment if warranted	If all fish are still actively searching and nothing has settled, add a small second amount	Whether the first portion was genuinely insufficient or the fish are simply still enthusiastic	Conditioned begging continues past satiety — the signal is food landing uneaten, not fish still interested
5. Stop before accumulation	Stop at the point where food begins reaching the substrate and staying there	The upper bound for this tank	This is a ceiling, not a target
6. Record it	Note the portion in a repeatable unit — number of pellets, a marked scoop, a fraction of a cube	Reproducibility	“A pinch” is not a record. A countable or marked measure is
7. Repeat consistently for two to four weeks	Same portion, same time, same delivery	Stability	Do not adjust during this period unless something is clearly wrong
8. Assess body condition	Species by species, individual by individual, against the photographic baseline	Whether the animals are actually getting what they need	This is the measurement that governs the direction of adjustment
9. Check the waste trend	Nitrate accumulation rate; filter cleaning interval; detritus accumulation	Whether the system is absorbing the input	Rising trend with stable stocking and maintenance points to input
10. Adjust one variable	Change portion, or frequency, or food form — not two at once	Attribution	Then return to step 7

How do the signals combine into a decision?

Body condition	Uneaten food	Nitrate trend	Most likely situation	Adjustment
Good across all species	None	Stable	Correctly calibrated	None — keep recording
Good across all species	None	Rising	Type 2 overfeeding, or an export problem (reduced water changes, declining plants, source water)	Check maintenance and source water first; then reduce portion modestly
Good in most, thin in benthic or nocturnal species	None	Stable or rising	Type 3 — misdirected food	Change delivery: sinking food, after-dark feeding, feeding station. Do not simply increase the total

Body condition	Uneaten food	Nitrate trend	Most likely situation	Adjustment
Thin across all species	None	Low and stable	Genuine underfeeding	Increase portion; reassess in two to four weeks
Rounded in dominant fish, thin in subordinates	None	Rising	Type 3 with a competition component	Multiple feeding locations; visual barriers; consider stocking
Rounded across all species	None	Rising	Type 2 — excess consumed food	Reduce portion; reduce energy density; substitute rather than add treats
Good	Food on substrate after feeding	Rising	Type 1 — uneaten food, possibly with form mismatch	Reduce portion; check food form against feeding zones
Variable	Food on substrate	Stable	Form mismatch — food is not reaching the intended animals and is not being eaten by anyone	Change food form before changing quantity

Table CII-1. Combining the three signals into a diagnosis. Note how often the correct adjustment is to delivery rather than to quantity.

Is this approach supported?

[REASONABLE INFERENCE] The components are. Incremental adjustment with observation is standard aquaculture feeding practice — commercial operations adjust feed supply every one to two weeks based on measured growth, and the FAO manual explicitly frames the correct feeding level as something to be determined by observation and results rather than by a fixed instruction. Body condition monitoring is recommended in aquatic veterinary reference. Nutrient accumulation as a measure of input-export balance follows directly from nutrient mass balance. The combined protocol is a synthesis, presented as a structured method rather than a validated one.

The answer to “how much should I feed my fish?”

Short version: Less than you think, in a form that reaches every fish in the tank, and then adjust based on what their bodies look like a month later.

Why there isn't a number: because the right amount depends on which species, how big, how old, how warm the water is, how energy-dense the food is, and how many of your fish can actually get to it. Every one of those varies between tanks, and several of them vary within a single tank.

What replaces the number: start small, watch the feeding, check body condition monthly, watch the nitrate trend, and change one thing at a time. That is slower than being handed a scoop size, and it is the only approach the evidence actually supports.

Part CIII — Scenario Library

Thirty-two scenarios drawn from the questions hobbyists actually ask, structured identically for retrieval. Each separates what is known from what is not, and each names the observation that would actually resolve it.

How to use this library

Every scenario follows the same seven fields. The two that matter most are **Best observation** — the single thing the keeper should look at before changing anything — and **Common mistakes**, which is usually where the hobby's conventional answer goes wrong.

Several scenarios resolve to the same underlying cause. That repetition is deliberate: it reflects how often distribution, conditioned behaviour, and the uneaten-versus-consumed distinction explain apparently unrelated symptoms.

Scenarios A–H: quantity, behaviour, and the basics

Scenario A — “How much food do I give my fish?”

Field	Assessment
What we know	Requirement scales with biomass, body size, life stage, temperature, and food energy density. Measured maintenance requirements for small ornamentals are very small — under 4 mg of dry feed per day for a neon tetra. Aquatic-veterinary guidance places adult maintenance near 3 % of body weight daily.
What we don't know	The correct quantity for this specific tank. No published figure covers mixed communities, and almost no keeper knows their fish's weight.
Questions to ask	Which species, at what sizes? What temperature? Which feed at the surface, midwater, and bottom? Which are nocturnal? Is anything growing or breeding?
Likely issue	Not an issue yet — this is the baseline question. The underlying problem is that the questioner expects a number.
Best observation	Watch a full feeding: what is consumed, how fast, and by whom. Then body condition at four weeks.
Possible adjustment	Start deliberately below instinct, in a form that reaches each feeding zone, and calibrate per Part CII.
Common mistakes	Giving a universal number; using volume rather than a repeatable measure; calibrating on the fastest feeders; forgetting to reduce as fish mature.

This is the document's central question. The full answer is Parts C, CI, and CII.

Scenario B — “My fish always act hungry.”

Field	Assessment
What we know	Food-anticipatory activity is a documented circadian phenomenon that persists even when food is withheld — in one zebrafish study, in 78 % of shoals. Fish also condition rapidly to the keeper's approach, and many are opportunistic feeders whose feeding reflex triggers on food presence rather than on satiety.
What we don't know	Whether this particular tank is also genuinely underfed. Begging behaviour cannot distinguish the two.
Questions to ask	Does the behaviour occur mainly at the usual feeding time, or continuously? Is anyone losing condition? Is anyone foraging constantly outside feeding times?
Likely issue	Almost always conditioned anticipation rather than hunger.
Best observation	Body condition — the only signal that separates learned behaviour from genuine need. Also whether searching behaviour persists hours away from feeding time.

Field	Assessment
Possible adjustment	Usually none. If condition is declining, increase; if it is good, the behaviour is information about your routine's consistency, not about hunger.
Common mistakes	Feeding in response to begging; adding a third feeding; interpreting a missed-meal reaction as evidence of starvation.

Scenario C — “I feed a pinch twice a day. Is that too much?”

Field	Assessment
What we know	A pinch is not a unit — it varies by hand, by food form, and by intent, and flake and pellet differ several-fold in mass at equal volume. Twice daily is a well-supported frequency for many adult community fish.
What we don't know	What the pinch actually weighs, and what the tank's biomass is. Without both, the question is unanswerable as asked.
Questions to ask	What is the total fish biomass? What food form? Is any food reaching the substrate uneaten? What is the nitrate trend between water changes?
Likely issue	Frequency is fine; quantity is unmeasurable and quite possibly excessive.
Best observation	Whether food reaches the substrate and stays there, plus the nitrate accumulation rate.
Possible adjustment	Convert the pinch to a repeatable measure — a counted number of pellets or a marked scoop — then calibrate downward per Part CII.
Common mistakes	Assuming frequency is the problem when quantity is; continuing to use an unmeasurable unit; adding a fasting day instead of measuring the portion.

Scenario D — “Food hits the bottom before my fish eat it.”

Field	Assessment
What we know	Food form determines where food goes; a mismatch between food buoyancy and the residents' feeding zone produces exactly this. Uneaten food on the substrate is the clearest Type 1 overfeeding signal.
What we don't know	Whether the cause is quantity, food form, particle size, or palatability.
Questions to ask	Are the fish surface or midwater feeders? Is the pellet too large? Is there anything living on the substrate that should be eating it? Is the food fresh?
Likely issue	Either too much food, wrong food form for the residents, or particle size mismatch — often two of the three.
Best observation	Watch whether fish approach the food and reject it (size or palatability) or simply never reach it (quantity or form).
Possible adjustment	Reduce portion first; if food still sinks uneaten, switch to a floating or slower-sinking form and check particle size.
Common mistakes	Adding bottom feeders to eat it — this converts a feeding error into a stocking decision. Assuming the fish will 'find it later.'

Scenario E — “My tetras eat everything before my corydoras get any.”

Field	Assessment
What we know	This is Type 3 misdirected feeding; the total may be correct and the distribution is not. It is one of the most common failure modes in community aquariums and it produces simultaneous over- and underfeeding in one tank.
What we don't know	Whether the corydoras are getting anything at all, since the absence of leftovers looks identical to correct feeding.
Questions to ask	Is the corydoras group in good body condition? Is the food sinking at all? Are they active and foraging at feeding time? How many midwater fish are intercepting?
Likely issue	Distribution, not quantity.
Best observation	Corydoras body condition — specifically the profile behind the head and along the abdomen — assessed monthly.

Field	Assessment
Possible adjustment	Sinking food delivered after the tetras are fed at the surface (decoy method); a feeding dish; feeding after lights-out. Do not simply increase the total.
Common mistakes	Increasing the overall portion, which feeds the tetras more and the corydoras the same; relying on 'they'll clean up the leftovers.'

Scenario F — “My kuhli loaches never come out to eat.”

Field	Assessment
What we know	Kuhli loaches are strongly nocturnal benthic feeders that are often entirely hidden during daylight. Daytime feeding frequently reaches them not at all. Fish entrain readily to dark-phase feeding schedules.
What we don't know	Whether they are eating anything at night, since the keeper is not present to observe.
Questions to ask	Is sinking food being offered at all? Is anything left in the morning? Is the substrate suitable for them to forage in? Are they in a group large enough to behave confidently?
Likely issue	Timing mismatch, compounded by food form.
Best observation	Place a known quantity of sinking food in a dish after lights-out and check it the following morning. Something either ate it or did not.
Possible adjustment	Sinking protein food a few minutes after lights-off; a dish to concentrate it and make remnants visible; live blackworms if available.
Common mistakes	Assuming they scavenge adequately; feeding algae wafers, which they are not equipped to use as a primary food; concluding they are fine because they are not visibly thin from across the room.

Scenario G — “My betta looks bloated.”

Field	Assessment
What we know	Bettas are small, singly kept, enthusiastically fed carnivores, and overfeeding is common. But abdominal distension has many possible causes: gut fill, constipation, egg development in females, internal parasites, organ enlargement, tumour, and dropsy secondary to systemic disease.
What we don't know	Which of those is occurring. The presentations overlap substantially, and veterinary sources note X-ray as the primary diagnostic route for buoyancy and distension cases.
Questions to ask	Are the scales raised or smooth? Is the fish still eating and active? Is it female? How much and what is it being fed? Has the distension developed over hours or weeks? Water parameters?
Likely issue	Most commonly overfeeding or gut fill. But pineconing scales, lethargy, and appetite loss point elsewhere.
Best observation	Scale profile viewed from above, plus whether the fish is still feeding and active.
Possible adjustment	If the fish is bright and eating with smooth scales: reduce portion, consider a short fast, review whether bloodworms have become the staple. If scales are raised or the fish is lethargic and off food: this is not a feeding question.
Common mistakes	Reaching for peas — bettas are carnivores with limited carbohydrate capacity; assuming every distension is constipation; extended fasting of a small fish with limited reserves; delaying assessment of a systemic problem.

Scenario H — “My fish have long strings of poop. Am I overfeeding?”

Field	Assessment
What we know	Faecal character reflects diet composition and digestion. Increased faecal output does follow increased intake. Pale, stringy, or trailing faeces are commonly associated in hobby practice with diet change, some internal parasites, and reduced food intake.
What we don't know	The specific relationship between faecal appearance and overfeeding in aquarium fish is not documented in the literature. This is a widely used sign with a weak evidence base.

Field	Assessment
Questions to ask	What colour and consistency? Is the fish eating normally? Has the food changed recently? Is body condition maintained? Are other fish affected?
Likely issue	Often a recent diet change. Sometimes increased intake. Persistent pale stringy faeces with weight loss is a different problem entirely.
Best observation	Body condition alongside faecal appearance. Faeces alone is a weak signal; faeces plus declining condition is a strong one.
Possible adjustment	If body condition is good and the food changed recently: observe. If condition is declining: investigate rather than adjust portion.
Common mistakes	Treating faecal appearance as a definitive overfeeding diagnosis; reflexive medication; ignoring the combination of stringy faeces with weight loss.

Scenarios I–N: water quality and absence

Scenario I — “My nitrate keeps rising fast.”

Field	Assessment
What we know	Food is usually the dominant nitrogen input to a closed aquarium. Only about 25–40 % of feed nitrogen is retained in fish tissue in growing fish, and essentially none in adults at maintenance. Nitrate accumulates unless exported.
What we don't know	Whether the cause is increased input or reduced export — the trend alone cannot distinguish them.
Questions to ask	Has feeding changed? Has stocking changed? Has the water change schedule slipped in frequency or volume? Have plants been removed or declined? What is the nitrate level in the source water? Was the substrate recently disturbed?
Likely issue	Frequently reduced export rather than increased input — a water change schedule that quietly lengthened.
Best observation	Measure nitrate immediately after a water change and immediately before the next, at a consistent interval. The difference is the accumulation rate.
Possible adjustment	Check source water first; then review maintenance; then reduce feeding. In that order.
Common mistakes	Assuming overfeeding without testing tap water; cutting feeding as the first response; reading a single value instead of a trend.

Scenario J — “My tank has tons of pest snails.”

Field	Assessment
What we know	Snail populations in aquaria are primarily food-limited and respond to available organic matter with a lag of weeks. A population boom is reliable evidence of surplus food.
What we don't know	Whether the surplus comes from fish food specifically. Decaying plants, a dead animal, vegetables left too long, and reduced maintenance all supply it too.
Questions to ask	What changed several weeks ago? Are vegetables being left in? Has anything died unnoticed? Has maintenance slipped? Were the snails recently introduced on plants?
Likely issue	Surplus organic matter of some kind. Fish food is one candidate among several.
Best observation	Audit organic inputs across the whole tank, not just the feeding routine.
Possible adjustment	Correct whichever surplus is actually present. Manual removal reduces standing population but not the underlying food supply.
Common mistakes	Treating snails as proof of fish overfeeding; adding snail-eating fish as a solution to an input problem; chemical treatments that harm other invertebrates. Snails are livestock and a useful indicator, not a pest event.

Scenario K — “My tank gets cloudy after feeding.”

Field	Assessment
What we know	Heterotrophic bacteria multiply rapidly in response to available dissolved and particulate organic carbon, and can reproduce on a timescale of hours. A sudden organic input is exactly the substrate required.
What we don't know	Whether the cloudiness is bacterial, fine particulate, or phytoplankton — and whether the tank is fully cycled.
Questions to ask	Does it appear within hours of feeding and clear over a day or two? Is it milky white or green? Is the tank newly set up? Is there adequate mechanical filtration? What are ammonia and nitrite?
Likely issue	Bacterial bloom on excess organic input, if the timing is reproducible and the tank is established.
Best observation	The timing relative to feeding, and the colour.
Possible adjustment	Reduce portion; remove uneaten food promptly; strain the fluid from thawed frozen food; verify mechanical filtration is functioning.
Common mistakes	Adding clarifying products without addressing the input; assuming all cloudiness is bacterial; ignoring the possibility of an incomplete cycle in a new tank.

Scenario L — “I skip feeding one day per week. Is that right?”

Field	Assessment
What we know	Fasting cycles of two days per week in trout produced full compensatory growth with no differences in growth parameters or body composition, alongside lower feed consumption, improved FCR, higher protein retention, and about 11 % lower nitrogen discharge. An eight-week fast in salmon showed no welfare deficit.
What we don't know	Whether fasting provides any health benefit beyond reduced intake. No evidence supports the digestive-rest or constipation-prevention rationales in aquarium fish.
Questions to ask	Are there fry, juveniles, gravid livebearers, very small-bodied species, or sick fish in the tank? Is the fasting day being used to compensate for overfeeding on the other six?
Likely issue	Safe and mildly beneficial for healthy adults. The mechanism is reduced total input, not digestive rest.
Best observation	Whether the day after the fast involves a larger meal — which cancels the benefit and raises peak loading.
Possible adjustment	Continue if the tank contains healthy adults. Do not fast fry, gravid, sick, or nano species. Do not compensate afterward.
Common mistakes	Believing it is physiologically required; fasting fry; escalating to multiple fasting days as a water-quality tool; feeding extra the day after.

Scenario M — “I'm leaving for five days. Do I need a feeder?”

Field	Assessment
What we know	Healthy adult fish tolerate short periods without food; reserves are mobilised and metabolic rate down-regulates. Overfeeding by a well-meaning sitter is more commonly reported as a vacation problem than starvation.
What we don't know	This specific tank's tolerance, which depends on species, size, life stage, and temperature.
Questions to ask	Any fry, juveniles, gravid, sick, or very small-bodied fish? What temperature? Is the tank stable and recently maintained? Are there specialised feeders?
Likely issue	For a tank of healthy adults, five days without food is generally tolerable and probably safer than an untrained sitter.
Best observation	Body condition before leaving, and whether any animal in the tank falls into a higher-need category.
Possible adjustment	For healthy adults: water change and substrate clean before departure, no feeding, no auto-feeder. If higher-need animals are present: pre-portioned sealed meals with a sitter, or a calibrated auto-feeder set to underfeed.
Common mistakes	Feeding heavily before leaving to 'top them up' — it does not work and adds waste; leaving an open container with a sitter; installing an uncalibrated auto-feeder the day of departure.

Scenario N — “I'm leaving for two weeks.”

Field	Assessment
What we know	Two weeks exceeds what is reasonable for most tanks without feeding, particularly at tropical temperatures. Equipment failure risk also rises with duration, and no feeding strategy compensates for a failed heater or filter.
What we don't know	How the system will behave unattended, and whether the chosen feeding method will remain reliable for the full period.
Questions to ask	Who can check the tank physically? Is an auto-feeder available and tested? Are there fry or higher-need animals? Is equipment reliable?
Likely issue	This requires an intervention, and human oversight matters as much as feeding.
Best observation	Auto-feeder performance across several days of testing while still present.
Possible adjustment	Preferred: sitter with pre-portioned sealed meals and written instructions that explicitly forbid adding extra. Alternative: calibrated auto-feeder set to underfeed, plus someone to check physically at least once. Vacation blocks are the least preferred option.
Common mistakes	Relying on a vacation block for two weeks; an untested auto-feeder; a sitter with a container and vague instructions; assuming fish tolerance extends indefinitely.

Scenarios O–V: food choice and species

Scenario O — “Should I feed flakes or pellets?”

Field	Assessment
What we know	There is no evidence that either form is nutritionally superior; both are manufacturing formats and content depends on formulation. Flake leaches water-soluble nutrients very rapidly — large proportions of vitamin C, B ₁₂ , choline, and pantothenic acid within 30 seconds. Pellets portion more precisely and can be engineered to reach a specific feeding zone.
What we don't know	Whether the leaching difference produces a measurable nutritional outcome in a well-fed aquarium fish.
Questions to ask	Where do the fish feed? How large are their mouths? Do you need food to reach the substrate? Do you use an automatic feeder?
Likely issue	Not a quality question at all — a delivery and portion-control question.
Best observation	Whether the food currently used reaches every feeding zone in the tank.
Possible adjustment	Usually both: a staple appropriate to the majority feeding zone plus a genuine sinking food for the bottom. Pellets if precise portioning or an auto-feeder matters; flake crumb for very small mouths.
Common mistakes	Treating flakes as inferior; using a single food form in a tank with multiple feeding zones; buying a large container of either.

Scenario P — “Are frozen bloodworms healthy?”

Field	Assessment
What we know	Chironomid larvae are consistently high in protein — around 76 % in one comparison — but variable in lipid content. They are a single unsupplemented invertebrate species, not a formulated diet. Environmentally collected samples have shown excess lead and low zinc.
What we don't know	Whether the widely repeated bloat claim has any basis. No direct evidence supports it.
Questions to ask	How often are they fed? Are they replacing a meal or being added to it? Are they the staple or a supplement? Is the fish rounding out over time?
Likely issue	Not harmful as a supplement. The real risk is frequency and the habit of adding them on top of the normal ration.
Best observation	Body condition over months, and whether bloodworm days replace or supplement the staple.
Possible adjustment	Keep as an occasional substitution — once or twice weekly at most, replacing that meal's staple, for species that take invertebrate prey.

Field	Assessment
Common mistakes	Feeding daily; using as a staple, especially for bettas; adding on top of normal feeding; handling with bare hands repeatedly — chironomid proteins are a documented human allergen.

Scenario Q — “Do fish need dietary variety?”

Field	Assessment
What we know	Supplementation of formulated feed with live feed improved growth in commercial ornamental production, and specific benefits are documented: <i>Daphnia</i> supplementation significantly increased swordtail broodstock fecundity; <i>Artemia</i> increased juvenile angelfish growth. Requirement data for ornamental species are thin, so any single formulation is an approximation.
What we don't know	Whether variety is <i>necessary</i> when a good complete diet is used, or merely beneficial. This has not been tested directly.
Questions to ask	Is the current staple nutritionally complete and appropriate for the species? Are supplements substituting for meals or adding to them? Is anyone rounding out?
Likely issue	Variety is insurance and enrichment more than necessity — and the most common route to accidental overfeeding.
Best observation	Whether weekly food input increased when variety was introduced.
Possible adjustment	Two or three complementary quality foods, rotated by substitution rather than addition.
Common mistakes	Adding treats to the normal ration; letting a rich supplement become the expected default so the staple is refused; assuming live or frozen automatically means better.

Scenario R — “Can fish live only on flakes?”

Field	Assessment
What we know	A nutritionally complete flake formulated for the species can meet known requirements, and formulated diets were developed specifically to address the deficiency syndromes documented in fish. Classical deficiency disease is uncommon on a modern complete diet.
What we don't know	Whether known requirements are complete requirements for a given ornamental species — vitamin requirements in particular have been studied in very few ornamentals.
Questions to ask	Is the flake complete and species-appropriate? Is it fresh and well stored? Does it reach every fish? How old is the container?
Likely issue	Probably yes for most community fish, with storage and delivery being larger practical risks than formulation.
Best observation	Body condition, colour, and activity over months; and the date the container was opened.
Possible adjustment	Acceptable. Small complementary additions provide insurance at little cost, and a sinking food is needed regardless if benthic fish are present.
Common mistakes	Using a large old container; treating 'complete' as covering species with genuinely different requirements — a strict herbivore or a specialised grazer needs an appropriate formulation.

Scenario S — “Can fish live only on pellets?”

Field	Assessment
What we know	Same as Scenario R. The format is not the determining variable; formulation, freshness, particle size, and delivery are.
What we don't know	Same uncertainties as R.
Questions to ask	Is the pellet size right for every mouth in the tank? Does it reach the substrate if needed? Is it species-appropriate?
Likely issue	Particle size and delivery zone are the practical risks, not nutritional adequacy.
Best observation	Whether any fish repeatedly takes and spits the pellet — a size complaint.
Possible adjustment	Acceptable. Consider two particle sizes in a size-mixed community, and a sinking type if benthic fish are present.

Field	Assessment
Common mistakes	One pellet size for a size-mixed community; floating-only pellets in a tank with bottom dwellers; assuming a 'tropical' pellet suits a strict herbivore or a specialised carnivore.

Scenario T — “Should I feed vegetables?”

Field	Assessment
What we know	Herbivores and omnivores benefit; most plant material is very high in water and low in nutrient density, so the value is largely grazing behaviour and bulk. No fish digests cellulose efficiently.
What we don't know	How much nutritional contribution vegetables actually make relative to prepared herbivore foods.
Questions to ask	Are the species present herbivores, omnivores, or carnivores? Is there a grazing surface already? How long do remnants stay in?
Likely issue	Appropriate for grazers and omnivores; of little value to carnivores.
Best observation	Whether the vegetable is actually being grazed, and whether the remnant is being removed.
Possible adjustment	Blanched zucchini, spinach, or plain nori, weighted or clipped, portioned to the grazers present, removed within about 24 hours, and counted as part of the ration.
Common mistakes	Feeding vegetables to carnivores; leaving remnants in for days — a documented driver of snail booms; seasoned or salted produce; treating vegetable feeding as free.

Scenario U — “My algae eater has no visible algae. What should I feed it?”

Field	Assessment
What we know	'Algae eater' is a commercial category, not a dietary one. Most species sold under it are omnivores or detritivores. A tank with no visible algae is a tank with no food for a genuine algivore.
What we don't know	How much biofilm and microalgal growth the tank actually supports below the threshold of visibility.
Questions to ask	Which species exactly? Is the tank mature with established biofilm? Is the animal's body condition holding? How long has it been in the tank?
Likely issue	The animal is likely underfed and declining slowly. <i>Otocinclus</i> in particular starve quietly in clean tanks.
Best observation	Body condition — abdominal profile from below or from the side against the glass.
Possible adjustment	Species-appropriate supplementation: algae wafers and blanched vegetables for genuine algivores; protein-containing sinking food for carnivorous loricariids such as <i>Hypancistrus</i> ; wood for wood-grazing species. Feed deliberately, not incidentally.
Common mistakes	Assuming the animal is fed by the tank; feeding algae wafers to a carnivorous pleco; buying an algae eater for an algae problem and then eliminating the algae.

Scenario V — “Can bottom feeders just eat leftovers?”

Field	Assessment
What we know	No. Benthic species are deliberate foragers with their own dietary requirements, not a disposal mechanism. The model contains an internal contradiction: if there are enough leftovers to sustain them, the tank is being overfed at the top.
What we don't know	How much incidental food benthic fish actually obtain in a well-managed tank — probably very little.
Questions to ask	What does their body condition look like? Is sinking food being offered deliberately? Are they active and foraging?
Likely issue	The bottom dwellers are underfed, or the tank is overfed, or both in sequence.
Best observation	Body condition of the benthic species specifically, assessed monthly.
Possible adjustment	Deliberate sinking food of appropriate composition, delivered where and when the animals forage. See Parts LIII and XLIII.

Field	Assessment
Common mistakes	Framing any animal as a cleanup crew; buying bottom dwellers to manage waste; assuming an absence of visible leftovers means the bottom is fed.

Scenarios W–Z and additional cases

Scenario W — “My fish spits food out.”

Field	Assessment
What we know	Particle size relative to gape is the most common cause. Aquaculture guidance is to feed the largest particle the fish can consume without exceeding it, and warns that differential access by size causes divergent growth and mortality of smaller fish.
What we don't know	Whether it is size, palatability, unfamiliarity, texture, or normal manipulation behaviour.
Questions to ask	Is the food new? How large is the particle relative to the mouth? Does the fish take it and drop it, or take it and re-take it? Is the food old or oxidised? Is the fish otherwise well?
Likely issue	Usually particle size, especially in bettas, small anabantoids, and nano species.
Best observation	Whether the fish takes and abandons (size or palatability) or takes, repositions, and swallows (normal manipulation in many cichlids and loricariids).
Possible adjustment	Offer a smaller particle, crush the pellet, or soak briefly to soften. Check the food's age.
Common mistakes	Assuming illness; switching foods repeatedly without testing size; concluding the fish is 'fussy.'

Scenario X — “One fish gets fat while the others stay skinny.”

Field	Assessment
What we know	This is the definitive Type 3 signature: the total ration is likely adequate and the distribution is not. Competition, feeding speed, zone, timing, and hierarchy all contribute.
What we don't know	Whether the thin fish are being outcompeted, are ill, or have a different dietary requirement entirely.
Questions to ask	Which fish is fat and which are thin — fast or slow feeders? Surface or benthic? Dominant or subordinate? Is there aggression at feeding? Are the thin fish otherwise healthy?
Likely issue	Distribution failure, possibly with a competition or hierarchy component.
Best observation	Watch a full feeding and record who eats first, who eats last, and who does not appear.
Possible adjustment	Multiple simultaneous feeding locations; broadcast rather than point feeding; target feeding the subordinates; visual barriers; feeding in two stages. Do not increase the total.
Common mistakes	Increasing the ration — which feeds the fat fish more; assuming the thin fish are ill without checking access; adding a feeding ring, which concentrates rather than distributes.

Scenario Y — “My fish won't eat.”

Field	Assessment
What we know	Anorexia is an early, sensitive, non-specific sign of stress, disease, or adverse conditions. Its value is in prompting investigation rather than in identifying a cause.
What we don't know	The cause. Anorexia alone does not narrow the differential.
Questions to ask	What are the water parameters? What is the temperature and is it stable? Is the fish new? Is it being harassed? Has the food changed? Are there other clinical signs — respiration, skin, fins, faeces, buoyancy, posture? Is it spawning or brooding?
Likely issue	Water quality first, then stress, then disease. In that order of prior probability.

Field	Assessment
Best observation	Water test, then temperature, then direct observation at feeding to establish whether the fish refuses food or is prevented from reaching it.
Possible adjustment	Correct whatever the investigation finds. Continue to offer small quantities of an accepted food and remove remnants; a returning appetite is the clearest recovery signal available.
Common mistakes	Changing food repeatedly before testing water; medicating without a diagnosis; withholding food as a treatment; waiting more than a few days without investigating.

Scenario Z — “Should I soak pellets before feeding?”

Field	Assessment
What we know	Dry pellets do absorb water and expand — that much is physically certain. Soaking definitely changes buoyancy and softens the particle. Water-soluble nutrients leach into water rapidly.
What we don't know	Whether pellet expansion causes clinical harm at normal feeding quantities. No controlled study in aquarium fish has been located. The claim is untested rather than disproven.
Questions to ask	Why do you want to soak — to make it sink, to soften it, or because of the expansion claim? How much are you feeding?
Likely issue	Probably a non-issue at appropriate portion sizes. The expansion concern scales with quantity eaten.
Best observation	Whether the fish is being fed an appropriate portion in the first place — which addresses the concern regardless of whether the mechanism is real.
Possible adjustment	Soak briefly if you want the food to sink or to soften it for a fish that struggles with hard particles. Do not soak for long, since nutrients leach out. Aquatic-veterinary guidance advises against letting pellets dissolve before being eaten.
Common mistakes	Extended soaking, which leaches nutrients; treating soaking as a substitute for portion control; applying the same logic to flake, which leaches far faster.

Additional scenarios discovered during research

Scenario AA — “I've had this tank for two years and suddenly my nitrate and algae are worse, but nothing changed.”

Field	Assessment
What we know	Ration as a percentage of body weight falls steeply as fish grow. A tank fed correctly for juveniles is overfed once they are adults, and the transition is invisible because the keeper's routine never changed.
What we don't know	Whether the drift is feeding, reduced export, or accumulated substrate phosphorus being released.
Questions to ask	How big were the fish when you set the portion? How big are they now? Has the water change schedule actually stayed the same? When was the substrate last cleaned thoroughly?
Likely issue	Requirement fell while input stayed constant — a very common and rarely diagnosed pattern.
Best observation	Compare current fish size to the size at which the portion was set, and audit the actual maintenance interval against the intended one.
Possible adjustment	Recalibrate the portion downward per Part CII; increase substrate maintenance, since phosphorus accumulates there in particulate form.
Common mistakes	Looking for a new cause when the cause is a slow drift; blaming the food brand; escalating algae treatments.

Scenario BB — “I added an automatic feeder and now the tank is a mess.”

Field	Assessment
What we know	Auto-feeder timing is reliable; dispensed quantity is not. Quantity varies with food form, humidity, hopper fill level, and mechanism wear, and flake dispenses particularly poorly. A single dispense point also concentrates food in one place.

Field	Assessment
What we don't know	How much is actually being dispensed per cycle, which almost nobody measures.
Questions to ask	What food form is in the hopper? Was the feeder calibrated before use? Where is it mounted relative to the water surface? How many cycles per day?
Likely issue	Over-dispensing, humidity clumping, or both.
Best observation	Dispense several cycles onto a dry surface and look at the actual quantity, compared with what you would have fed by hand.
Possible adjustment	Recalibrate to underfeed; switch from flake to pellet; mount further from the surface; reduce cycles; add a manual sinking feeding for benthic species.
Common mistakes	Trusting dial settings; using flake; setting it and not checking for weeks; assuming an auto-feeder solves distribution.

Scenario CC — “My shrimp used to be fine and now they're dying off since I started feeding more.”

Field	Assessment
What we know	Shrimp are sensitive to water quality and to some treatments. Increased food input raises organic loading and can support heterotrophic bacterial activity and oxygen demand. Shrimp also compete poorly with fish for food.
What we don't know	Whether the deaths relate to water quality, a specific food ingredient, a treatment, or a parameter shift unrelated to feeding.
Questions to ask	What changed exactly, and when? Water parameters including TDS and GH/KH? Any copper-containing food or treatment? Is there uneaten food accumulating? Temperature?
Likely issue	Possibly organic loading and oxygen demand, possibly an unrelated parameter shift. Shrimp die-offs frequently have non-feeding causes.
Best observation	Full water testing, plus whether uneaten food is accumulating in low-flow zones where shrimp congregate.
Possible adjustment	Reduce portion; remove accumulated detritus gently; confirm stable parameters. Shrimp are livestock with their own requirements, not cleanup animals.
Common mistakes	Assuming feeding is the cause without testing; large corrective water changes that shift parameters sharply; adding more food specifically for the shrimp on top of fish feeding.

Scenario DD — “I bought a big tub of food because it was better value.”

Field	Assessment
What we know	Vitamin C is heat-, oxygen-, and light-labile, with stability ranking freezer > refrigerator > opaque room temperature > clear room temperature. Aquatic-veterinary guidance advises discarding a room-temperature container two months after opening. Fish feeds have high protein and oil content that deteriorates rapidly.
What we don't know	How much nutritional degradation a healthy adult community fish actually experiences from moderately old food.
Questions to ask	How much food does this tank use per month? When was the container opened? Where is it stored? Is it clear or opaque?
Likely issue	The container will outlast its nutritional quality by a wide margin.
Best observation	Estimate monthly consumption and divide it into the container size. The result is often measured in years.
Possible adjustment	Buy the smallest container available. If the large one is already purchased: decant into small sealed portions, freeze the reserve, and keep only a small working portion at room temperature.
Common mistakes	Judging food quality by appearance and smell — vitamin loss is invisible; storing food above the tank where it is warm and humid; refrigerating a container that is opened daily, which causes condensation.

Scenario EE — “My cichlid tank has recurring bloat problems.”

Field	Assessment
What we know	Aufwuchs-grazing cichlids such as mbuna and specialised herbivores such as <i>Tropheus</i> have digestive physiology adapted to bulky, low-fat, plant-inclined material taken near-continuously. Hepatic lipidosis is reported clinically in African cichlids. High-fat diets produce hepatic steatosis and immune suppression across many finfish.
What we don't know	The specific aetiology of any individual case. 'Bloat' in cichlid keeping covers several distinct conditions and is not a diagnosis.
Questions to ask	What is the natural diet of these specific species? What protein and fat level is the food? Are high-protein or high-fat treats being fed? How frequently and in what portions? Water parameters and temperature?
Likely issue	Dietary mismatch is a plausible contributor for grazing species fed carnivore-oriented food, alongside water quality and stress.
Best observation	Match the food's protein and fat level against the species' natural dietary strategy, and review treat frequency.
Possible adjustment	Move grazing species to an appropriate plant-inclined, moderate-protein, low-fat formulation; feed smaller portions more frequently; provide grazeable surfaces; eliminate high-fat treats.
Common mistakes	Feeding one 'cichlid food' across a mixed-trophic cichlid tank; treating bloat as a single disease; high-protein feeding for aufwuchs grazers; assuming more protein is better.

Scenario FF — “The fish are fine but my filter needs cleaning twice as often as it used to.”

Field	Assessment
What we know	Mechanical filter loading integrates particulate production over time. Uneaten food, food dust, and faecal solids are the feeding-related contributors. A shortening cleaning interval with unchanged stocking points to increased particulate input.
What we don't know	Whether the increase is from food, substrate disturbance, plant debris, media degradation, or reduced flow.
Questions to ask	Has feeding changed in quantity, form, or type? Has stocking changed? Have plants been trimmed or declined? Is the media itself degraded? Has flow dropped?
Likely issue	Often an unnoticed feeding increase — frequently a switch to a food that crumbles more, or an added supplementary feeding.
Best observation	The cleaning interval itself, tracked over time. This is one of the few overfeeding signals that does not require catching a feeding in progress.
Possible adjustment	Reduce portion; switch away from a food that produces heavy dust; check whether supplements were added rather than substituted.
Common mistakes	Treating it as purely a filter maintenance issue; replacing media unnecessarily; ignoring that reduced flow from a clogged filter also reduces oxygenation.

Part CIV — Myths, Rules and Controversies

Forty-two claims evaluated against the evidence assembled in this document. The classification column states how much support the claim has; the explanation column states what a better formulation would be.

Reading the classifications

A claim classified as folklore is not necessarily harmful, and a claim classified as supported is not necessarily useful. Several of the most persistent rules survive precisely because they fail safe — they are wrong about the mechanism and right about the direction.

Where the evidence is genuinely mixed, the claim is marked contested rather than being resolved artificially in either direction.

Quantity and frequency claims

Claim	Classification	What the evidence says
“Feed only what fish eat in two minutes”	[USEFUL RULE OF THUMB] / [CONTESTED CLAIM]	No primary source has been located, and the same rule circulates as 30 seconds, 1, 2, 3, and 5 minutes. It measures the consumption speed of the fastest feeders, then infers a quantity meant to feed every fish. Works acceptably in a single-species tank of moderate feeders; fails structurally in a mixed community with benthic and nocturnal species. Its real function is to provide a stopping rule, which is genuinely valuable.
“Feed a pinch”	[USEFUL RULE OF THUMB]	Not a unit. Varies by hand size, food form (flake and pellet differ several-fold in mass at equal volume), and tank population. Useful only as a lower bound, and only because it errs small.
“Fish should be fed once per day”	[COMMON AQUARIUM PRACTICE]	Defensible for many adult fish. In one snapper trial, once-daily feeding produced no significant difference in weight gain versus twice daily, with higher nitrogen retention efficiency, lower FCR, and lower total nitrogen waste output. The weakness is distribution — a single delivery window disadvantages slow, shy, benthic, and nocturnal fish.
“Fish should be fed twice per day”	[COMMON AQUARIUM PRACTICE]	Also defensible, and optimal in several published trials including <i>Betta splendens</i> and juvenile snapper. Better for distribution in mixed communities. Both this and once-daily are reasonable; the failure is adding a second feeding without halving the portion.
“More frequent feeding produces better growth”	[CONTESTED CLAIM]	True only up to a species- and stage-specific optimum, after which growth plateaus and then declines. A meta-analysis found the relationship quadratic. In <i>Betta splendens</i> , three meals daily produced both less intake and less gain than two.
“A fish's stomach is the size of its eye”	[UNSUPPORTED AQUARIUM FOLKLORE]	No reliable source supports the anatomical claim, and roughly a quarter of teleost fishes — including all cyprinids — have no stomach at all. As a conservative portion heuristic for a small insectivore it is harmless and points in the right direction.
“One pinch per tank”	[UNSUPPORTED AQUARIUM FOLKLORE]	Requirement scales with total biomass and body size, not with the existence of a tank. Ten small tetras and one large cichlid differ by more than an order of magnitude in required ration.
“Fish will stop eating when they are full”	[CONTESTED CLAIM]	Many aquarium fish are opportunistic feeders whose suction-feeding reflex triggers on food presence rather than being gated by satiety, having evolved under unpredictable food availability. Interest reliably outlasts need.
“Fish are always hungry”	[UNSUPPORTED AQUARIUM FOLKLORE]	The behaviour is predominantly conditioned food-anticipatory activity — a circadian phenomenon that persists when food is withheld, in one zebrafish study in 78 % of shoals — combined with rapid conditioning to the keeper's approach.

Fasting and going without food

Claim	Classification	What the evidence says
“Fish need one fasting day per week”	[CONTESTED CLAIM]	Safe and well tolerated: two fasting days per week in trout produced full compensatory growth with no differences in growth or body composition, plus lower feed consumption, improved FCR, higher protein retention, and about 11 % lower nitrogen discharge. But 'need' is unsupported — the benefit is reduced input and waste, not a physiological requirement.
“Fasting gives the digestive system a rest”	[UNSUPPORTED AQUARIUM FOLKLORE]	No study demonstrates a requirement for periodic digestive rest in fish. The phrase imports a mammalian intuition that has not been tested in this context.
“Fasting prevents constipation”	[UNSUPPORTED AQUARIUM FOLKLORE]	No published evidence links a weekly fasting day to reduced digestive impaction in aquarium fish.
“Fasting prevents obesity”	[REASONABLE INFERENCE]	Plausible and probably true, but the mechanism is simply reduced weekly energy intake — roughly 14 % — rather than anything specific to fasting as a practice.
“Healthy fish can go a week without food”	[CONTESTED CLAIM]	Broadly true as a survival statement for healthy adults of moderate size. It is a poor husbandry statement, because survival is not the standard being asked about, and it does not hold for fry, juveniles, nano species, gravid livebearers, or sick fish.
“All fish can be fasted safely”	[UNSUPPORTED AQUARIUM FOLKLORE]	Early-stage feed restriction produced long-term growth inhibition in turbot, and reviews caution that restriction at juvenile stages must be carefully evaluated. Fry, gravid, sick, and very small-bodied fish are poor candidates.
“A hungry fish is a healthy fish”	[UNSUPPORTED AQUARIUM FOLKLORE]	Confuses a conditioned feeding response with a health indicator. A fish that is genuinely energy-deficient is losing condition, which is the opposite of healthy.
“Underfeeding is safer than overfeeding”	[CONTESTED CLAIM]	Directionally sound for healthy adults, where reserves buffer a short deficit. Actively wrong for fry and juveniles, where restriction can cause permanent stunting; for very small-bodied species with minimal reserves; and in any tank with a distribution problem, where reducing the total starves the subordinate fish first.

Food type claims

Claim	Classification	What the evidence says
“Flakes are junk food”	[UNSUPPORTED AQUARIUM FOLKLORE]	Flake and pellet are manufacturing formats; nutritional content depends on formulation. Flake does leach water-soluble nutrients faster — large proportions of vitamin C, B ₁₂ , choline, and pantothenic acid within 30 seconds — which is an argument for feeding food that is eaten quickly, not for the format being inferior.
“Pellets are always better”	[CONTESTED CLAIM]	Pellets portion more precisely, leach more slowly, and can be engineered for a specific feeding zone. Those are advantages of control, not of nutrition. No general nutritional superiority is demonstrated.
“Frozen food is better than dry food”	[CONTESTED CLAIM]	Frozen invertebrates are single ingredients and therefore not complete diets; prepared foods are formulated against a nutrient specification. Frozen foods offer palatability, natural form, and enrichment, and supplementation demonstrably improved growth and fecundity in some ornamental species. Excellent supplements, poor staples.
“Live food is always best”	[CONTESTED CLAIM]	Live feed improves growth in commercial ornamental production and provides genuine enrichment, but it is described in the same literature as often nutritionally deficient and as a potential transmitter of parasitic, bacterial, and viral disease. <i>Artemia</i> nutritional value varies geographically and temporally, which is why commercial enrichment exists at all.
“Bloodworms should be fed every day”	[UNSUPPORTED AQUARIUM FOLKLORE]	A single unsupplemented invertebrate species with variable lipid content is not a complete diet for any fish. Best treated as an occasional substitution.
“Bloodworms cause bloat”	[CONTESTED CLAIM]	No direct evidence. The association is plausibly confounded: bloodworms are highly palatable, energy-dense, and typically fed generously and in addition to the normal ration. A very large meal of anything produces distension.

Claim	Classification	What the evidence says
“Freeze-dried food must always be soaked”	[CONTESTED CLAIM]	No controlled study has been located. Soaking does make the food sink, which is a real reason to do it, and it does leach water-soluble nutrients, which is a real reason not to do it for long. The expansion rationale is untested.
“Dry pellets swell inside fish and cause problems”	[CONTESTED CLAIM]	Pellets certainly absorb water and expand; whether this causes clinical harm at normal feeding quantities has not been demonstrated in aquarium fish. Portion control addresses the concern regardless of whether the mechanism is real.
“Peas cure constipation”	[CONTESTED CLAIM]	The fibre mechanism does not hold: shelled green peas contain about 5 % fibre, the same as many prepared fish foods, and no fish digests cellulose efficiently. Aquatic-veterinary commentary notes goldfish metabolism is not robust enough to process large amounts of fibre. The plausible effect is substitution for a heavy meal — closer to a short fast. Inappropriate for carnivores, which handle carbohydrate poorly.
“Daphnia relieves constipation”	[CONTESTED CLAIM]	No controlled evidence located. The mechanism is at least coherent — chitin is poorly digested and adds indigestible bulk, and carnivores will actually eat it, unlike a pea. Low-energy, palatable, and unlikely to do harm; the specific therapeutic claim is untested.
“Garlic cures parasites”	[CONTESTED CLAIM]	Garlic has a genuine research literature with reported antiparasitic, antimicrobial, and immunostimulatory effects at dosages of roughly 0.05–40 g/kg feed. But antiparasitic demonstrations are predominantly immersion baths, effect sizes are partial (14–20 % reduction in one carp study), and dietary effects required 60 days of conditioning in one trial. Garlic extract also has measurable toxicity. Reasonable as an appetite stimulant; not a treatment for an active outbreak.
“Colour-enhancing food really works”	[STRONG EVIDENCE]	One of the few marketing claims the evidence supports. Fish cannot synthesise carotenoids and depend on dietary supply. Astaxanthin, canthaxanthin, and paprika oleoresin all significantly improved red colouration in goldfish trials. But source matters — goldfish metabolise very little beta-carotene and no lutein to astaxanthin — deposition saturates, and growth and feed efficiency were unaffected.
“Spirulina is a superfood for fish”	[CONTESTED CLAIM]	A legitimate carotenoid source and a good plant protein component for herbivore formulations, with demonstrated pigmentation effects in fancy carp. The general health and immune claims attached to it in marketing exceed the evidence.

Species, behaviour and cleanup claims

Claim	Classification	What the evidence says
“Floating food causes swim bladder disease”	[CONTESTED CLAIM]	Partly supported and species-specific. Goldfish and koi are physostomous, and veterinary sources state air swallowed during surface feeding can enter the swim bladder, with sinking food sometimes helping in mild cases. But buoyancy disorders have many causes — polycystic kidneys, gut gas, fluid, rupture, infection, neoplasia, injury, and the globoid body shape of fancy varieties — X-ray is the main diagnostic route, and many cases are untreatable.
“Bottom feeders eat the leftovers”	[UNSUPPORTED AQUARIUM FOLKLORE]	Contains an internal contradiction: if there are enough leftovers to sustain them, the tank is overfed at the surface; if the surface is fed correctly, the bottom dwellers are starved. Both branches are husbandry failures.
“Algae eaters don't need feeding”	[UNSUPPORTED AQUARIUM FOLKLORE]	Aquarium algae supply is variable, usually declining because the keeper is removing it, and most animals sold under this label are omnivores or detritivores rather than obligate algivores. <i>Otocinclus</i> in particular starve slowly in clean tanks.
“Plecos eat fish waste”	[UNSUPPORTED AQUARIUM FOLKLORE]	Faeces are the residue of digestion. A grazing pleco may incidentally ingest detritus containing faecal material along with biofilm and microorganisms; it is grazing the microbial community, not consuming waste as food.
“Fish will eat poop”	[UNSUPPORTED AQUARIUM FOLKLORE]	Same reasoning. No aquarium animal reduces the system's nutrient load by eating waste — nutrients passing through a second animal remain in the system.

Claim	Classification	What the evidence says
“Plecos need wood because they digest it”	[UNSUPPORTED AQUARIUM FOLKLORE]	Wild <i>Panaque</i> gut contents are up to about 75 % wood, but studies of transit time, enzyme activity, and fermentative end products concluded they do not obtain energy from digesting it. Wood passed through the gut in under four hours with no significant cellulose assimilation, and the fish did not thrive on a woody diet in the laboratory. No fish species has been shown to digest cellulose efficiently. Wood supports natural grazing behaviour, not nutrition.
“Corydoras clean the bottom”	[UNSUPPORTED AQUARIUM FOLKLORE]	They are benthic carnivores and omnivores that forage for invertebrates and organic particles, and they require deliberate feeding with sinking food.
“Cleanup crew animals reduce bioload”	[UNSUPPORTED AQUARIUM FOLKLORE]	Adding animals adds metabolic waste. Detritivores redistribute and process organic matter and make some of it more available to filtration; they do not remove nitrogen or phosphorus from the system. Only export removes nutrients.
“Mollies will clear your algae”	[CONTESTED CLAIM]	Mollies are omnivores with a substantial plant fraction that will graze algae opportunistically. They are not algae control equipment and require a full omnivore diet.
“A planted tank can feed its own fish”	[CONTESTED CLAIM]	A mature, densely planted, lightly stocked tank supplies meaningful food for biofilm grazers, shrimp, snails, and fry. It supplies very little for midwater particulate feeders and essentially nothing for carnivores — which covers most community fish. Natural food production is also partly a recycling of organic input the keeper provided.

Nutrition and water quality claims

Claim	Classification	What the evidence says
“More food makes fish grow faster”	[CONTESTED CLAIM]	Only up to a plateau. In mandarin fish, growth increased to 2.5 % BW/day with no further gain above, while feed efficiency and protein retention <i>fell</i> at 3.0 %. In largemouth bass, the lowest rate tested gave the best FCR with no difference in final biomass. Above the plateau, extra food becomes waste.
“High protein is always better”	[UNSUPPORTED AQUARIUM FOLKLORE]	Protein above requirement is deaminated for energy, producing ammonia. Sales and Janssens warn specifically that high dietary protein levels needed for growth of juveniles in ponds present nitrogen breakdown problems with adult fish in small home aquaria. Requirements range from about 29 % for juvenile goldfish to about 50 % for discus.
“Fish cannot digest carbohydrates”	[UNSUPPORTED AQUARIUM FOLKLORE]	Warmwater omnivores and herbivores use carbohydrate well — about 40 % dietary carbohydrate has been recommended for tilapia and goldfish carbohydrate digestibility of around 70 % reported. Carnivorous and coldwater species use it poorly, tolerating roughly 20 % or less. The species determines the answer.
“Plant ingredients are filler”	[CONTESTED CLAIM]	Digestible carbohydrate is a protein-sparing energy source for omnivores and herbivores, which reduces ammonia output. It is genuinely poorly used by carnivores. Same ingredient, opposite verdict depending on the fish.
“The first ingredient must be whole fish”	[UNSUPPORTED AQUARIUM FOLKLORE]	Ingredient order is by as-added weight and is dominated by moisture content. A product listing whole fish first may contain less fish-derived protein than one listing fish meal first.
“Fish meal means low quality”	[UNSUPPORTED AQUARIUM FOLKLORE]	‘Meal’ describes drying and grinding, not a quality tier. Fish meal is the benchmark against which replacement ingredients are measured, and the research programme to replace it is driven by cost and sustainability rather than by nutritional inadequacy.
“Variety is always better”	[CONTESTED CLAIM]	Variety provides nutritional insurance against thin ornamental requirement data, and enrichment, with demonstrated fecundity and growth benefits in specific cases. It is also the single most common route to accidental overfeeding, because supplements are added to rather than substituted for the ration.
“One complete pellet is all a fish ever needs”	[CONTESTED CLAIM]	Probably adequate for most community fish, given that formulated diets were developed specifically to prevent documented deficiency syndromes. But vitamin requirements have been studied in very few ornamental species, and one food cannot serve a community containing herbivores, omnivores, and carnivores at different feeding zones.

Claim	Classification	What the evidence says
“Overfeeding causes algae”	[CONTESTED CLAIM]	Overfeeding raises nitrogen, phosphorus, and dissolved organic carbon availability, which removes one brake on algae. Whether algae expresses depends on light, CO ₂ , competing plant biomass, and stability. Reducing food alone frequently fails to resolve an outbreak.
“Overfeeding causes pest snail explosions”	[REASONABLE INFERENCE]	Snail populations are food-limited and respond to surplus organic matter with a lag. But the surplus may come from decaying plants, a dead animal, vegetables left too long, or reduced maintenance. A boom is a prompt to audit, not a verdict.
“Overfeeding is the number one cause of aquarium problems”	[CONTESTED CLAIM]	Asserted constantly with no supporting data. Plausible that it is a common contributor, but inadequate cycling, overstocking, insufficient water changes, unstable temperature, and incompatible stocking are all also common, and no one has ranked them. The practical danger is single-variable diagnosis.
“Feeding less means fewer water changes”	[CONTESTED CLAIM]	Arithmetically true — less input means slower accumulation — but it inverts the dependency. Feeding is determined by what the animals need; maintenance is then set to handle the resulting waste. Water changes also dilute dissolved organics and replenish minerals, which feeding reduction does not address.

Additional claims identified during research

Claim	Classification	What the evidence says
“Old fish food is fine as long as it looks and smells OK”	[UNSUPPORTED AQUARIUM FOLKLORE]	Vitamin degradation is invisible and odourless. Vitamin C is heat-, oxygen-, and light-labile, and aquatic-veterinary guidance recommends discarding a room-temperature container two months after opening.
“Buying a big container of food is better value”	[UNSUPPORTED AQUARIUM FOLKLORE]	A small community tank may use 15–30 g of dry food per year; a 250 g canister would take most of a decade to consume. It is not thrift; it is buying years of progressively oxidising food.
“Feeder fish are a natural diet for predators”	[UNSUPPORTED AQUARIUM FOLKLORE]	Feeder goldfish and rosy red minnows are cyprinids and contain thiaminase, which degrades vitamin B ₁ . Historically, predatory fish maintained on live feeders showed poor health and premature death from parasites, bacterial infection, and thiamine deficiency.
“Thiaminase forms after death, so live fish are safe”	[UNSUPPORTED AQUARIUM FOLKLORE]	Thiaminase content is a property of the species. It does not change with whether the animal is alive, dead, fresh, or frozen, and it does not sort by freshwater versus marine origin.
“Herbivores need to eat constantly”	[CONTESTED CLAIM]	Directionally sound for wild herbivores processing bulky low-energy plant material, but it does not transfer to a fish fed concentrated prepared food with five to ten times the energy density. The defensible version is continuous access to grazeable material plus a modest measured ration.
“Carnivores should eat less often”	[CONTESTED CLAIM]	True for large-prey predators, where veterinary guidance permits once or twice weekly feeding. False for insectivores such as bettas, which are also carnivores and take many tiny prey items. The operative variable is meal size relative to gut capacity, not whether the prey was animal or plant.
“Feeding at the same time every day over-conditions fish”	[UNSUPPORTED AQUARIUM FOLKLORE]	Consistent scheduling produces entrainment, which improves consumption efficiency and gives the keeper a reliable daily health check. Random feeding disrupted energy homeostasis in goldfish in chronodisruption work.
“You should feed fish right when the lights come on”	[USEFUL RULE OF THUMB]	No supporting study located. Some sources recommend the lights be on 30 minutes before feeding so fish are fully active and the keeper can observe. Sensible, but presented with more confidence than the evidence warrants.
“Vacation blocks are a reliable way to feed while away”	[CONTESTED CLAIM]	Blocks add a slowly dissolving organic and mineral mass with no mechanism to stop if the fish ignore them. Formulations vary and independent testing is not available for most products. They rank last in the practical hierarchy of vacation options.
“Automatic feeders are set and forget”	[UNSUPPORTED AQUARIUM FOLKLORE]	Timing is reliable; quantity is not. Dispensed amount varies with food form, humidity, hopper fill, and wear. Flake dispenses particularly poorly. Calibration across several days before reliance is essential.

Claim	Classification	What the evidence says
“Insect-based foods are always better and more sustainable”	[CONTESTED CLAIM]	Black soldier fly larvae meal is a well-evidenced partial replacement with real sustainability advantages and a demonstrated result in goldfish at up to 15 % inclusion. But chitin content (3–10 %) limits inclusion in a species-dependent way — 75 % replacement caused intestinal damage in Jian carp, and turbot growth fell at 1.47 % dietary chitin.
“If there's no food left over, you fed the right amount”	[UNSUPPORTED AQUARIUM FOLKLORE]	This detects only uneaten food. A tank in which fish eat everything but eat more than they need still delivers all that surplus nitrogen and phosphorus to the water — via the fish. It is also equally consistent with bottom dwellers receiving nothing.

Part CV — Current Hobbyist Questions and Pain Points

Drawn from public aquarium forums, community discussion, retail and manufacturer educational content, and hobby publications. Community discussion identifies what confuses people; it is not evidence about fish.

How this section should be read

Everything below describes **what hobbyists are confused about**, not what is true. Forum and comment consensus is used here strictly as a map of the confusion, and the “evidence” column in each entry points back to the research sections that address it.

The recurring structural cause of the confusion is worth naming once: the hobby answers a *distribution* question with a *quantity* answer, and answers an *uncertain* question with a *confident* number.

The question	Why hobbyists are confused	Conflicting advice they encounter	What the evidence suggests	What remains uncertain
How much should I actually feed?	No source gives a usable unit. “A pinch” and “two minutes” are not measurements, and the person asking usually cannot tell whether their current amount is right.	Two minutes, three minutes, four to five minutes, five minutes, “three flakes per fish”, “a bite or two”, and “as much as they’ll eat” all appear in mainstream sources. Some experienced keepers explicitly say the five-minute rule is far too much.	Requirement scales with biomass, size, stage, temperature, and food density. Measured maintenance for a neon tetra is under 4 mg dry feed per day; veterinary guidance places adult maintenance near 3 % of body weight. Parts III–V.	The correct quantity for any specific mixed community. No published figure covers it.
Am I overfeeding? There’s no food left over.	The main taught signal — leftover food — only detects one of three overfeeding modes. Keepers with clean substrates and rising nitrate have no framework for what is happening.	“If they eat it all, you’re fine” versus “overfeeding is the number one problem in the hobby.” Both are widely repeated and they contradict each other in this exact case.	Type 2 overfeeding — excess consumed food — delivers surplus nitrogen and phosphorus via the fish rather than via decomposition. Part XIII.	The quantitative relationship between a given degree of surplus intake and a given rate of nitrate rise in aquaria.
My fish act starving all the time.	The behaviour is extremely convincing and the standard answer (“they’re always hungry, ignore it”) feels dismissive and offers no way to tell the difference.	“Fish are opportunistic, ignore begging” versus “if they’re hungry, feed them — skinny fish are worse than a dirty tank.”	Food-anticipatory activity is a circadian phenomenon that persists when food is withheld; fish condition rapidly to keeper approach. Body condition is the signal that separates learned behaviour from need. Parts XXI–XXII.	Nothing substantial — this is one of the better-evidenced answers available.
Should I fast my fish once a week?	The practice is near-universal and the rationale offered varies wildly, from digestive rest to constipation prevention to “it’s natural.”	“Essential for digestive health” versus “completely unnecessary if you feed properly” versus “cruel.” All three are stated confidently by experienced keepers.	Fasting cycles are well tolerated and demonstrably reduce feed use and nitrogen discharge, but no evidence supports the digestive-rest or constipation rationales. It is an anti-overfeeding tool. Part X.	Whether any health benefit exists beyond reduced intake; the safe duration for very small-bodied ornamental species.
My corydoras / kuhli loaches / pleco aren’t getting food.	The keeper is doing everything the general advice says and the bottom dwellers are still thin. The advice they have received (“they’ll eat the leftovers”) is actively wrong.	“Bottom feeders clean up” versus “you must target feed them.” Retail advice and experienced-keeper advice diverge sharply here.	Benthic species require deliberate sinking food delivered where and when they forage. The leftovers model contains an internal contradiction. Parts XLIII, LIII–LIV.	How much incidental food benthic species actually obtain in a well-managed tank. Probably very little, but unquantified.
One fish is fat and the others are thin.	Quantity advice offers no help at all, because the total may already be correct. The keeper’s instinct — feed more — makes it worse.	“Feed more so everyone gets some” versus “you’re overfeeding, that’s why one is fat.” Both responses are common and both miss the mechanism.	This is the definitive distribution-failure signature. The fix is delivery — multiple locations, staged feeding, target feeding — not quantity. Parts XX, LV, XCIV.	Nothing major; the mechanism is well understood, just poorly communicated.

The question	Why hobbyists are confused	Conflicting advice they encounter	What the evidence suggests	What remains uncertain
Flakes or pellets? Which is better?	Framed as a quality question when it is a delivery question. Keepers spend a lot of attention here and very little on whether food reaches every zone.	“Flakes are junk, pellets are premium” is close to received wisdom in enthusiast spaces, with little supporting argument beyond leaching.	Both are manufacturing formats; content depends on formulation. Flake leaches faster; pellets portion better and can be engineered for a feeding zone. Parts XXX–XXXI.	Whether the leaching difference produces a measurable nutritional outcome in practice.
Are bloodworms bad for my fish?	Bloodworms are the most popular supplementary food and simultaneously the most warned-against, with no clear guidance on frequency.	“Bloodworms cause bloat and should be rare” versus “the best food for bettas, feed them often.”	High protein (around 76 %) but variable lipid content; not a complete diet; no direct evidence for the bloat claim. The real risk is frequency and additive feeding. Part XXXIV.	Whether the bloat association has any mechanism beyond very large meals; the actual lipid variability of commercial product.
Do my fish need variety?	Variety is presented as unambiguously good, and keepers implement it by adding foods rather than substituting, which quietly increases weekly intake substantially.	“Variety is essential” versus “a good staple is all they need” versus “variety causes overfeeding.”	Variety is insurance against thin ornamental requirement data plus enrichment, with specific demonstrated benefits in some species. It is also the most common accidental overfeeding route. Substitute, don't add. Part XXIII.	Whether variety is necessary against a genuinely complete species-appropriate staple.
How do I feed while I'm away?	Advice ranges from “they'll be fine” to elaborate arrangements, with no framework for matching the method to the absence length or the livestock.	“A week is nothing” versus “always use a feeder” versus “never trust an auto-feeder.”	Healthy adults tolerate short absences; overfeeding by a well-meaning sitter is the more commonly reported failure. Pre-portioning is the highest-leverage single change. Part XII.	Tolerance duration for specific small-bodied ornamental species.
My algae eater has nothing to eat.	Keepers buy the animal to solve algae, succeed, and then have an animal whose food supply they eliminated. The retail framing creates the problem.	“They'll find enough” versus “otos starve in most tanks.” The second is increasingly the enthusiast consensus and rarely reaches new keepers.	Most animals sold as algae eaters are omnivores or detritivores; genuine algivores need continuous soft algae and biofilm. All require supplementation. Part XLI.	How much biofilm a mature tank supports below the threshold of visibility, and what that means for a small grazer's requirement.
How long can I leave food in the tank?	No consistent guidance, and the answer clearly differs by food form — which almost no source acknowledges.	“Remove after two minutes” versus “leave wafers overnight” versus “vegetables for 24 hours.” All can be right; they are rarely presented together.	Dwell time depends on leaching rate, structural integrity, and which animal it is for. Veterinary guidance advises pellets should not be allowed to dissolve before being eaten. Part LXIII.	Decomposition and leaching rates for specific commercial products.
Should I soak pellets or freeze-dried food?	Stated as a rule with a confident mechanism that turns out to be untested, and with a real cost (nutrient leaching) that is never mentioned.	“Always soak, or it swells inside them” versus “unnecessary, and it washes out the vitamins.”	Expansion is real; harm at normal quantities is undemonstrated. Soaking does make food sink and does leach nutrients. Portion control addresses the concern regardless. Parts XXXVII, LXI.	Whether expansion causes clinical problems at feeding-relevant quantities in any species.
My nitrate keeps climbing and I don't know why.	Feeding is blamed first because it is the most-discussed cause, while source water and a quietly lengthened maintenance interval go unchecked.	“You're overfeeding” is the default forum response regardless of the details supplied.	Food is usually the dominant input, but reduced export is frequently the actual change. Source water nitrate is the single most overlooked test. Part LXVII.	Nothing conceptually — the failure is diagnostic sequence, not missing knowledge.

Structural patterns in the confusion

1. **Quantity answers to distribution questions.** The most frequently asked question (“how much?”) and the most frequently occurring problem (“some fish aren't getting any”) are different problems, and the hobby's stock answer addresses only the first.
2. **Confident numbers for uncertain quantities.** The proliferation of incompatible versions of the two-minute rule is direct evidence that the number was never measured, yet each version is stated without hedging.

3. **Mechanisms invented to justify practices that work for other reasons.** Fasting (digestive rest), peas (fibre), and soaking (expansion) all have plausible-sounding mechanisms that the evidence does not support, attached to practices that are broadly harmless or mildly beneficial for different reasons.
4. **Commercial categories treated as biological ones.** “Algae eater”, “bottom feeder”, “cleanup crew”, and “cichlid food” are retail groupings that encode husbandry assumptions which are often false.
5. **Single-variable diagnosis.** Rising nitrate, algae, cloudiness, and snails are each attributed to overfeeding without investigating maintenance, stocking, source water, or plant health.
6. **Underfeeding treated as costless.** “Err on the side of less” is repeated without the exceptions — fry, nano species, gravid fish, and tanks with distribution problems — where it is wrong.

Part CVI — Search Intent Findings

Search phrasings observed in public discussion, question sites, and content targeting. No search volume figures are given, because none were independently verified.

A note on this section's limits

The phrasings below are collected from how the questions are actually written in public forums, question sites, and existing content titles. They are a reasonable guide to natural language phrasing and question intent.

They are **not** accompanied by search volume, difficulty, or ranking data, because that data was not independently obtained and inventing it would be worse than omitting it.

Cluster	Observed phrasings	Intent type	Content implication
Quantity	how much should I feed my fish · how much food should I feed my fish · how much to feed aquarium fish · how much should I feed tropical fish · how much food for betta · how many pellets to feed betta	Informational, immediate, expects a number	The honest answer disappoints this intent unless it is reframed fast. Lead with what makes the number impossible, then give the method
Frequency	how often should I feed aquarium fish · how often should I feed my fish · how often feed betta · should I feed my fish twice a day · how often should I feed bloodworms	Informational, expects a schedule	Better evidenced than quantity; once or twice daily is defensible and can be stated with reasonable confidence
Overfeeding	can you overfeed fish · how do I know if I am overfeeding fish · signs of overfeeding fish · what happens if you overfeed fish · overfeeding fish symptoms	Diagnostic, often triggered by a visible problem	High-value cluster. The three-types framing (Part XIII) is the differentiator most existing content lacks
Underfeeding and hunger	how do I know if my fish are hungry · why are my fish always hungry · are my fish underfed · signs fish are not getting enough food · skinny fish aquarium	Diagnostic, often anxious	The FAA explanation is genuinely satisfying and well evidenced; pair it with body condition assessment
Fasting	should I fast my fish · fish fasting day · do fish need a fasting day · is it bad to skip feeding fish	Informational, seeking permission or validation	Reframing fasting as portion control rather than digestive rest is a differentiated angle
Absence	how long can aquarium fish go without food · aquarium fish vacation feeding · leaving fish for a week · automatic fish feeder · vacation feeder block review	Transactional and time-pressured	Distinguish survival from husbandry; the sitter-overfeeding point is counterintuitive and memorable
Variety	do fish need variety in their diet · can fish live on flakes only · best food for aquarium fish · do I need more than one fish food	Informational, comparison-oriented	Substitute-don't-add is the practical takeaway most content misses
Food type comparison	flakes vs pellets fish · are frozen bloodworms good for fish · freeze dried vs frozen fish food · best betta food · are algae wafers good	Commercial investigation	Delivery-not-quality reframing; avoid declaring a winner
Specific foods	should I feed fish vegetables · can fish eat peas · do peas help constipation fish · is garlic good for fish · what vegetables can fish eat	Informational, often remedy-seeking	The pea and garlic sections are the strongest myth-correction material in the document
Bottom dwellers	what do bottom feeders eat · do corydoras eat leftovers · what do plecos eat · how to feed kuhli loaches · what to feed otocinclus · sinking food for corydoras	Informational, husbandry-specific	High-value and underserved. The 'leftovers' contradiction is a strong hook
Community feeding	how to feed shy fish · one fish eats all the food · how to feed bottom feeders in community tank · feeding nocturnal fish	Problem-solving	The distribution framing is the document's most practically useful contribution

Cluster	Observed phrasings	Intent type	Content implication
Water quality link	does overfeeding cause algae · overfeeding cloudy water · overfeeding and nitrate · too much food snails aquarium · does feeding less keep tank clean	Diagnostic, system-level	Connects directly to the secondary theme and to the existing cloudy-water cluster
Storage	does fish food expire · how long does fish food last · should I refrigerate fish food · old fish food safe	Informational, low-competition	Genuinely underserved, with clear veterinary guidance available — a likely high-value segment
Labels	how to read fish food label · what should be in fish food · is fish meal bad · best fish food ingredients	Commercial investigation	The ingredient-order correction is contrarian and defensible

Question phrasings the eventual content should answer verbatim

These are the natural-language forms most likely to be typed or spoken, and the document should contain each as an explicit heading or answerable statement so that retrieval matches directly.

- How much should I feed my fish? · How often should I feed my fish? · Is the two-minute rule real?
- How do I know if I'm overfeeding? · How do I know if I'm underfeeding? · Why do my fish always act hungry?
- Should fish have a fasting day? · How long can fish go without food? · What do I do about feeding on vacation?
- Do fish need variety? · Can fish live on one food? · Are flakes worse than pellets?
- Are bloodworms healthy? · Do peas cure constipation? · Does garlic treat parasites? · Should I soak pellets?
- Do corydoras eat leftovers? · What do plecos actually eat? · What should I feed an algae eater with no algae?
- How do I make sure the shy fish gets food? · How do I feed nocturnal fish?
- Does overfeeding cause algae? · Does overfeeding cause cloudy water? · Does feeding less mean fewer water changes?
- Does fish food go bad? · Should I refrigerate fish food? · How do I read a fish food label?

Part CVII — Additional Questions Discovered

Questions this research surfaced that the original brief did not name, each of which materially affects the eventual episode.

Question	Why it matters	What the evidence suggests	Uncertainty
How much of the food's nutritional value survives the journey from factory to fish?	Reframes food quality from a purchasing decision into a handling decision. A keeper cannot change a formulation but can change storage and feeding speed.	Substantial losses at every stage: approximately 75 % of supplemented vitamin C was lost during processing in one shrimp feed study; storage degrades further; a large proportion of several water-soluble vitamins left commercial flake within 30 seconds of water contact.	Cumulative loss has not been measured end to end for a specific aquarium product.
Does feeding frequency affect water quality independently of total quantity?	If yes, frequency is a water-quality lever and not only a nutrition one — which changes the advice.	Yes. In a snapper trial, once-daily feeding gave higher nitrogen retention efficiency, lower FCR, and lower total nitrogen waste output than twice-daily or alternate-day regimes at comparable intake.	Whether this transfers to small ornamental species in closed systems.
What happens to feeding requirements as a tank's fish mature?	Explains the extremely common pattern of a stable tank that 'suddenly' develops nutrient problems after a year or two with no change in routine.	Ration as a percentage of body weight falls steeply with size — roughly 3.6 % to 0.7 % across the size range in a standard feeding chart. A tank fed correctly for juveniles is overfed once they are adults.	The magnitude of this drift in a typical mixed community over typical timescales.
Is the community aquarium a fundamentally unsolvable nutritional problem?	If the honest answer is 'approximately yes,' that changes how confidently any feeding advice should be delivered.	Sales and Janssens concluded that rather than satisfying individual species' needs, the practical approach is an intermediate feed that minimises deviation from the crude requirements of all tank inhabitants — an explicit acknowledgement that every community diet is a compromise.	How large the compromise cost actually is in welfare terms.
Does a fish's appetite reliably track its nutritional needs?	The entire 'feed until they stop' family of rules depends on this, and it appears to be false for many aquarium species.	Opportunistic feeders evolved under unpredictable food availability; the suction-feeding reflex triggers on food presence. Fish will feed past requirement, and satiation feeding produced higher body lipid than restricted feeding in tilapia.	Species variation in satiety regulation among ornamental fish.
Is body condition scoring formalised for ornamental fish?	It is the most useful hobbyist-accessible measurement and there is no standard for it.	Assessment is standard in aquatic veterinary and zoological practice and is explicitly recommended, but no standardised numerical scoring system for ornamental species was located.	This is a genuine gap and arguably an opportunity for the brand — a species-calibrated visual condition reference does not appear to exist for hobbyists.
How much do fish actually obtain from a mature tank's natural production?	Determines whether ecosystem-tank claims have any quantitative basis.	Meaningful for biofilm grazers, shrimp, snails, and fry; negligible for midwater particulate feeders and carnivores. Natural production is also partly a recycling of the keeper's own organic input rather than an independent source.	No quantification exists for aquarium-scale systems.
Does the phosphorus fraction of food behave differently from nitrogen in aquarium maintenance?	If phosphorus is largely particulate, water changes alone do not export it — which changes maintenance advice.	Yes. Most nitrogen waste is dissolved (60–90 % as ammonia) while a larger share of phosphorus is particulate in faeces (reported 25–85 %), accumulating in substrate and detritus.	Aquarium-specific partitioning has not been measured.
Are freeze-dried and frozen foods equivalent nutritionally?	Keepers treat them as interchangeable and choose on convenience.	Both are typically single-ingredient invertebrate products and neither is a complete diet. Freeze-drying is shelf-stable and removes living organisms but is not a validated sterilisation step; frozen product retains more palatability for some fish.	Direct nutritional comparison of matched commercial products was not located.

Question	Why it matters	What the evidence suggests	Uncertainty
Does feeding technique change competitive behaviour, or only food access?	If technique changes behaviour, feeding management becomes a welfare tool rather than only a nutrition one.	Multiple simultaneous feeding locations reduce a dominant fish's ability to monopolise, and scarcity intensifies competition. Both are behavioural effects rather than pure access effects.	Not studied systematically in ornamental community contexts.
Should coolwater species kept at tropical temperatures be fed differently?	Very common situation — goldfish, white clouds, hillstream loaches in heated community tanks.	Elevated temperature raises metabolic rate, appetite, and oxygen demand, which increases feeding requirement while accelerating the animal's overall metabolic pace.	Long-term consequences for lifespan and condition in ornamentals.
Is there any evidence for the widely repeated betta 'stomach the size of its eye'?	It is the single most repeated portion rule for the most commonly kept fish.	No reliable source for the anatomical claim was located. As a conservative portion heuristic it errs safe and points in the right direction.	The original source of the claim could not be traced.

Part CVIII — Additional Topics Discovered

Adjacent material that would strengthen the eventual episode and does not fit cleanly under any brief heading.

Topic	Why it is worth including	Where it connects
The maintenance-requirement figures as a hook	The measured maintenance ration for a neon tetra — under four milligrams of dry feed per day — is the single most arresting number in this research. It converts an abstract warning about overfeeding into something a viewer can visualise.	Part III; opening of any episode segment on quantity
The three types of overfeeding	The clearest structural contribution available. It explains why keepers with spotless substrates still have nitrate problems, and why 'feed only what they can eat' is incomplete rather than wrong.	Part XIII; central to the overfeeding segment
Food as nutrient import into a sealed box	Reframes feeding from a nutrition decision to a system decision without moralising, and connects directly to the brand's system-level thinking.	Parts XIV, XCIX; the secondary cleanliness theme
The distribution-versus-quantity distinction	Resolves the largest category of real-world feeding problems and is almost absent from mainstream content.	Parts XX, LIII–LV; likely the most useful practical takeaway
Vitamin leaching within 30 seconds	A concrete, surprising, well-sourced fact that simultaneously argues for smaller portions, faster consumption, against extended soaking, and against letting food sit.	Parts XXIV, XXX, LXI, LXIII
Food-anticipatory activity as a circadian rhythm	Converts a dismissive answer ('they're just greedy') into an interesting one ('they've entrained a clock to your schedule'), which is both more accurate and more engaging.	Part XXII
The 'no fish digests cellulose' finding	A single fact that corrects pleco folklore, undercuts the pea mechanism, and reframes fibre advice — unusually high-leverage for one sentence.	Parts II, XXVIII, XL, XLII
Container size as a nutritional variable	Actionable, cheap, counterintuitive, and almost never discussed. A likely underserved search segment.	Parts LXX–LXXII
Body condition assessment as a teachable skill	No standardised ornamental scoring system appears to exist. A species-calibrated visual reference would be a genuinely novel brand asset and integrates with the Daily Tank Journal.	Part XXI; potential standalone product
The aquaculture translation problem	Explains why so much confidently cited fish nutrition data does not apply to a home aquarium, and models the honest handling of extrapolated evidence.	How to Read This Document; Part IV
Feeding as enrichment and welfare	Sales and Janssens explicitly identified welfare-correlated nutrition measurement as unexploited in aquatic species. Framing feeding as the largest available enrichment lever is both accurate and consistent with the brand's welfare-first position.	Part XCIII
Filter cleaning interval as a feeding diagnostic	An overfeeding signal that does not require catching a feeding in progress, which makes it usable by keepers who are not present at feeding time.	Part LXVI
Nitrate accumulation rate rather than nitrate level	A methodological correction with broad application beyond feeding, and one the Daily Tank Journal format directly supports.	Part LXVII; Part CII
The 'why is there no number' explanation itself	Audiences respond well to being told why a question is hard, provided they are then given a method. This is the episode's structural spine.	Parts C–CII

Glossary

Terms used in this document, including those introduced in the terminology section and those appearing later.

Term	Definition
Agastric	Lacking a true stomach and therefore lacking an acid digestion phase. Approximately a quarter of teleost fishes, including all cyprinids.
Allometric scaling	The non-linear relationship between body size and a physiological rate. Metabolic rate scales with body mass at an exponent below one, which is why small fish need a higher percentage of their body weight in food.
Amylase (α -amylase)	The starch-digesting enzyme. Activity separates dietary groups in fish more reliably than protease activity does.
Anorexia	Loss of appetite. In fish, an early, sensitive, and non-specific sign of stress, disease, or adverse environmental conditions.
Apparent satiation	The point at which fish visibly stop actively taking food. The standard observable endpoint in feeding trials; not a physiological measure of fullness.
Astaxanthin	A red carotenoid pigment responsible for red colouration in goldfish and common carp. Readily metabolised from zeaxanthin; goldfish convert very little beta-carotene and no lutein to it.
Aufwuchs	The mixed surface community of algae, biofilm, microfauna, and trapped detritus growing on submerged surfaces. Not a synonym for algae — the animal fraction matters nutritionally.
Benthic	Living or feeding at or near the substrate.
Bioload	Loose hobby term for the waste-generating burden on a system. Used in this document only where the underlying quantity — nitrogen input, oxygen demand, organic loading — is specified.
Biofilm	The bacterial and microalgal surface layer forming on any wetted surface in an established aquarium. A genuine food source for grazers, invertebrates, and fry.
Body condition	Visual or measured assessment of an animal's energy reserves and muscle mass relative to its skeletal frame.
Canthaxanthin	A carotenoid pigment used in colour-enhancing feeds; produced significant red colour enhancement in goldfish trials.
Carotenoid	A class of natural pigments fish cannot synthesise and must obtain from the diet to express their natural colouration.
Chitin	The structural polysaccharide of arthropod exoskeletons. Partially digestible by some fish via chitinase; digestibility falls at higher dietary inclusion. Relevant to insect-based feeds and crustacean prey.
Chironomid	The midge family whose aquatic larvae are sold as bloodworms.
Compensatory growth	Accelerated growth following a period of feed restriction, often accompanied by hyperphagia on refeeding. May be full, partial, or absent depending on species, life stage, and conditions.
Crepuscular	Most active at dawn and dusk.
Crude protein	Total nitrogen multiplied by 6.25. A ceiling on protein content, not a measure of digestible or available protein.
Decapsulated cysts	<i>Artemia</i> cysts with the outer shell chemically removed, usable directly as a food without hatching.
Detritivore	An animal feeding on decomposing organic matter and its associated microbial community.
Detritus	Particulate organic matter in decomposition together with its bacteria, fungi, and microfauna. The microbial load is usually the nutritional value, not the dead matter itself.
DHA (docosahexaenoic acid)	A long-chain omega-3 highly unsaturated fatty acid (22:6n-3). Essential in the diet for marine fish, which cannot synthesise it; freshwater fish can generally elongate shorter-chain precursors.
Dissolved organic compounds (DOC)	Organic material in solution rather than in particulate form. Accumulates from leached nutrients, decomposition, mucus, and metabolic products. Not routinely tested by hobbyists.

Term	Definition
EPA (eicosapentaenoic acid)	A long-chain omega-3 highly unsaturated fatty acid (20:5n-3).
Ectotherm	An animal whose body temperature and therefore metabolic rate is set by its environment. All aquarium fish.
Enrichment (of live food)	Holding live prey in a suspension of oil emulsion, microalgae, yeast, or lecithin before feeding, to raise its content of essential nutrients. Standard practice because <i>Artemia</i> nutritional value varies by strain and batch.
Essential fatty acid	A fatty acid the animal cannot synthesise and must obtain from the diet. Freshwater fish generally require linoleic acid (18:2n-6), linolenic acid (18:3n-3), or both.
FCR (feed conversion ratio)	Feed offered divided by weight gained. Lower is more efficient. In adult fish at maintenance, effectively infinite — no food is being converted to new tissue.
Feeding rate	Ration expressed as a percentage of body weight per day (% BW/day). The standard scientific unit.
Feeding zone	The vertical stratum where a species preferentially takes food: surface, midwater, or benthic.
Food-anticipatory activity (FAA)	Increased locomotor activity before an expected feeding time. A documented circadian phenomenon in fish that persists when food is withheld. Commonly misread by keepers as hunger.
Gastric evacuation	The emptying of the stomach after a meal. Its rate governs the return of appetite and therefore the sensible interval between meals in stomached species.
Guaranteed analysis	The declared minimum or maximum levels of crude protein, crude fat, crude fibre, moisture, and sometimes ash on a feed label. Describes gross composition, not nutritional value.
Hepatic lipidosis / steatosis	Fatty liver. Accumulation of lipid in liver tissue, produced by excessive energy or fat intake. Reported clinically in ornamental fish, particularly African cichlids and rainbowfish.
HUFA	Highly unsaturated fatty acids, principally EPA and DHA.
Hyperphagia	Elevated feed intake, characteristically seen on refeeding after a period of restriction.
Insectivore	An animal feeding primarily on insects and their larvae. A subset of carnivores with a small-prey feeding rhythm.
Leaching	Loss of water-soluble nutrients from food into the water before ingestion. Fastest for water-soluble vitamins and for finely divided or long-immersed food.
Maintenance ration	The intake at which body weight is held constant — no growth, no loss. Far below satiation.
Microdiet	A formulated feed manufactured at very small particle size for larval and early juvenile fish.
Nauplii	Newly hatched larval stage of <i>Artemia</i> and other crustaceans.
Physostomous	Retaining a pneumatic duct connecting the swim bladder to the oesophagus, allowing air to be swallowed into or vented from the swim bladder. Goldfish and koi are physostomous.
Piscivore	A fish-eating predator. Characteristically takes large infrequent meals.
Planktivore	An animal feeding on suspended plankton, typically via fine gill rakers and continuous small-particle intake.
Proximate analysis	Measurement of a food's gross composition: moisture, crude protein, crude fat, crude fibre, and ash.
Ration	The quantity of food offered over a defined period, usually per day. Distinct from intake.
Relative gut length (RGL)	Gut length divided by body length. Broadly higher in herbivores and detritivores, with wide overlap between dietary categories.
Target feeding	Deliberate delivery of food to a specific animal or location using a pipette, tube, tongs, or dedicated station.
Thiaminase	An enzyme that degrades thiamine (vitamin B ₁). Present in cyprinid fishes, mussels, crustaceans, and certain forage fishes. A property of the species, not of whether the prey is alive or dead.
Xylivory	Wood-eating. Practised by <i>Panaque</i> and <i>Panaqolus</i> catfishes, which consume large quantities of wood without efficiently digesting its cellulose.
Zeitgeber	An environmental cue that entrains a biological rhythm. Scheduled feeding acts as a zeitgeber in fish.

Key Evidence Summary

The quantitative anchors and the strongest findings, collected in one place for rapid retrieval.

Quantitative anchors

Figure	Value	Context	Section
Neon tetra maintenance feed	3.8 mg/day (68 J DE/day) at 0.18 g, 26 °C	Measured maintenance requirement	Part III
Moonlight gourami maintenance feed	<28.5 mg/day at 1.87 g, 26 °C	Roughly 7× the neon tetra figure	Part III
Goldfish maintenance feed	11.5–25.8 mg/day at 3.6–11.7 g, 20 °C	Measured maintenance requirement	Part III
Adult maintenance ration	~3% of body weight per day	Aquatic-veterinary guidance	Part IV
Fry and fingerling ration	Up to 5% of body weight per day	Aquatic-veterinary guidance	Parts IV, VII
Feeding rate by size	~3.6% BW/day at 0.9 g vs ~0.7% BW/day at 450 g	Standard aquaculture feeding chart, same temperature	Part IV
Optimal feeding rate range in trials	Commonly 1.6–2.5% BW/day for growing fish	Starry flounder, mandarin fish, meagre	Part IV
Return of appetite after satiation	~4 hours in Nile tilapia at 28 °C	Gastric evacuation study; 2–3 h intervals caused gastric overload	Parts II, VI
Optimal feeding frequency, betta	2 meals/day; declined at 3	1000 mg feed → 279 mg flesh at 2 meals vs 886 mg → 242 mg at 3	Part VI
Nitrogen retention in fish tissue	~25–40% typical (range ~10–49%)	Growing production fish; approaches 0% in adults at maintenance	Part XIV
Phosphorus retention in fish tissue	~17–40%; dietary P excretion often 69–86%	Aquaculture nutrient balance	Parts XIV, LXVIII
Form of nitrogen waste	60–90% dissolved, mainly ammonia	Exported principally by water change	Parts XIV, LXVII
Form of phosphorus waste	25–85% particulate in faeces (species-dependent)	Accumulates in substrate; exported by substrate cleaning	Part LXVIII
Water-soluble vitamin leaching	Large proportions of vitamin C, B ₁₂ , choline and pantothenic acid lost within 30 seconds	Commercial flake diets	Parts XXIV, XXX
Vitamin C loss in processing	~75% of supplemented vitamin C lost during processing	Shrimp feed study	Parts LXX, CVII
Vitamin C storage stability ranking	Freezer > refrigerator > opaque at room temp > clear at room temp	Ascorbic acid stability study	Parts LXX, LXXII
Dry feed shelf life after opening	Discard 2 months after opening at room temperature	Aquatic-veterinary guidance	Part LXXI
Frozen feed shelf life	Commercial feeds in freezer 6 months; frozen invertebrate foods 1 year; never refreeze	Aquatic-veterinary guidance	Part LXXI
Fasting cycles, rainbow trout	2 days/week: full compensatory growth; feed 74.8 vs 87.9 g/fish; FCR 1.00 vs 1.08; protein retention 30.5 vs 27.6%; N discharge 45.5 vs 51.1 kg/t	Feed restriction pulse trial	Part X
Fasting, Atlantic salmon	8-week fast: no welfare reduction short or long term; catch-up growth took several months	Production-scale trial	Part X

Figure	Value	Context	Section
Protein requirement range, ornamentals	~29% juvenile goldfish to ~50% discus; 53% goldfish larvae	Dose-response growth trials	Parts VII, XXV
Carbohydrate digestibility	~70% goldfish; ~50% moonlight gourami; carnivores tolerate ≤~20% digestible carbohydrate	Comparative digestibility data	Part XXVII
Relative gut length ranges	Carnivores 0.2–2.5; omnivores 0.6–8.0; herbivores 0.8–15.0 times body length	Overlapping ranges, not clean categories	Part II
Wood transit in Panaque	<4 hours; no significant cellulose assimilation; up to ~75% of gut contents is wood in the wild	Wood-eating catfish digestive physiology	Part XLII
Invertebrate protein content	Chironomid larvae ~76%; <i>Artemia</i> nauplii ~52.5%; <i>Daphnia</i> ~39.7%	Comparative live-food analysis	Parts XXXIII–XXXVI
Pea fibre content	~5% crude fibre — comparable to many prepared fish foods	Aquatic-veterinary commentary	Parts XXVIII, XL
Particle size effect on survival	~59% survival on rotifers/microdiet vs ~2% on <i>Artemia</i> alone in small-gape angelfish larvae	Larval feeding trial	Parts VII, LIX
FAA persistence under food withdrawal	78% of zebrafish shoals retained anticipatory activity at the old clock time	Circadian entrainment study	Part XXII
Carotenoid pigmentation	Astaxanthin 75 mg/kg, canthaxanthin 75 mg/kg, paprika oleoresin 180 mg/kg all improved red colour in goldfish; growth unaffected; deposition saturates	Five-month pigmentation trial	Part LXXIX
BSFL meal in goldfish	Up to 15% inclusion partially replaced fish meal with no negative effects	Ornamental insect-protein trial	Part LXXVI

The strongest and weakest conclusions in this document

Conclusion	Evidence strength
Requirement scales with biomass and body size, not fish count	[ESTABLISHED SCIENTIFIC PRINCIPLE]
Food is the dominant nutrient input to a closed aquarium and most of it is not retained in fish tissue	[ESTABLISHED SCIENTIFIC PRINCIPLE]
Roughly a quarter of teleost fishes have no stomach; all cyprinids are agastric	[ESTABLISHED SCIENTIFIC PRINCIPLE]
No fish species has been shown to digest cellulose efficiently	[STRONG EVIDENCE]
Feeding frequency beyond a species-specific optimum does not improve growth and can reduce it	[STRONG EVIDENCE]
Water-soluble vitamins leach from food within seconds of water contact	[STRONG EVIDENCE]
Fasting cycles are well tolerated by healthy adult fish and reduce feed use and nitrogen discharge	[STRONG EVIDENCE]
Begging behaviour is predominantly conditioned anticipation rather than hunger	[STRONG EVIDENCE]
High-fat feeding produces hepatic steatosis and immune effects across many finfish	[STRONG EVIDENCE]
Storage conditions materially degrade the nutritional value of fish food	[STRONG EVIDENCE]
A tank with no uneaten food can still be substantially overfed	[REASONABLE INFERENCE]
Distribution failure is a more common real-world problem than total quantity error	[REASONABLE INFERENCE]
Variety functions as nutritional insurance and enrichment rather than as necessity	[REASONABLE INFERENCE]
Body condition is the most useful hobbyist-accessible feedback signal	[REASONABLE INFERENCE]
Overfeeding contributes to algae pressure	[CONTESTED CLAIM]

Conclusion	Evidence strength
Freeze-dried food and pellets must be soaked before feeding	[CONTESTED CLAIM]
Peas relieve constipation in aquarium fish	[CONTESTED CLAIM]
<i>Daphnia</i> relieves constipation	[CONTESTED CLAIM]
Garlic treats parasites at hobby dosing and delivery	[CONTESTED CLAIM]
Fasting provides digestive rest or prevents constipation	[UNSUPPORTED AQUARIUM FOLKLORE]
Bottom feeders can live on leftovers	[UNSUPPORTED AQUARIUM FOLKLORE]
Algae eaters do not need supplementary feeding	[UNSUPPORTED AQUARIUM FOLKLORE]
Plecos eat fish waste; cleanup crews reduce bioload	[UNSUPPORTED AQUARIUM FOLKLORE]
The first ingredient must be whole fish; fish meal indicates low quality	[UNSUPPORTED AQUARIUM FOLKLORE]
Fish cannot digest carbohydrates	[UNSUPPORTED AQUARIUM FOLKLORE]
Optimal feeding frequency for adult <i>Corydoras</i> and most ornamental species	[INSUFFICIENT EVIDENCE]
Nutritional requirements of senescent ornamental fish	[INSUFFICIENT EVIDENCE]
Quantitative contribution of natural food production in a mature aquarium	[INSUFFICIENT EVIDENCE]

Claims that could not be traced to any reliable source

In accordance with the research brief, the following widely repeated recommendations were investigated and **no primary or reliable source was located**. They should be presented as untraced hobby convention rather than as established practice.

- The origin of the “feed what they eat in two minutes” rule, in any of its variants.
- The origin and anatomical basis of “a fish's stomach is the size of its eye.”
- Any controlled test of peas as a treatment for digestive impaction in aquarium fish.
- Any controlled test of whether dry pellets or freeze-dried food cause clinical harm through expansion inside aquarium fish.
- Any evidence that a scheduled weekly fasting day prevents constipation or provides digestive rest in aquarium fish.
- Any data supporting the ranking of overfeeding as the leading cause of aquarium problems.
- Any quantification of how much food benthic species incidentally obtain in a well-managed community aquarium.
- Any standardised body condition scoring system for ornamental freshwater fish.

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Sources consulted, grouped by category. All web sources accessed 12 September 2026. Entries marked *[secondary]* were not read in the original and are cited as reported in the review indicated; they should be verified against the primary publication before being quoted on air.

One attribution caveat that must be resolved before broadcast

The rainbow trout feed-restriction result cited in Part X (two fasting days per week; feed 74.8 vs 87.9 g per fish; FCR 1.00 vs 1.08; protein retention 30.5 vs 27.6 %; nitrogen discharge 45.5 vs 51.1 kg per tonne) was retrieved from a bibliographic record whose displayed title referred to an Atlantic salmon eight-week fasting study, while the abstract text described a five-week rainbow trout trial. These appear to be two separate publications and the record may conflate them.

The *findings* are reported accurately here. The *attribution* of the trout figures to a specific paper is not confirmed, and this should be resolved before the numbers are attributed to a named study in published content.

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Note on citation practice in this document

No citation in this document was invented. Where a figure comes from a review rather than from the primary publication, the review is named in text and the primary entry is marked [*secondary*] above. Where a popular recommendation could not be traced to any reliable source, that is stated explicitly in the relevant section and collected in the “Claims that could not be traced” list in the Key Evidence Summary. Two items — the rainbow trout fasting figures and the Shiau and Hsu processing-loss figure — carry attribution caveats and should be verified before being cited by name in published content.

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